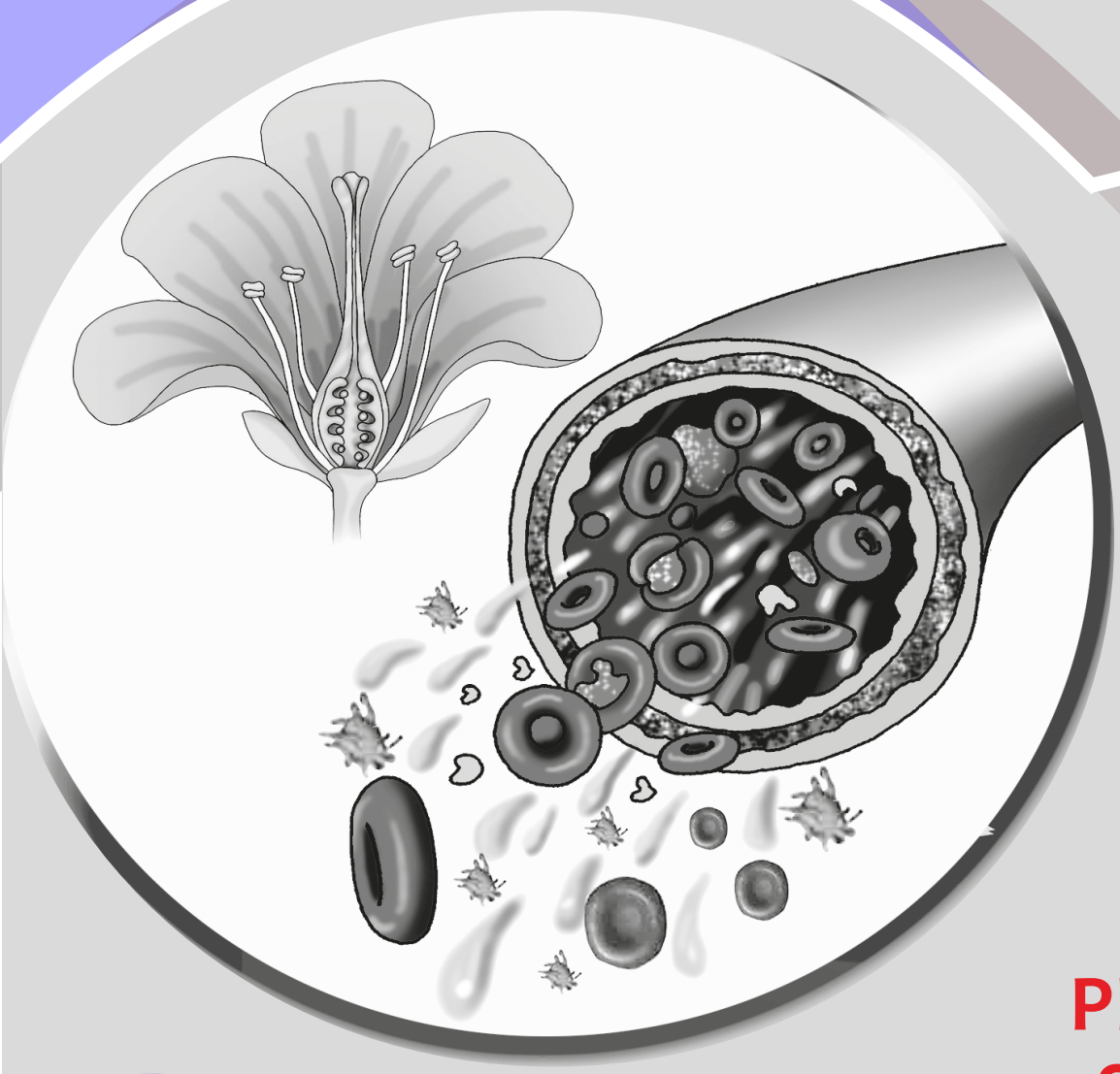
Science and Elementary Technology



PRIMARY SIX (P6)

Teacher's Guide

REB RWANDA BASIC EDUCATION BOARD

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SCIENCE AND ELEMENTARY TECHNOLOGY (SET)

PRIMARY SIX

TEACHER'S GUIDE



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FOREWORD

The Rwanda Basic Education Board (REB) is honored to present the Science and Elementary Technology Teacher's guide for Primary six. This guide is designed to support the effective implementation of the Competence-Based Curriculum (CBC), ensuring consistent and pupil-centered teaching practices that align with Rwanda's vision of empowering pupils' s to reach their full potential, actively participate in society, and pursue meaningful careers.

As Rwanda advances toward building a knowledge-based economy, education plays a pivotal role. Teachers are central to this transformation, guiding pupils to connect classroom knowledge with real-life experiences, thus fostering individual growth and national development.

This Teacher's Guide serves as a comprehensive resource for teaching Science and Elementary Technology at Primary six. The subject is an integral part of the CBC, aimed at equipping pupils with essential knowledge, skills, values, and attitudes necessary for becoming responsible and competent citizens. Through this subject, pupils are encouraged to develop problem-solving abilities and contribute positively to their communities.

This guide places special focus on engaging pupils in meaningful, activity-based learning experiences, whether individually or collaboratively where they can explore, discover, and construct knowledge. Unlike traditional approaches where learning is often passive and teacher-centered, the CBC promotes active, pupil-driven engagement. Teachers encourage pupils to participate in activities that allow them to apply what they learn, to develop key competences, and build values that empower them to positively impact their lives and society.

To support this active learning approach, teachers are encouraged to:

- Plan lessons effectively and prepare relevant materials;
- Facilitate group discussions and collaborative activities;
- Apply active teaching strategies such as inquiry, research, investigations, and both group and individual tasks;
- Create opportunities for pupils to develop competences like critical thinking, problem-solving, creativity, innovation, communication, and teamwork;
- Recognize and encourage pupils' contributions;
- Guide pupils in refining their ideas;
- Promote self, peer, and group assessments using appropriate CBC-based assessment methods.

This teacher's guide is divided into 4:

Part I: Highlights the structure of this book and gives you general methodological guidance;

Part II: Sample lesson plans as reference for lesson preparation;

Part III: Content map showing unit name, key unit competence and number of periods per each unit.

Part IV: Detailed the teaching guidance for each concept given in the pupil's book.

I sincerely thank everyone who contributed to the development and refinement of this teacher's guide. Special recognition goes to the REB team who coordinated the entire process, as well as to education development partners, teachers, and all individuals who supported the process from start to the end. Your continued suggestions and feedback are highly valued and will help improve future editions of this textbook.



Dr. MBARUSHIMANA Nelson

Director General, REB.

ACKNOWLEDGEMENT

I would like to extend my sincere appreciation to everyone who contributed to the refinement of the Science and Elementary Technology Teacher's guide for Primary six. This achievement would not have been possible without the active involvement of various education stakeholders.

Special thanks go to the Rwanda Basic Education Board (REB) staff and subject experts for their valuable work on this teacher's guide.

I am also deeply grateful to the education development partners whose support was instrumental throughout the process. In particular, I acknowledge University lectures, Teachers, Partners and all stakeholders for their contributions in editing of this guide's development.



MURUNGI Joan

Head of Curriculum, Teaching and Learning Resources Department/REB

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PART I. INTRODUCTION

1. About the teacher's guide

This book is a teacher's guide for the Primary Six for Science and Elementary Technology subject for schools in Rwanda. It is to be used with the Primary Six Pupil's Book for the same subject, to help teachers to implement the competence-based curriculum for the Science and Elementary Technology Subject syllabus.

As the name says, this is a guide that teachers can refer to when preparing their lessons. Teachers may use the guidance provided, but they should also be creative and consider their own class contexts when preparing their lessons.

This teacher's guide has three main parts:

i. Methodological guidance for the teacher

This section guides the teacher on:

- Developing pupils' competences,
- Integrating cross-cutting issues,
- Teaching learners with special educational needs,
- Using active methods and techniques for teaching Science and Elementary Technology,
- Carrying out appropriate assessment of learning.

ii. A sample lesson plan

The sample lesson plan is a guide for teachers in developing their own lesson plans.

iii. Unit development

This is the main part of the guide, because it shows the teacher how to teach the content of each unit and lesson.

Each unit begins with the unit title, and the key unit competence that are taught in the unit. (For example, the key competence of Unit 2 is to "identify various ICT tools and their uses" and "operate a radio and a television effectively"). Then the rest of the unit is presented:

- The knowledge, skills, attitudes and values that the pupils should successfully learn in the unit.
- A list of the lessons in the unit, the learning objectives of each lesson, and the number of class periods to be used for each lesson.
- A list of key vocabulary words for the unit.
- The recommended approach to teaching each lesson in the unit, including a list of teaching and learning materials for the lessons.
- Recommended questions and activities for the end of unit assessment. A list of

additional activities that can be given to less able pupils (remedial activities), normally-learning pupils (consolidation activities), and advanced pupils (extended activities). These activities could be carried out during class time, or as after-school activities.

2. Methodological guidance for the teacher

2.1. Developing values and competences

In 2015, Rwanda moved from a knowledge based to a competence-based curriculum (CBC) for pre-primary, primary and general secondary education. The CBC requires shifting the teaching and learning approach from being teacher-centred to being learner-centred. Teachers are responsible for developing their pupils' learning achievements, and for creating a safe and supportive learning environment for all pupils.

The CBC approach of teaching and learning also focuses on the students gaining specific skills, instead of just learning information that they can repeat in an exam. This approach focuses on what pupils can do, not only what they know. The pupil should be able to demonstrate the ability to use the knowledge, skills, values and attitudes that he or she has learned in a given lesson, in new or different situations.

In this approach, pupils develop basic competences with three specific learning objectives: gaining knowledge, learning skills and adopting appropriate attitudes. The teacher uses a pupil-centred approach, not the traditional teacher-centred approach, to teach each unit and lesson. The pupil can then be assessed based on whether he or she meets the standards set for those competences.

In addition to subject competences, pupils also develop generic competences which are useful in many learning areas and situations.

In the Science and Elementary Technology curriculum, the generic competences include:

Generic competences	**Ways of developing generic competences**
Critical thinking	Learning activities must require pupils to; calculate; interpret, analyze, compare and contrast (numbers, simple texts, objects, manipulative, storybooks, flash cards, charts etc.) to develop critical thinking.
Creativity and innovation	Learning activities should encourage pupils to be creative - such as building with construction blocks, painting, reading, or storytelling and writing while finding solutions to problems and sharing new ideas. Pupils learn to apply what they already know as basis for gaining new new knowledge and solving challenges both in the classroom and in everyday life.

Research and problem solving	Pupils are expected to find and use sources of information including books, the internet, and other knowledgeable people and to apply their knowledge to solve a given challenge or problems in the classroom and in real-life situations
Communication (literacy and Numeracy	All learning activities must require pupils to discuss in groups, write and present findings to the class to express their ideas to others.
Co-operation, interpersonal relations and life skills	All activities must require pupils to; work and share ideas in pairs or in groups to develop cooperation and interpersonal life skills.
Lifelong learning	Pupils should see the value of learning throughout their lives. They should make a practice of constantly gaining and updating knowledge and skills to be able to adapt to the fast-changing world.
Professional skills	The Teacher is encouraged to use range of teaching skills e.g. asking as many learners as possible to reinforce communication, probing, explaining, illustrating with examples, using blackboard, silence and non-verbal cues, using audio – visual aids, recognizing pupils' behavior to improve learning engagement in the classroom

2.2. Integrating cross-cutting issues in the curriculum

There are 10 cross cutting issues in the CBC

Competency-based learning also addresses specific *cross-cutting issues* into the teaching and learning process. Cross-cutting issues are aspects of life that are relevant to many curriculum subjects, and they should be made part of those subject lessons.

Some cross-cutting issues may seem specific to particular learning areas or subjects, but the teacher needs to address all of them whenever they come up. The pupils should also be encouraged to talk about these cross-cutting issues, in the classroom and outside the classroom, so that they can develop positive attitudes and values related to these issues. The teacher must introduce them at the most appropriate age and grade Level.

Below are examples of how cross cutting issues can be addressed:

Cross-cutting issue	Ways of addressing cross-cutting issues
Comprehensive Sexuality Education: • The Comprehensive Sexuality Education program aims to provide children, adolescents, and youth with age-appropriate, culturally and gender-sensitive knowledge, skills, and values. It helps them make responsible decisions about sexual and social relationships. It also supports understanding and expressing feelings, values, and attitudes to promote behaviors that reduce risks and enhance well-being.	• The teacher can lead pupils to discuss the following situations: "Alcohol abuse"; "unwanted pregnancies"; "home violence" etc. to advise pupils on how they can fight against them. • Give learners an opportunity to give examples of cases of abuse or violence they know • Give the students contacts where to report in case they are abused.
Genocide studies • The study of the 1994 Genocide against the Tutsi helps children understand it as the worst crime committed against the Tutsi in the1994. It teaches why the genocide happened and emphasizes the importance of preventing such atrocities in the future. The story of recovery and national unity in Rwanda serves as a foundation for rebuilding the nation. Educating children on fighting genocide denial and its ideology is crucial. This knowledge equips learners to help prevent future genocides.	• Teachers use simple methods to teach primary pupils about the 1994 genocide against the Tutsi, starting by asking what they know about the 1994 genocide against the Tutsi. • Upper primary pupils discuss 1994 genocide against the Tutsi in pairs or groups, share their ideas, and ask questions. • Teachers clearly explain what the 1994 genocide against the Tutsi is, why it happened in 1994, and why it must never happen again. • Lessons include Rwanda's recovery, national unity, and the importance of peace, with possible visits to 1994 genocide against the Tutsi memorial sites for older pupils.
Environment and Sustainability: • Integration of studies around the Environment, climate Change and Sustainability in the curriculum make pupils understand the need to protect the environment.	• Teachers should use real-life examples and pupils' experiences to explain environmental issues like soil erosion, effects of tree cutting and flooding, while encouraging responsible waste disposal and care during practical activities.

Financial education: • The integration of Financial Education into the curriculum is to teach young children the use of money; teach them to develop the culture of saving and how to use money responsibly.	• Teachers should guide pupils to discuss the importance of money, saving through methods like saving boxes or bank accounts, and spending wisely on essential needs. Pupils should also learn to use materials and tools carefully to make them last longer and avoid unnecessary replacements.
Gender: • At school, gender roles should be understood as roles that complement each other in a family, in a community or the nation.	• Teachers should address gender issue through assigning leading roles in groups to both girls and boys; provide equal participation in the lesson to avoid any gender stereotype in the learning process. • Give learners time to talk about different gender roles they know from their homes and how they complement each other. Let pupils discuss why these roles are beneficial for everyone.
Inclusive education: • Inclusion is based on the right of all pupils to a quality and equitable education that meets their basic learning needs and understands the diversity of backgrounds and abilities as a learning opportunity.	• Teachers should identify pupils with special needs and use adapted teaching and learning resources when available while conducting a lesson. • Teachers must encourage inclusive participation and foster peer support for learners with disabilities.
Peace and values education: • Peace and Values Education (PVE) is the education that promotes social cohesion, and positive values, including the culture of peace personal responsibility, empathy, critical thinking to build a more peaceful society.	• Through a given lesson, a teacher should set a learning objective which address positive attitudes and values, e.g. encourage pupils to discuss positively, listen to each other's ideas, take turns and be able to come up with meaning full positive ideas. • The teacher leads pupils to discuss on the meaning peace and why it is important to have peace at home, at school and in the community. • Encourage pupils to respect each other and to create good peaceful and friendly relations

Standardization culture: • Standardization Culture is about introducing a positive culture around the classroom that promotes respect of instructions, classroom rules; good health, cleanliness, self-development self-respect; dignity and the respect of other people and other people's culture.	• Lessons that involve making things should emphasize quality work and using the right materials properly. • Engaging pupils with practical activities and word problems helps them to understand their role and recognize how a culture of standardization contributes to health, the economy, industry, and overall well-being.
Impact of social media: • Teaching the impact of social media fosters digital literacy, creativity, and collaboration while providing access to diverse learning resources. It promotes critical thinking and responsible online behavior. However, balanced use is essential to avoid distractions and mental health issues. This prepares pupils for effective digital engagement in their academic and social lives.	• To make digital safety learning fun and engaging, the activities should be interactive, visual, and story-based, while being culturally relevant and age-appropriate. • Digital safety can be taught through Child Online Protection clubs; edutainment and interactive multimedia textbooks; storytelling and games. Edutainment makes learning engaging and practical. Role-playing and puzzles help children practice safe internet habits and handle online risks. • To set clear, age-appropriate rules for internet use, start by having open conversations about which websites and apps they want to use and explain the importance of safety.
Disaster risk reduction and management: It builds essential awareness and skills for preparedness, helping to reduce vulnerabilities and foster a culture of safety. It nurtures critical thinking, problem-solving and social responsibility, empowering children as agents of change. It aims to breaks cycles of vulnerability and supports sustainable development. This prepares future generations to adapt effectively to evolving risks like climate change.	• Engaging pupils in projects analyzing past local disasters and developing preparedness plans. • Using storytelling and role-playing based on community events to teach risks in a relatable way. • Forming disaster management clubs that promote peer learning and community outreach. • Collaborating with local experts and authorities to provide hands-on, culturally relevant education.

2.3. Attention to pupils' special educational needs in teaching and learning

Pupils in the classroom may have different ways of learning new information. Some pupils have special needs, such as impaired sight, impaired hearing, or other disabilities; these special needs mean that they will learn differently from other pupils. The teacher must be able to adapt the teaching methodologies and approaches that he or she uses, to be able to help each pupil to learn well in the classroom.

For example:

- Teachers need to use a variety of learning activities when appropriate, such as role-play, music and singing, games and quizzes, and outdoor activities if possible.
- The classroom should be well organized, in order to limit distractions. This will help pupils with special needs to stay on track and follow instructions during the lessons.
- The pace of teaching should not always be slow, or always be fast; this is because pupils learn at different speeds from each other, and they process information differently.
- Use clear, simple language to explain the meaning of new words or concepts, and use those new words more than once during the class time.
- Use multi-sensory strategies whenever possible. Since pupils learn in different ways, it is important to include listening, speaking, hands-on activities, and visual aids when teaching.

The teacher should work closely with the Special Educational Needs coordinator for the school, to help pupils with specific disabilities to learn well.

For pupils with *developmental difficulties (such as difficulties in cognitive or language development, or difficulties in behaviour)*, the teacher should:

- Use simple words and sentences when speaking to the class.
- Use objects that the pupil can see, feel and handle, instead of just talking or reading about them.
- Break tasks down into small steps or learning objectives. The leaner should start with an activity that he or she already knows, before moving on to something that is more difficult.
- Begin by giving the learner more help, and then gradually give him or her less help.
- Be sure that learners with disabilities are included in groups of non-disabled learners.

For pupils with *visual impairment,* the teacher should:

- Help the pupils to use their other senses (hearing, touching, smelling and tasting) to learn and play.
- Where possible, provide objects that the learner can touch and feel to help explain a concept.
- Speak to the pupil's parents or caregivers, to learn how the pupil manages his or her lack of sight when he or she is at home. This can help the teacher to understand how to work well with the pupil in the classroom.
- Plan group learning activities as much as possible so that the learner is not left alone.

For pupils with *hearing impairment,* the teacher should:

- Make sure that the pupil is paying attention before the teacher begins to speak.
- Use gestures, body language and facial expressions to help communicate.
- Use pictures and objects as much as possible.
- seat the pupils with hearing difficulties at the front of the classroom if possible.
- Speak to the pupil's parents or caregivers, to learn how they communicate with the child at home. Then use those same ways of communicating with the learner in the classroom. Also, find out whether the pupil has had a hearing test.

For pupils with *physical disabilities* or *difficulty in moving around*, the teacher should:

- Adapt activities so that pupils who have difficulty in moving can participate.
- Ask parents/caregivers to assist with adapting furniture to make it easier for the pupil to use it.
- Get advice from parents or a health professional about assistive devices for the pupil.

2.4. Using the active learning approach

Children in the classroom have different ways of learning new content. The main learning styles are:

- *Active learning* or *reflective learning:* learning by doing something, or learning by thinking about it.
- *Sensing learning* or *intuitive learning*: learning specific facts, or learning how ideas and facts are related.
- *Visual learning* or *verbal learning*: learning from what they see, or learning by what they hear or say;
- *Sequential learning* or *global learning*: learning one step after the other, or learning to see the larger picture all at once.

Each of these learning styles is effective for the child who uses it naturally. However, in a classroom with many pupils, the best way to teach the whole class is by using the *active learning* approach. This approach is the most effective in getting the most pupils to use their own experience and knowledge to help them learn new content.

In the active learning approach, the pupil is central to learning. Pupils are not just taught ideas and information that they passively accept; instead, they actively build on their own ideas and skills to learn new content. The pupil who is engaged in active learning can communicate with other pupils about the topic, do his or her own investigation into the topic, and develop knowledge and skills in his or her own ways.

The role of the teacher in active learning includes:

- Using pair and group discussions, pupil research and individual activities to help the pupils learn;

- Giving pupils tasks that help them grow in their ability to think critically, solve problems, be innovative, and cooperate with each other in finding answers.
- Supporting pupils' learning by recognizing their contributions to class activities.
- The Competence-Based Curriculum (CBC) approach encourages active learning methods and techniques. In the Science and Elementary Technology subject, these methods and techniques may include brainstorming by asking questions, demonstrations, role playing, discussion technique, scenario building, exploration and research, experiments, field visits, quiz technique. In science, all used active methods and techniques must be in 5 Es teaching-learning approach (Engage, Explore, Explain, Elaborate, Evaluate).

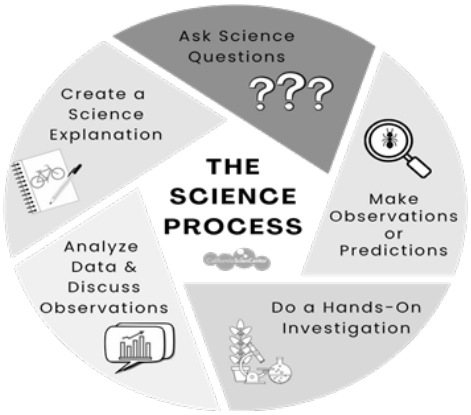
A lesson that is taught using an active learning approach should include steps like these:

- Introducing the new topic by connecting it with previous lessons;
- Developing the lesson to include discovery activities for the pupils, followed by a discussion of what they learned;
- Summarizing what was learned; and
- Exercises that link the lesson learned to the world outside the classroom.

2.5. Science process and engineering design process in teaching SET

a) The science process

Science learning focuses on inquiry and discovery. Pupils develop understanding by actively using their science knowledge and taking part in investigations. The Science Process helps learners think and act like scientists.



??? Introduce the Science Question:

Each lesson should begin with a clear science question. This question guides the lesson and keeps both the teacher and pupils

focused on the learning goal. By the end of the lesson,

pupils should be able to answer the question with increased understanding

Make Observations or Predictions:

Pupils bring prior knowledge from their experiences and previous learning. They can connect this knowledge to the science question by recording observations or making predictions in their exercise books.

Do a Hands-On Investigation:

Pupils explore the science question through hands-on investigations that build key science skills. These skills are called the Science Practices and include

- Asking questions
- Developing models
- Planning and carrying out investigations
- Collecting and analyzing data
- Using math to do science
- Creating explanations with evidence
- Obtaining, evaluating, and communicating information

During this time, the teacher supports inquiry by asking guiding questions and encouraging discussion. Pupils collect both quantitative and qualitative data. These firsthand observations provide evidence to answer the science question and may lead to new questions for further investigation.

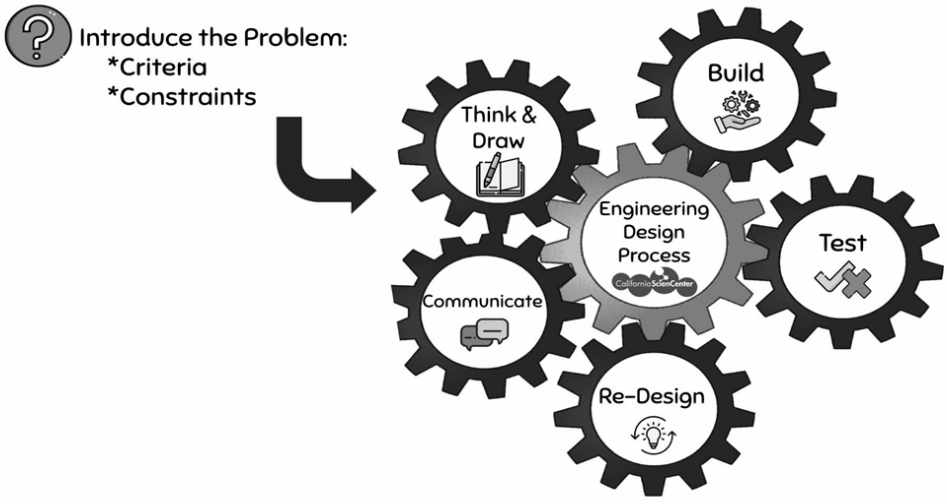
Analyze Data and Discuss Observations:

Pupils interpret their findings by examining data, comparing observations, and sharing results with peers. The teacher supports learning by introducing scientific vocabulary and addressing gaps in understanding. The goal is to develop a shared understanding of the scientific concept.

Create a Science Explanation:

The science question is revisited, and pupils explain what they have learned. Their explanations should include evidence from the investigation, analysis of data, observations, and appropriate scientific language.

b) Engineering design process



Engineering is all about solving problems! The Engineering Design Process allows pupils of all ages to think and behave .as engineers. Pupils who are solving a relevant problem will need time to think and draw, build, test, and communicate their solutions. In this process, they learn to collaborate with others and have opportunities to re-design and improve their solutions.

First, pupils must be introduced to the problem they are trying to solve. This is also the time that the criteria (what the design must include to be successful) and the constraints (what limitations are in place when solving the problem) are told to the pupils.

Pupils are then given time to think and draw. Both individually and in groups, pupils think through possible solutions to the problem and draw initial ideas for a solution. Once a possible solution is agreed upon, pupils will build their solution prototypes.

Pupils need to be given an opportunity to test their design to see if it meets the criteria and constraints that were introduced in the problem. After making observations and collecting data while testing, pupils are given a chance to re-design their solution to make it more successful. The Engineering Design Process continually flows from think and draw to redesign until the best solution is reached, or when time runs out.

Lastly, pupils communicate with their peers how and why they have designed their solution and reflect on the process.

2.6. Assessment

Assessment activities help the teacher to see how well the various pupils have learned lesson content, with the goal of improving teaching and learning in the classroom.

Two kinds of assessment are important in Rwandan education:

- *Continuous* or *formative assessment* aims to help improve the learning of the pupils and the teaching of the teacher. The pupils' learning is evaluated at the end of lessons and units; this is in order to see how the pupils are progressing, and also how well the teacher is teaching. If the assessment shows that some pupils are not learning well, the teacher must adjust his or her teaching approach to better support the pupils' learning. Formative assessment can identify slow, average and gifted pupils; different learning activities can then be given to match the speed of the pupils' learning.
- *Summative assessment* is usually done at the end of unit, end of a term or a school year. National examinations are summative. Summative assessment shows how well the entire class is learning, and the results can be used to improve school-level performance. The results of summative assessments show teachers, school leaders and parents how well the educational aims of the school and the education system are being met.

PART II. SAMPLE LESSON PLAN

School Name: xxxxxxxxxxxx **Teacher's name: xxxxxxxxxxxx**

Term	Date	Subject	Class	Unit N°	Lesson N°	Duration	Class size
3	xxxxx	SET	P6	11	2 of 7	80 min	45
Type of special educational needs to be catered for in this lesson and number of learners in each category				Hearing impairment (2): Let the pupils sit in front of the teacher so that they can hear the instruction clearly. Special attention to slow learners.			
Unit title			Human respiratory system				
Key unit competence			Pupils will be able to identify the parts of the respiratory system, explain how it works, and demonstrate healthy habits to keep it functioning well.				
Title of the lesson			Mechanism of respiration (Breathing in and out).				
Instructional objectives			Using appropriate locally available materials, pupils will be able to construct and explain correctly the mechanism of respiration (breathing in and out).				
Plan for this class (location: in / outside)			Inside the class in groups.				
Learning materials **(for all learners)**			Plastic bottles, balloons, rubber sheets and rubber band, rubber stoppers with a hole, y-shaped glass tubes, string video and animations.				
References			SET P6 Pupil's Book, SET P4 Teacher's Guide Book, Syllabus for Science and Elementary Technology, Upper primary (REB).				

Timing for each step	Description of teaching and learning activity		Generic competences and Cross cutting issues to be addressed + a short explanation.
	Guided by constructed model demonstrating breathing mechanism and working in groups, teacher helps pupils to practice and explain how the constructed model is similar to human breathing mechanism and compare the constructed model to human breathing system.		
	Teacher activities	**Pupils' activities**	
Introduction: **10minutes**	Ask questions related to human breathing system. Ask probing questions: Give examples of human respiratory organs. How does the human body get oxygen? Teacher breathes in and out and asks pupils what he /she is doing. Now we are going to learn the effect of breathing in and out to the movement of respiratory system parts. **Key question**: What happens to the respiratory organs and other body part during breathing mechanisms?	Pupils provide feedback to teacher's questions. Answers: Nose, trachea, bronchi, bronchioles, lungs, ... We get oxygen by breathing in fresh air, and we remove carbon dioxide from the body by breathing out stale air. Answer: the volume of the thorax increases.	**Communication skills:** through listening and answering teacher's question. **Critical thinking:** while thinking and predicting the answer of questions.

Lesson development: **60 minutes**	**Giving instructions:** - Arrange pupils in groups. - Assign roles in groups appropriately. - Ask pupils to follow the printed procedure to construct and use a model to demonstrate breathing mechanism. **Study questions:** a) What do you think the following parts represent in the respiratory system? a) Bell jar b) Balloons c) Rubber sheet d) Y-shaped tube b) Draw sketches to show the appearance of the balloons in what happens if you pull down the rubber sheet and if push it inside. c) Repeat the (b) steps several times as you note what happens. d) Explain your observation. - Move and guide pupils while finding out the answers to the study questions. - -Ask pupils to watch the displayed movie (link: https://www.youtube.com/watch?v=C5XGh5o9ZOU) and answer to other related questions.	In their groups: - construct a model to demonstrate breathing mechanism. Follow the procedures and answer the study questions: Answer: a)  a) Bell jar = thorax (chest cavity) b) Balloons =lungs c) Rubber sheet = diaphragm d) Y-shaped tube = trachea b) When we pull down the rubber sheet, the balloon gets inflated. Pushing the rubber sheet inside causes the balloon to collapse. The constructed model is like human respiratory system. When diaphragm goes down it makes the chest cavity larger like when pulling the rubber sheet, the space inside the bottle gets larger. Pupils answer to the asked question.	**Peace and values education:** through assigning roles in groups, respecting other's ideas during group discussions and presentations. **Gender education:** through assigning roles to both girls and boys in group discussion. **Critical thinking**: through analyzing the constructed model and answer related questions. **Inclusive education**: through providing clear instruction and procedure to everyone. Writing the keys words on chalkboard and use gestures to help pupils with hearing impairment.

<table>
<tr>
<td></td>
<td>Study question:

Explain how do you think the findings in this experiment relate to the working of the breathing system?

- Guide students to construct their explanation for each step.
- Ask pupils to present their findings
- Follow pupils' presentation and ask questions that help them to develop their critical thinking.
- Ask pupils to present the constructed model.
- Provide additional information to what pupils did not explain well.
- -Ask pupils to draw a table of two columns and describe what happens on the following organs during inhalation and exhalation: diaphragm, rib cage and thoracic cavity.
- -Guide pupils while solving the given exercise and complete the table.</td>
<td>Answer: The volume of the balloon increases during pulling down the rubber sheet. This is the same as what happen when the diaphragm contracts, the volume of the thoracic cavity increases due to the increase of lungs volume.

Pupils present their findings

The rest of the class listen to findings of the presenters and write down key notes

Pupils give answers to asked questions.

Pupils write down the additional information provided by the teacher.

Pupils use the acquired knowledge and answer the given exercise:

<table>
<tr><th>Breathing in (inhalation)</th><th>Breathing out (Exhalation)</th></tr>
<tr><td>muscles contract and diaphragm flattens</td><td>muscles relax and diaphragm forms a dome shape.</td></tr>
<tr><td>Rib cage moves upwards and outwards</td><td>Rib cage moves downwards and inwards</td></tr>
<tr><td>Thoracic cavity contract/ expands</td><td>Thoracic cavity relax</td></tr>
</table>
</td>
<td>Gender education: through allowing all boys and girls to lead groups, contribute equally and make presentation.

Cooperation: through helping each other and taking decision together.

Lifelong learning: through accepting and analyzing what happens during breathing in and out in breathing mechanism.</td>
</tr>
</table>

Conclusion: 10 minutes	Help pupils to draw conclusion by highlighting the key points and give a summary note. Give pupils exercises. Briefly explain a series of events that take place during breathing mechanism.	Pupils draw conclusion to check the achievement of objectives. Pupils do the exercise. The mechanism of breathing involves two main processes: inhalation and exhalation. Inhalation occurs when the diaphragm moves down and the chest gets bigger. Exhalation occurs when the diaphragm moves up and chest becomes smaller.	**Critical thinking**: through analyzing and providing answers to questions. **Gender education:** through allowing both boys and girls to contribute equally while drawing conclusion
Teacher self-evaluation	The lesson has been well conducted and the fixed objective has been well achieved as learners can explain correctly the mechanism of respiration.		

PART III. CONTENT MAP

SCIENCE AND ELEMENTARY TECHNOLOGY (SET), PRIMARY 6			4 periods per week
No	**Unit title**	**Key unit competence**	**Number of periods per unit**
1	Mechanics tools	Pupils will be able to identify and use mechanics tools correctly, take care of them, and act safely and responsibly when working with tools.	6
2	Simple machines	Pupils will be able to identify, use, and care for simple machines safely, and explain how they make work easier.	6
3	Air pollution	Pupils will be able to identify the causes and effects of air pollution, take action to reduce it, and show care for clean air in the environment.	6
4	Cows and goats management	Pupils will be able to explain how to keep and care for cows and goats properly and show good habits towards animals living in homes and communities.	12
5	Word processing and spreadsheet	Manipulate documents using a Word Processing program and perform mathematical calculations using a Spreadsheet program and format a spreadsheet document.	12
6	Introduction to emerging technologies	Explore, apply, and reflect on the use of emerging technologies to solve real-life problems and improve everyday activities.	6
7	Programming for Children	Use Scratch tod esign shapes, do calculations, and create simple animation with Sprites and backgrounds.	14
8	Plants reproduction	Pupils will be able to identify and explain plant reproduction, recognize its different types, and demonstrate care for plants.	8
9	Sustainable waste management	Pupils will be able to explain and sort different types of waste, use ways to manage waste safely, and show care for the environment by handling waste properly.	8

10	Human circulatory system	Pupils will be able to identify the parts of the circulatory system, explain how it works, and demonstrate healthy habits to keep it functioning well	16
11	Human respiratory system	Pupils will be able to identify the parts of the respiratory system, explain how it works, and demonstrate healthy habits to keep it functioning well.	8
12	Human reproductive system	Pupils will be able to identify the parts and functions of the human reproductive system and demonstrate appropriate hygiene practices, prevention and responsible behaviors toward sexually transmitted infections and diseases.	16
13	Energy management	Pupils will be able to identify forms and sources of energy, explain energy transformations, and demonstrate responsible use of clean energy to protect the environment.	7
14	Magnetism	Pupils will be able to explain how magnets work, identify their types and uses, make a temporary magnet, and demonstrate interest in exploring magnets in daily life.	7
TOTAL PERIODS			132

PART III: UNIT DEVELOPMENT

Unit 1

MECHANICS TOOLS



i. Key unit competence

Pupils will be able to identify and use mechanics tools correctly, take care of them, and act safely and responsibly when working with tools.

ii. Prerequisites

This unit is about mechanics tools. Remember that pupils at this level have already interacted with tools before. For example, in Primary 4, they learnt about agricultural tools, their safe use and maintenance. While in Primary 5, they learnt about carpentry and masonry tools. Take advantage of this and link what they learn here with their past experiences about these tools. Ask pupils some probing questions on the Prerequisites to assess their progress.

Further, inform pupils that the concepts in this unit can be applied in metal work. Let pupils understand that when they continue their education in this area, they may become mechanical or civil engineers.

Introductory activity and guidance

a) Guidance on the introductory activity

This topic is about mechanics tools, their uses and maintenance.

- Ask pupils to look at the picture on the introductory activity and let them discuss what they see.
- Let them brainstorm in five minutes to discover what is taking place in the picture of the introductory activity.
- Ask them questions about the picture and orient their answers to the right ones.
- Ask pupils to give the topics they think this unit will include based on the picture.
- Give time for brainstorming and after share the main sub-units.
- Guide the pupils to discover that it is important to learn mechanics as a way of getting out of such situations.

b) Possible answers to the introductory activity

1. It shows a van or mini bus with passengers with a punctured tyre. The passengers are becoming impatient with the driver.
2. The person near the car (driver) is trying to repair it but he is stuck and does not know what to do.
3. He is using a hammer. It is the wrong tool to remove the tyre.
4. Mechanics commonly use spanners, pliers, hammers, screw drivers, jacks, metal saws, drills and many others.

iii. Generic competences:

- Critical thinking: Developed through identifying appropriate mechanics tools for different tasks, analyzing the dangers of their misuse, and selecting suitable safety precautions.
- Research and problem solving: Developed through observing mechanics tools, visiting a garage, investigating their uses and maintenance, and applying safety measures during practical work.
- Co-operation, interpersonal relations and life skills: Developed through group discussions, role-plays, practical activities, and collaborative preparation of safety checklists.

iv. Basic values:

- Dignity and integrity: Developed through handling mechanics tools responsibly, following safety rules, and maintaining tools properly after use.
- Hard work: Developed through active participation in practical activities, proper maintenance of tools, and careful completion of assigned tasks.
- Commitment and resilience: Developed through practicing correct tool-handling techniques, learning from mistakes, and consistently applying safety procedures.

v. Cross- cutting issues to be addressed

1. Environment, climate change and sustainability

The oils and other chemicals that come from machines when not well disposed cause pollution. By causing pollution our environment gets spoilt. This may bring problems. Caution the pupils against disposing of these wastes anyhow.

2. Gender education

Highlight that Rwanda promotes equal opportunities for men and women in science, technology, engineering, and mathematics (STEM) fields and Technical education (eg.TVT, TSS). Emphasize to pupils that both men as well as women can earn an income by becoming a mechanic.

3. Peace and values education

Inform the pupils that to be successful in mechanics job, it requires to be honesty and trustful, cooperative and having teamwork spirit, to easily and effectively solve the conflict, to respect each other , to be responsible and disciplined and gender inclusive. Advise pupils not to use mechanics tools as weapons to hurt others, people should live in peace and harmony in order to develop. Inform them that they should be willing all the time to accommodate views of others. In a case, a child gets hurt accidentally others should provide first aid before taking him or her to the hospital.

4. Financial education

Let the pupils understand that well maintained tools last for long thereby reducing maintenance and replacement costs. Stress the fact being a mechanics can help them to earn money and live better. Tell them how to manage their income, grow their businesses, and make savings.

5. Standardization culture

Encourage pupils to use branded or certified tools that meet international or national standards for their durability and performance.

6. Inclusive learning

During practical activities, allow all pupils to participate equally irrespective of their gender, physical disability or mental challenges. During pairing, bright pupils should be mixed with slow pupils.

vi. List of lessons

#	Lesson title	Learning objectives	Number of periods: 6
1	The common Mechanics tools	• Identify and explore some mechanics tools	2
2	Use and maintenance of mechanics tools	• Explain the use and maintenance of mechanics tools • Explain how each tool is used and how to take care of mechanical tools • Develop interest in using tools to make and fix things • Show care and respect for mechanics tools.	1

3	Dangers of the misuse of mechanics tools	• Identify potential dangers of misusing mechanics tools • Recognize the dangers of using tools in the wrong way. • Take responsibility for their own safety and the safety of others.	1
4	Precautions when using mechanics tools	• Identify the ways of preventing and reducing the chances of dangers • Describe ways to keep safe and avoid accidents when using mechanics tools • Follow safety rules when handling tools. • Use tools in a safe and respectful way to protect themselves, others, and the environment.	1
5	End of unit assessment	• Assess the acquired competences • Demonstrate the achievement of the key unit competence	1

Guidance on different lessons

1.1 The common mechanics tools

a) Learning objectives

Identify and explore some mechanics tools

b) Teaching and learning resources

- Common mechanics tools including as spanner, pliers, screwdriver, hack saw, mechanics hammer, hand drill, and bench vice, photographs showing mechanics tools, videos on mechanics tools, VCD or DVD player, charts of common mechanics tools on manila papers to be used in case you do not have charts on the mechanics tools.
- The teacher may use locally available mechanics tools, models, pictures, or videos as appropriate to the lesson context to support learning.

c) Activities' guidance and answers

Guidance to the learning activity 1.1

i) Ask pupils to look at the picture on the learning activity and let them discuss what they see.

ii) In case you have a pupil, who has visual impairment or disability, let the other pupils support in the description of the image by telling them what they can see. For example: In the picture 1; the pupils will describe that there is a person who is repairing a car. In the picture 2, the pupils will describe that there are different materials or metallic

objects.

iii) Let them brainstorm for three minutes to discover what is taking place in the picture of the learning activity.

iv) Guide pupils to answer the questions asked on the pictures. **N.B:** Remember that there is no right or wrong answer here, the purpose of the learning activity is to introduce pupils' prior knowledge on the content to be learnt.

v) Build on the pupils' prior knowledge and introduce them to the common mechanics tools.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answer to activity 1.1

1. The person in the picture is repairing a car.
2. Some pupils will answer 'Yes, I have seen similar objects at home or garage, or in our community'. Some pupils might answer 'No, we have not seen similar objects at home or garage, or in our community'. There is no correct answer because this is a yes or no question that is to introduce pupils to the new content based on the prior knowledge of a pupil.
3. The tools in the pictures are found in the garages or in places where they are people repairing different kinds of machines; for example, a pupil might answer that she/he found such tools at home when the parents had called a **technician** at home to repair some objects. (We are expecting the pupil not to be aware of the name 'A mechanic' that is why you can expect the pupils of use the name 'Technician')

Guidance to the application activity 1.1

i. Lead the pupils to visit a mechanics workshop or garage nearby. When in the garage, the pupils should identify the various mechanics tools.

ii. Let one of the mechanics at the workshop guide them. You should be closer to guide them too in order to prevent accidents that may occur. In case there is no garage nearby, show pupils a video on mechanics at work. Let them watch the video carefully and list down the various tools being used and their uses.

iii. Let pupils ask probing questions such as: *Who is a mechanic? What do mechanics use to do their work? What do we call places where mechanics work from?*

iv. Let pupils observe how the various tools are being used. They can come up with a table like this:

Name of a tool	Tool (Drawing)

During the activity, keep on observing and hearing what pupils do and answer, and note their progress. At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarize the content.

Answers for application activity 1.1

Refer to the summary under the sub-heading 1.1 in the pupil's book for the names and the drawings of mechanics tools.

1.2. Use and maintenance of mechanics tools

a) Learning objectives

- Explain the use and maintenance of mechanics tools
- Store mechanics tools properly and practice safe maintenance of them.

b) Teaching and learning resources

- Common mechanics tools such as spanner, jacks, pliers, screwdriver, hack saw, toolbox, grease and oil, photographs showing mechanics tools and their maintenance
- Where necessary, the teacher may improvise mechanics tools or use suitable locally available alternatives to demonstrate their use and maintenance.

c) Activities' guidance and answers

Guidance to the activity 1.2

i) Ask pupils to look at the picture on the learning activity and let them discuss what they see.

ii) In case you have pupils, who have visual impairment or disability, let the other pupils support in the description of the image by telling them what they can see. For example: The pupils may describe that they can see a muslim lady who has a plier in her hands, and trying to bend or cut a wire in the picture (a), etc.

iii) Let them brainstorm for three minutes to discover what is in the pictures of the learning activity.

iv) Ask the pupils to answer the questions that follow the pictures. N.B: Remember that there is no right or wrong answer here, the purpose of the learning activity is to introduce pupils' prior knowledge on the content to be learnt.

v) Build on the pupils' prior knowledge and introduce them to the uses and maintenance of mechanics tools.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the learning activity 1.2

1. The lady in the picture (a) is cutting the wire using a plier.
2. The young man in (b) is greasing his plier after using it.
3. The man in (c) is driving the screws to hold a metal.

Guidance to the application activity 1.2

i) This application activity involves reading and the comprehension of the text so that pupils will be able to find the answers from the text.

ii) Ask the pupil to read the scenario carefully, and this must be engaging, and you have to make sure that all the pupils are following the one who is reading.

iii) Pupils will take turns in reading. The role of the teacher is to make sure that all pupils are accommodated in regard to their special educational needs. If needed, those who are advanced might need to read slowly for other to catch up.

iv) After reading, let the pupils write down the tools in their notebook they have found in the scenario.

v) Provide the answers to the reflection questions.

vi) During the activity, keep on facilitating how pupils are reading and answering, and note their progress. At the end of the lesson, ask again some questions on the lesson learnt to assess the learning and summarise the content.

Answer to the application activity 1.2

Tools to write down: Screwdriver, Hand drill, Hacksaw.

1. Amina used a screwdriver to drive the screws and hold together small metals that were separated. Their father used a hand drill to make more new holes for extra screws. Then Kamanzi used a hacksaw to cut a small, damaged piece of metal.
2. Kamanzi cleaned his hacksaw, Amina and his dad put the screwdriver and the hand drill in a toolbox for safe keeping.

1.3. Dangers of the misuse of mechanics tools

a) Learning objective

Identify potential dangers of misusing mechanics tools

b) Teaching and learning resources

- Toolbox containing some mechanics tools, photographs showing different dangers of mechanics tools, videos on the dangers of misusing the mechanics tools, charts of common dangers mechanics tools on manila papers to be used in case you do not have photographs showing the dangers on the mechanics tools.

c)Activities' guidance and answers



Guidance to the activity 1.3

iv) Ask pupils to observe the picture on the learning activity and let them discuss what they see.

v) In case you have pupils who have visual impairment or disability, support in the description of the image by telling them what they can see. For example: There is a little boy who was using a hammer and he is crying, or there is a nail in a wood and the boy has some blood on his hand, etc.

vi) Ask the pupils to answer the questions that follow.

vii)Build on the pupils' prior knowledge and introduce them to the possible dangers of misusing the mechanics tools.

Note: Ensure to develop the values and generic competencies related to this lesson.



Answers to the activity 1.3

1. The danger observed in the picture is that the boy's finger is bleeding. The boy has injured his finger when hammering / fixing the nail into the timber.
2. The boy was hitting a nail in the wood.
3. The boy was hurt by a mechanics hammer



Guidance to the application activity 1.3

i) This activity explores the acquisition of pupils' knowledge to the possible dangers of the misuse of mechanics tools.

ii) Let the pupils observe carefully the pictures, the pictures include those that are safe and those that are not safe.

iii) Ask pupils to observe carefully without talking for one minute, and then answer the questions that follow.

iv) Ask pupils to answer the questions that follow the picture and make sure that the pupils answer one by one.

v) During the activity, keep on facilitating how pupils are answering, and note their progress. At the end of the lesson, ask again some questions on the lesson learnt to assess the learning and summarise the content.

Answers to the application activity 1.3

1. b. He used the tool without asking.

2. c. He got hurt.

3. Because you can get injured or cause accidents if you don't know how to use the tools properly.

4. Two dangers of misusing mechanics tools:

(a) Injuries (cuts, bruises, or broken bones)

(b) Damaging tools or equipment

1.4. Precautions to take when using mechanics tools

a) Learning objectives

- List different measures to be taken to prevent and reduce the dangers when using mechanics tools
- Identify common hazards associated with the use of mechanical tools in a workshop environment

b) Teaching and learning resources

- Protective clothing such as gloves, mouth masks, goggles, overall, first aid kit, photographs showing different precautions to take into considerations when using mechanics tools, videos on the precautions when using the mechanics tools.
- The teacher may improvise appropriate teaching and learning resources based on the lesson context and locally available materials.

c) Activities' guidance and answers

Guidance to the activity 1.4

i) Ask pupils to examine the picture on the learning activity and let them discuss what they see.

ii) In case you have a pupil who has visual impairment or disability, support in the description of the image by telling him/her what they can see. For example: In the picture, I can see a man wearing a helmet in his head, or I can see a car, or I can see a mechanic using an electric saw to cut metals, or I can see a mechanic not wearing a coverall properly.

iii) Ask pupils to observe carefully for one minute to answer the questions that follow.

iv) Allow pupils to answer the questions one by one to assess each pupil's acquisition of knowledge and competence on the lesson learnt.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 1.4

1. Answers from pupils may vary from one another. A pupil may reply that: 'I can see a man wearing a helmet in his head, or I can see a car, or I can see a mechanic using an electric saw to cut metals, or I can see a mechanic not wearing a coverall properly' etc. (The last point is the best).
2. The person is wearing a helmet, gloves, gumboots, and a coverall.
3. No, he should also put the arms in the coverall to protect the whole body.

Guidance to the application activity 1.4

i) This activity is much similar to the application activity 1.3, but it emphasizes on safety precautions.

ii) Let pupils observe carefully the pictures that include the safe and unsafe use of mechanics tools.

iii) Allow pupils to answer the asked questions, and ensure that each pupil participates fully to assess their acquisition of the new knowledge and competence

iv) During the activity, keep on facilitating how pupils are answering, and note their progress. At the end of the lesson, provide summative questions and summarise the lesson learnt.

Answers to the application activity 1.4

1. Yes, we have safe and unsafe use of mechanics tools in pictures
2. **A. Safe use:**
   - A person mechanic wearing goggles for safety (Picture d).
   - A child using a screwdriver with gloves (Picture a).

   B.Unsafe use:
   - A mechanic being cut by a sharp hacksaw (Picture b).
   - A mechanic wearing open shoes in the garage (Picture a).

3. A mechanic being cut by a sharp hacksaw: We should keep the mechanics tools safely after use so that people cannot be hurt by misplaced sharp materials.

   A mechanic wearing open shoes in the garage: The mechanic should wear protective gumboots in order to be safe.

1.5. End unit assessment

I. Guidance to end unit assessment

1. This part provides guidance and answers for end of unit assessment with cross reference to the textbook.
2. Tell pupils that the last part of the unit is the end of unit assessment comprising of the project where they apply what they learnt and questions that check pupils' knowledge, understanding, skills, attitudes, and values related to the whole content of the unit.

3. For the project:

a) Request every pupil to do the project either at school (in small groups of 4 to 5 pupils) or at home. As a teacher, you will decide if it is possible to do the project at school or home and give them guidance. They will use materials available in their community.

b) Share the problem as stated in the pupil's book.

c) Explain to pupils that their toolbox will need to pass the following tests:

i) It must be large enough to hold the following tools:

1. 1 open ended spanner
2. 1 hacksaw
3. 1 pliers
4. 1 screwdriver

ii) It must be strong enough to carry these tools when picked up.

iii) It must have a divider inside, so that each tool has its own space.

iv) Each tool space must have a label to show where the tool belongs.

d) Encourage pupils to follow the steps of the Engineering Design Process by:

i) Drawing their individual designs in their exercise books before discussing with others. Remind them to think about the criteria (passing the tests) as they design their toolbox.

ii) Using the learning resources to build their toolboxes

iii) Requesting pupils to submit their work, organize a class display table or mini exhibition and allow pupils to test their tool boxes by asking themselves or others to demonstrate the following:

1. Does the toolbox have a labeled divider for each tool?
2. Can the toolbox be picked up and carried?

e) Reward the objects made or work of pupils by providing feedback and marks.

f) Select some of the good toolboxes made by pupils and display them in class.

4. For the quiz:

a) Choose some questions for the quiz depending on available time for the assessment. The quiz must be done individually whereby pupils will submit their responses on papers to be marked.

b) All questions may not be done during the period(s) allocated to the end of the unit assessment, request them to answer the questions during their self-study time or as homework.

c) Mark their answers and keep records of every student's achievement.

d) Provide the remedial, consolidation and extend activities as necessary.

II. Answers to the end of unit assessment

Possible answers for the project

The toolboxes do not need to be the same design or use the same learning resources. Accept all innovations.

1. We used ____ to make our toolbox because ____.
2. A revision we would make to our toolbox would be ____ so that it would ____.
3. It is important to keep tools clean and in the right place because _____.
4. Tools that are left on the ground _____.
5. A toolbox protects people and tools by _____.

A. Knowledge and understanding

1. B. Open-ended spanner
2. C. Making holes
3. C. Jack
4. Hacksaw
5. Screwdriver
6. Pliers
7.

Tools	Use
1. Hand drill	Making holes into wood or metals
2. Bench vice	Immobility or holdding metals
3. Spanner	Tightening of nuts and bolts

B. Skills

8. D. Clean and dry them before storing
9. B. Gloves and closed shoes
10. Toolbox
11. Oiling
12. False

C. Attitudes and values

13. True
14. False
15. C. Tell a teacher or help them use it safely
16. C. Stop work immediately and report it

1.6. Additional activities

A. Remedial activities

1. Some mechanics tools that I know are……and……
2. What are the two precautions to take in a workshop that you know?
3. We should…… all tools after use.
4. Who is a mechanic?

❖ **Answers to remedial activities**

1. Spanner, screwdriver, pliers, hack saw, bench vice, etc.
2. Wearing protective clothes and using tools carefully.
3. Clean
4. Somebody who repairs or maintains tools/machines.

B. Consolidation activities

1. What is the importance of mechanics in our society?
2. Mention different methods used to maintain mechanics tools?
3. What would happen to tools if the mechanics tools were not properly maintained?
4. Mention different dangers that we face while working in a mechanics workshop?
5. What should be done in the garage to safeguard the lives of people working in the garage?

❖ **Answers to consolidation activities**

1. Mechanics repair vehicles, machinery, tools, and equipment keeping them in good working condition and help in the economic growth.
2. The tools can be maintained by oiling, greasing and cleaning and wiping.
3. If the tools are not properly maintained, they rust and get damaged
4. These include:
   a) Being injured by sharp tools.
   b) Cuts or burns on our skin while using the tools.
   c) Dangerous chemicals getting into our eyes, nose or mouth.
5. Such measures include:
   - Workers should always wear appropriate safety gear such as gloves, safety goggles, overalls, ear protection, and safety boots.
   - Using the tools carefully to avoid injury
   - Keep the garage clean and organized:
   - Dispose of dangerous chemicals safely so they cannot affect the lives of people.

C. Extended activities

1. Explain how mechanics interfere with the environment.

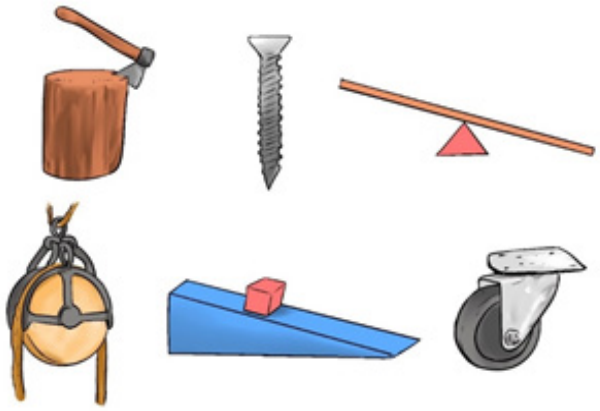
2. Discuss the advantages of mechanics in the society.

❖ **Answers to extended activities**

1. Oil and grease from garages pollute the environment. Used vehicle and machines parts are a danger to the environment in general.
2. - Mechanics ensure that cars, buses, trains, and planes are safe and functional, supporting mobility and economic activity.
   - Mechanics create and provide jobs to the people.
   - Mechanics enhance safety and improve quality of life.

Unit 2

SIMPLE MACHINES



i. Key unit competence:

Pupils will be able to identify, use, and care for simple machines safely, and explain how they make work easier.

ii. Prerequisitess

The content in this unit is about types of simple machines, their categories and safety precautions to take while using simple machines. It may help to remind pupils what they learnt in previous unit on mechanics tools. Also, remind them about what they learnt under agricultural tools, carpentry and masonry tools in primary 4 and 5.

Let pupils answer some questions on the Prerequisites to help their understanding that the majority of these tools are examples of simple machines. It may also help to differentiate between these simple machines and complex or heavy machinery used in agriculture and industry.

Finally let pupils understand the fact that concepts learnt in this unit can be applied in physics and other areas such as woodwork and metalwork.

Let pupils know that when they continue their education in this area, they may become mechanical or civil engineers.

Introductory activity and guidance

a) Guidance on the introductory activity

This topic is about simple machines and all the related concepts of simple machines.

- Ask pupils to look at the picture on the introductory activity and let them discuss what they see.
- Let them brainstorm for five minutes to discover what is being done in the picture of the introductory activity.
- Ask them questions under the picture and try to orient their answers to the right ones.
- Guide the pupils to discover and suggest some of the simple machines that are used in the picture.

- Let pupils compare situations when simple machines are used and when they are not used and make conclusions on which is better.
- Ask pupils whether they could know what topics they think will be covered in the unit based on the picture.
- Give time for some brainstorming and after share the main sub-units.

b) Possible answers to the introductory activity

1. No, it was not easier to move materials.
2. The answer is no, because the workers are doing many tasks that involve manual labor like lifting, carrying, and repetitive motions, which can strain muscles and joints, causing physical fatigue.
3. Workers can use wheelbarrows to move the cement and pulleys to take the cement from the ground floor to the second floor.

iii. Generic competences:

- Critical thinking: Developed through distinguishing simple machines from other tools, classifying them into different types, and explaining how each type makes work easier.
- Research and problem solving: Developed through observing, exploring, and building simple models of machines, investigating how they work, and applying them to solve everyday problems.
- Co-operation, interpersonal relations and life skills: Developed through group discussions, station rotation activities, sorting games, and collaborative practical work involving simple machines.

iv. Basic values:

- Dignity and integrity: Developed through using simple machines responsibly, following safety procedures, and taking proper care of tools and classroom models.
- Hard work: Developed through active participation in practical investigations, model construction, and correct use of simple machines to accomplish tasks.
- Commitment and resilience: Developed through repeated practice in using different simple machines safely, learning from mistakes, and improving practical skills.

v. Cross cutting issues to be addressed

1. Environment, climate change and sustainability

Most of the simple machines can be made by cutting down trees; pupils should be cautioned not to practice deforestation. Used oils and other chemicals from simple machines when not disposed of well, cause environmental pollution. Caution the pupils against disposing of wastes and their management.

2. Gender education

Inform pupils that both men as well as women can earn a living by making and using simple machines.

3. Peace and values education

Make pupils aware that simple machines should not be used to hurt others. People should live in peace and harmony in order to develop. Inform them that they should be willing all the time to accommodate views of others. In case a child gets hurt accidentally others should provide first aid before taking him or her to the hospital.

4. Financial education

When we make and use our local simple machines, our work becomes easy and cheap. Encourage pupils to develop a habit of making their own simple machines where possible. Let the pupils understand that, well maintained tools last for long hence reduces costs of replacement. Also, let pupils know that when they grow up, they can earn a living from making and fabricating simple machines.

5. Standardization culture

Warn pupils against using counterfeit materials as this may lead to frequent breakdowns which end up being very costly. Also encourage pupils to always buy and use original products/tools.

6. Inclusive learning

Allow all pupils to participate during activities irrespective of their gender, physical disability or mental challenges. Also during pairing, bright pupils should be mixed with slow pupils.

vi. List of lessons

#	Lesson title	Learning objectives	Number of periods: 6
1	Definition of simple machine and difference between simple machine and other tools	• Define simple machines • Differentiate simple machines from other types of tools. • Draw and label different types of simple machines. • Build simple models or patterns of levers and other machines	2
2	Types of simple machines	• Identify different types of simple machines. • Describe the three classes of levers. • Group simple machines based on their classes.	2

3	Dangers and safety in using simple machines	• Explain possible dangers of using of simple machines andhow to prevent them. • Safe handling of different simple machines. • Be aware of safety when working with simple machines. • Act responsibly when using machines to avoid harming self or others. • Use simple machines carefully and correctly.	1
4	End of unit assessment	• Assess the acquired competences • Demonstrate the achievement of the key unit competence	1

Guidance of different lessons

2.1. Definition and difference between simple machine and other tools

a) Learning objectives

- Define simple machines
- Differentiate simple machines from other types of tools.

b) Teaching and learning resources

Common simple machines such as saw, hammer, bottle opener, wheelbarrow, pulleys, photographs or charts showing simple machines, if you do not have the photographs, you can improvise or use videos.

c) Activities' guidance and answers

Guidance to the activity 2.1

iv) Ask pupils to examine the pictures on the learning activity and let them discuss what they see.

v) In case there is a pupil who has visual impairment or disability, let other pupils support in the description of the image by telling her/him what they can see. For example: In the picture A; there is a little boy who is trying to lift and carry a sack of potatoes using hands. In the picture B; pupils might describe that there is a girl who is wearing glasses and she is happily carrying a sack of potatoes on a wheelbarrow.

vi) Allow the pupils to answer the questions asked in the picture.

vii) Build on the pupils' prior knowledge and introduce them to the definition of simple machines and their differences from other tools.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 2.1

1. The boy in A is trying to lift and carry a sack of potatoes using his hands.
2. The girl in B is moving a sack of potatoes using a wheelbarrow.
3. The girl in picture B finds the work easier by moving a sack of potatoes using a wheelbarrow because she is using less effort.

Guidance to the application activity 2.1

i) This application activity is a project work that pupils will have to complete to assess their understanding on the difference between simple machines and other tools.

ii) Read and introduce the application activity to the pupils and make sure that all the pupils are able to complete the activity.

iii) Give time to the pupils to ask questions before taking the activity at home so that they can do the project work as required. N.B: If you do not make the instructions clear, the pupils will be misled, and they might have difficulty in completing the work.

iv) Pupils with special need should also be accommodated according to their needs. For example, if you have pupils who have physical disability, tell them that they can find support from parents or siblings, or you can pair them with advanced pupils from the same class.

v) Inform the pupils the time to present their work so that they can bring their developed simple machines to school.

vi) During the activity, keep on facilitating how pupils are answering, and note their progress. At the end of the lesson, provide summative questions and summarise the lesson learnt.

Answers to the application activity 2.1

The answer to this application activity is the project results (Simple machines developed) that pupils will demonstrate at the preferred time that you have set. Refer also to the content of subheading 2.1

2.2. Types of simple machines

a) Learning objectives

- Identify different types of simple machines.
- Go through the engineering design process to design a solution to lift a model of a fallen tree.

b) Teaching and learning resources

- Common simple machines such as saw, hammer, bottle opener, wheelbarrow, pulleys, etc
- Photographs or charts showing simple machines.

- Scissors, scotch or glue, wires or string, learning resources to create with (e.g., wood, cloth, paper, plastic, foil, cardboard, etc.), learning resources to test designs (a pencil, stick or metal bar to model a fallen tree, and a ruler).

c)Activities' guidance and answers



Guidance to the activity 2.2

i) Tell pupils to observe the pictures on Activity 2.2 in the pupil's book and let them discuss what they see. When there is a pupil who has a visual impairment or disability, let other pupils support in the description of the image by telling him/her what they can see.

ii) Ask the pupils to answer the questions that follow and harmonise their answers.

iii) Build on the pupils' prior knowledge and introduce them to the problem as stated in part B of engineering a solution in Pupil's book.

iv) Explain to pupils that in real life, large problems sometimes need to be thought about at a smaller scale before building. Show pupils a model of the school yard with the fallen tree model. Explain to pupils that they are going to design a model (representation) to solve this problem.

v) Explain to pupils that their solution will need to pass the following tests:

a) Lift the fallen tree to at least 13cm off their desks

b) Use at least 2 different kinds of simple machines

c) Fit on their desktop space

vi) Divide the class into small groups of 3-4 pupils to complete this task.

vii) Show pupils the learning resources that will be available to them to make their solution.

viii) Tell pupils to draw their individual design in their exercise books before discussing with others. Encourage them to think about the criteria (passing the tests) as they design their solution.

ix) Allow pupils to share their individual ideas with their group afterwards.

x) As a group, they will decide which design to build together. It is okay to combine ideas as well. The groups' solutions do not need to be the same. Accept all innovations.

xi) Give groups the learning resources to build their designs.

xii) As groups finish building, gather them around the model of the school yard and fallen tree to begin testing. To test the design solutions, groups should attempt to use their device to lift the model of the fallen tree, and make observations of how their device performs during the test.

xiii) Give groups time to re-design and re-build next class period if desired or during the free period.

xiv) Have pupils explain their design to another group.

xv) Prompt pupils to talk about the following:

a) How did the group decide which simple machine to use

b) How the simple machines make the work of lifting the tree easier

c) How they decided which learning resources to use

d) What happened when they tested their device

e) What re-designs they need (or needed) to make and why

Answers to the activity 2.2

1. Possible answers on what pupils might see: a screwdriver in A, a wood/hand saw in B, a ladder in C , and a wheelbarrow in D.
2. - A screwdriver drives screws to hold together the metals,
   - A wood/hand saw cuts trees easier,
   - A ladder helps people to climb on different levels at the construction sites,
   - A wheelbarrow helps us to carry loads easily.
3. No, they do not do the same work. Each machine has its specific task; that is why there are different types of simple machines.

- For the solutions designed in Part B, the solutions do not need to be the same design or use the same learning resources. Accept all innovations.

Possible answers in discussions when engineering a solution:

- We decided to use the simple machines ____and ____ because ____.
- Using the simple machine of ____ makes work easier by ____.
- We used the following learning resources ____ because ____.
- When we tested our device, it ____.
- If we were to re-design, we would ____ because ____.

Guidance to the application activity 2.2

i) This application activity involves doing a small hands-on experiment with some reflection questions on the lesson learnt.

ii) Guide pupils on the first question of the application activity 2.2 step-by-step, and make sure to engage all the pupils according to their special needs.

iii) Provide the answers to the reflection questions.

iv) During the activity, keep on facilitating how pupils are answering, and note their progress. At the end of the lesson, provide summative questions and summarise the lesson learned.

Answer to the application activity 2.2

1. a. After a small experiment of lifting a stone using a strong piece of wood, pupils will be able to move the stone;

b. To move the stone without a piece of wood will be hard;

c. Because the piece of wood was acting as a simple machine/lever;

2. Three examples of simple machines: Levers, inclined plane, wheel and axle, wedges.

3. Levers: bottle opener; inclined planed: stairs/staircase; wheels and axle: bicycle; wedges: knife.

2.3. Dangers and safety in using simple machines

a) Learning objectives

- Explain possible dangers of using of simple machines and how to prevent them.
- Safe handling of different simple machines.

b) Teaching and learning resources

Common simple machines such as saw, hammer, bottle opener, wheelbarrow, pulleys, photographs or charts showing simple machines, if the photographs are not available, improvise or use videos.

c) Activities' guidance and answers

Guidance to the activity 2.3

i) Ask pupils to look at the picture on the activity 2.3 and let them discuss what they see.

ii) If there is a pupil who has visual impairment or disability, let the other pupils support in the description of the image by telling him/her what they can see. For example: In the picture A, a man got injured by a needle while sewing using hand, in picture B, a carpenter was cut by sharp wood plane, and in picture C, a young boy feels off the bicycle.

iii) Ask the pupils to answer the questions that follow and harmonise their answers.

iv) Build on the pupils' prior knowledge and introduce them to the possible dangers and safety in using simple machines.

Note: Ensure to develop the values and generic competencies related to this lesson.

❖ **Answers to the activity 2.3**

Answers to the activity 2.3

1. A man got **injured** by the needle.
2. Machines with sharp edges can **cut** the body and cause injury.
3. People can **fall** and get **hurt** while using machines like ladder or bicycle.
4. No, the things that happened to these people are not good.

Guidance to the application activity 2.3

i) The application activity involves matching the pictures with the safety methods.

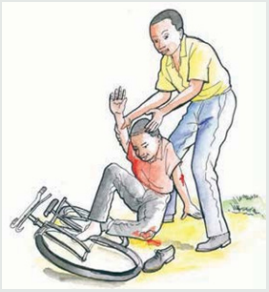
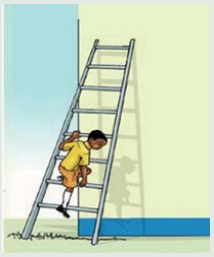
ii) Guide the pupils to match the dangers with the correct safety measures .

iii) While pairing pupils, make sure that they are paired according to their needs and abilities so that they could help one another. N.B: Be gender responsive and inclusive in your pairing.

iv) Provide the correct answers to the pupils.

v) During the activity, keep on facilitating how pupils are answering, and note their progress. At the end of the lesson, provide summative questions and summarise the lesson learnt.

Answer to the application activity 2.3

Danger	Safety
	Move slowly and wear a helmet when you are on a bicycle.
	Ask a friend to hold the ladder you when you are going to climb it.
	Wear gloves when using a hammers.
	Be careful when using a needle.

2.4. End unit assessment

I. Guidance to end unit assessment

1. This part provides the guidance and answers for end of unit assessment with cross reference to the textbook.
2. Tell the pupils that the last part of the unit is the end of unit assessment comprising the project where they apply what they learnt and questions that check pupils' knowledge, understanding, skills, attitudes and values related to the whole content of the unit.
3. **For the project:**

- Request every pupil to do the project either at school or at home. As a teacher, you will decide if it is possible to do the project at school or home and give them guidance. They will use materials available in their community.

- Request pupils to submit their work, organize a class display table or mini exhibition and allow pupils explain how they made their objects and how it is useful.
- Reward the objects made or work of pupils by providing feedback and marks.
- - Select some of good objects made by pupils and display them in class.

4. **For the quiz,** choose some questions for the quiz depending on available time for the assessment. The quiz must be done individually whereby pupils will submit their responseson papers to be marked.
5. As all questions may not be done during the period(s) allocated to the end of unit assessment, request them to answer the questions during their self-study time or as homework.
6. Mark their answers and keep records of every student's achievement.
7. Provide the remedial, consolidation and extended activities when necessary.

II. Answers to the end of unit assessment

A. Knowledge and understanding

1. Possible answers: Bottle opener: Levers; Stairs/staircase: inclined planed; Bicycle: Wheels and axle; Knife: Wedges.
2. Simple machines are important because they help people to do work more easily.
3. Door handle, a steering wheel, bicycle and a windlass.
4. Wheel and axle.
5. The main parts of a lever are Effort, Load and a Fulcrum
6. Screws and bolts are used to hold and join pieces of metals or wood together; screws are also used in car jacks. They are under the type of screws.
7. B. Wheelbarrow
8. C. Fulcrum
9. B. Lift objects
10. Inclined plane
11. Pulley
12. Wedge

B. Skills

13. B. Use it carefully and safely
14. B. Ask a friend to hold it
15. Helmet
16. Balance
17. False
18. True
19. False

20. B. Keep them clean and store safely
21. Responsibility
22. I learned that a small force can lift a heavy object when using a lever. It was fun building a mini seesaw and testing it.
23. - Always check that the machine is working properly before using it.
    - Do not put your hands or fingers near moving parts while the machine is working.
24. Simple machines help us do work more easily. At home, we use a knife (wedge) to cut fruits, and a broom (lever) to clean the floor. At school, we use scissors (lever) to cut paper. They save time and energy and make our work easier and faster.

2.5. Additional activities

A. Remedial Activities

1. A…… makes work easier.
2. Some examples of simple machines include……..and……..
3. Which are the six classes of simple machines?
4. A windlass is an example of a ……. machine.

❖ **Answers to remedial activities**

1. Simple machine.
2. Wheelbarrow, tin opener, bottle opener, nut cracker, pair of scissors, fishing road, etc.
3. Levers, wheel & axle, pulley, screw, wedge, inclined plane.
4. Wheel and axle.

B. Consolidation activities

1. What is the importance of the invention of simple machines?
2. Name the three types of pulleys.
3. What is a wheel and axle?

❖ **Answers to the consolidation activities**

1. The invention of simple machines by early human beings helped us to become civilized and live a simpler life than before.
2. Single fixed, movable and block and tackle.
3. A wheel and axle is a simple machine that consists of a circular object (the wheel) connected to a shaft (the axle) in a way that both rotate together. When one part rotates, the other part also rotates, allowing for force transfer and potential amplification.

C. Extended activities

1. Why is it not recommended to open a bottle using your teeth?
2. Compare and contrast a windlass and a staircase.
3. What is the difference between block and tackle pulley and a movable pulley?

❖ Answers to extended activities

1. It may break the teeth.
2. A windlass is an example of a wheel and axle; a staircase has steps and is an inclined plane.
3. Movable moves; block and tackle is fixed.

Unit 3

AIR POLLUTION



i. Key unit competence:

Pupils will be able to identify the causes and effects of air pollution, take action to reduce it, and show care for clean air in the environment.

iii. Prerequisites

Pupils were already introduced the concept of air and its components in primary four. They also learnt about the uses of the main components of air i.e. nitrogen, oxygen and carbon dioxide. In this lesson, they are supposed to learn about air pollution, its causes, consequences and how it can be controlled. Further, it is also important to highlight the fact that the concepts in this unit can be applied in environmental management in social studies. Also, pupils should be made aware when they continue with their education in this area, they may become environmentalists. Ask some questions on the Prerequisites lessons to assess their progress.

Due to increased human activities that include industrialization, transportation, use of fuel for power generation, cooking, cigarette smoking and many others; such activities lead to air pollution. When air is polluted, it becomes harmful to human beings, animals and plants.

Introductory activity and guidance

A. **Guidance on the introductory activity:**

- During this lesson, air pollution, its causes, effects and control/ prevention measures will be explained
- Use the pictures in pupil's book on the introduction activity to guide the pupils on what they are going to discover in this lesson.
- Let pupils observe the pictures carefully. There are pictures of a moving car, factories releasing smokes and fumes and a place where garbage are burnt releasing smokes to the environment. Another picture shows a mother and her children cooking using charcoal in a closed room.
- If there is any pupil with disability, make sure to cater for him/her too by providing the needed assistance or by pairing them with advanced leaners.

- Ask the pupils to answer the questions that follow. N.B: Remember that there is no right or wrong answer here, the purpose of the learning activity is to introduce pupils' prior knowledge on the content to be learnt.
- Build on the pupils' prior knowledge and introduce them to the concepts of air pollution that concern this unit.

B. Possible answers for introductory activity

1. Burning garbage and release of smokes
2. An industry/factory releasing fumes
3. A mother and her children cooking food in a closed room.
4. In picture (D) there is a moving car exhausting fumes. The similarity in all the pictures is that either garbage, the factory, the charcoal stove, or the car are all releasing toxic smokes.

iii. Generic competences:

- Critical thinking: Developed through identifying sources, causes, and effects of air pollution, analyzing real-life situations, and proposing appropriate measures to reduce pollution.
- Research and problem solving: Developed through field observations, experiments, case studies, and investigating practical ways to prevent and control air pollution in the community.
- Communication (Literacy and Numeracy): Developed through presenting findings, creating posters, participating in discussions, and communicating solutions to promote clean air.

iv. Basic values:

- Dignity and integrity: Developed through taking personal responsibility for protecting air quality by avoiding activities that pollute the environment and encouraging responsible behavior.
- Patriotism: Developed through appreciating the importance of protecting Rwanda's environment and actively participating in keeping the country clean and healthy.
- Commitment and resilience: Developed through consistently adopting environmentally friendly practices and encouraging others to reduce air pollution despite challenges.

v. Cross cutting issues

1. Gender education

Both boys and girls should participate in conserving our environment. It is not a preserve of the boy or girl gender. Also, emphasize to pupils that anybody irrespective of their gender can pursue a carrier in environmental conservation. Give examples of role models who are successful in the area where the pupils come from.

2. Standardization culture

Emphasize the need to use certified equipment and machines which do not pollute the environment. Caution pupils against using counterfeits which may lead to accidents or injury of users or frequent breakdowns

3. Inclusive education

All pupils should be encouraged to participate during lessons and practicals. Special arrangement should be made to take care of pupils with special needs. For example, provide brail for blind pupils, large print text for those with sight problems and allocate physically challenged pupils to others to assist them during field trips and practical activities. Further, this category should be given tasks that they can manage during the practical sessions.

4. Financial education

Pupils should be made aware of the fact that a clean environment ensures healthy living devoid of diseases hence helps to reduce health related costs. Also, buying quality equipment will lead to saving money in the long run as they stay for long and costs on repairs are reduced.

5. Environment and sustainability education

This is a topic that directly affects the environment. Emphasize the need to reduce air pollution and other forms of pollution in general. Caution the pupils against activities that cause air pollution like cigarette smoking, bush burning among others.

6. Peace and values education

People should learn to live in harmony and they should be ready to help other people, especially those who are affected by natural disasters like floods or landslides.

vi. List of lessons

#	Lesson title	Learning objectives	Number of periods: 6
1	Definition of air pollution and common air pollutants	• Define the term air pollution and recognize a polluted air. • Identify common air pollutants.	2
2	Sources of common air pollutants	• Identify the sources of common air pollutants • Apply the knowledge of recognizing air pollutants	1

3	Consequences of polluted air	• Identify causes and dangers associated with air pollution • Explain the causes and dangers of air pollution.	1
4	Protection of air against air pollutants	• Explain how to protect air against air pollutants • Develop positive attitude towards avoiding air pollution • Describe how to keep the air clean and safe. • Appreciate the value of clean and fresh air. • Show interest in keeping the air clean. • Act responsibly by avoiding activities that pollute the air. • Speak out and encourage others to stop polluting the air.	1
5	End of unit assessment	• Assess the acquired competences • Demonstrate the achievement of the key unit competence	1

Guidance on different lessons

3.1. Definition of air pollution and Common air pollutants

a) Learning objectives

- Define the term air pollution and recognize a polluted air.
- Identify air pollutants

b) Teaching and learning resources

- Textbooks, or charts showing air pollution, photographs of polluted cities or towns, computers connected to the Internet, any other improvised materials that can demonstrate the concepts of air pollution.

c) Activities' guidance and answers

Guidance to the activity 3.1

i) Ask pupils to look at the picture on the activity 3.1 and let them discuss what they see.

ii) In case of a pupil who has visual impairment or disability, let the other pupils support in the description of the image by telling him/her what they can see.

iii) Ask the pupils to answer the questions that follow and harmonise their answers.

iv) Build on the pupils' prior knowledge and introduce them to air pollution and its common pollutants.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 3.1

1. There some burning charcoal, the factory releasing smokes and dirty water, cars exhausting fumes, and pupils burning garbage.
2. It is very bad to the environment
3. Toxic substances such as carbon dioxide, carbon monoxide and unburnt fuel are emitted to the atmosphere when fuels such as gasoline and diesel oil are burned in automobiles. These substances cause a variety of environmental problems such as air pollution and global warming leading to the damage of plants, human and animal health.

Guidance to the application activity 3.1

i) The application activity 3.1 will involve parents and pupils should be provided with clear instructions on how to approach their parents.

ii) The activity is made of four questions that are open-ended and the questions require to take short summary of answers.

iii) The answer might vary in the ways of formulations, but they should be the same in regard to what was learnt on in this lesson.

iv) This activity makes the Prerequisites of the next lesson. Therefore, pupils should submit their completed activities before the start of the next lesson.

v) Inform the pupil when to submit their findings.

vi) During the activity, keep on facilitating how pupils are answering, and note their progress. At the end of the lesson, provide summative questions and summarise the lesson learnt.

Answer to the application activity 3.1

These are the possible answers to the activity

1. Charcoal is made by burning wood slowly with little air. First, wet or dry wood is stacked in a pit or kiln and covered to stop too much air from getting in. Then, the wood is burned slowly, which removes water and gases. After cooling, the black charcoal is collected and used for cooking or heating
2. No, making charcoals is not good to the environment, because it involved cutting down trees and burning them, which releases harmful smokes to the environment.
3. The plastic wastes are disposed with other household waste.
4. Yes, they are bad to the environment. Because they require to be burnt when being disposed, they do not rot.

3.2 Sources of common air pollutants

a) Learning objectives

- Identify the sources of common air pollutants
- Apply the knowledge of recognizing sources of air pollutants

b) Teaching and learning resources

- Petroleum jelly, paper, glass jar, materials to burn (paper, wood, charcoal, strips of cloth), lighter or matches.

c) Activities' guidance and answers

Guidance to the activity 3.2

i) Tell pupils that they will do two investigations to answer this science question: What are sources of air pollution we find in our daily lives?

ii) Remind pupils of what they learned about air pollution.

iii) Let volunteers define what air pollution is. Build on this and ask whether the pupils know what causes air pollution.

iv) Let pupils look at the pictures of activity 3.2 in the pupil's book and share what they think is happening. For pupils with visual impairment assist them or pair them with the advanced learners to help them in the description of the pictures.

v) Let them suggest how the activities in the picture are going to affect the air.

vi) Guide the pupils to reach the intended answers.

vii) Build on pupils' prior knowledge and introduce them to air pollution and its common pollutants.

Then, proceed as follows for Part B:

viii) Remind pupils that they will do the second investigation to answer this science question:

What are sources of air pollution we find in our daily lives?

ix) Ask pupils what were the sources of air pollution they found in pictures previously observed.

x) Instruct pupils to think about the following materials the teacher has chosen for this investigation (e.g., paper, wood, charcoal, strips of cloth). Ask pupils to predict what they think will happen if these materials are burnt. What might happen to the air around them? Will they find the same air pollutants as they observed on pictures. Why or why not?

xi) Divide the class into small groups of 3-4 pupils to carry out this investigation.

xii) Give each group a glass jar, a material to burn (vary the materials given to each group), and a lighter or matches.

If it is not possible to do this investigation in small groups, the teacher can do it as a pupil-led demonstration. Allow pupils to assist with the demonstration, making predictions, and observations of what is occurring.

xiii) Instruct the pupils to place their materials in the glass jar. The groups will light the materials on fire, and gently close the jar – not tightly, some oxygen must be able to get into the jar.

xiv) Instruct pupils to make observations of what they see happening to the air in the jar.

xv) After the investigation is complete, ensure all jars have no burning materials left in them that could pose a safety hazard to the pupils.

xvi) Discuss with pupils the observations they made.

a) Ask pupils to share what they think caused the black soot on the inside of the jar to form.

b) Ask pupils if this is a form of air pollution, if so, which one.

xvii) Restate the science question that was introduced at the beginning of the lesson: What are sources of air pollution we find in our daily lives? Have pupils answer the science question in their own words. Encourage them to use information from the lesson.

Answers to the activity 3.2

1. This is a complex picture with different possible sources of air pollution. Pupils' answers will vary depending on what they see.
2. Yes, they are the sources of air pollution.
3. Some sources include burning charcoal, fireplaces, burning garbage, volcanic eruption, fuels from transformers or power plants, dust from sweeping, etc.

Possible answers in discussions:

- I think the black soot forms on the inside of the jar because ____.
- I think this is a form of air pollution because ____ (smoke, sulfur dioxide, carbon dioxide and carbon monoxide is given off when we burn materials).

Possible written response to science question:

Sources of air pollution we find in our daily lives are dust, carbon dioxide, carbon monoxide, smoke, and sulfur dioxide. I know this because we saw the dust particles on the petroleum jelly we left outside, and we saw the smoke in the jar when we burned (material).

Guidance to the application activity 3.2

i) Application activity 3.2 will involve some materials from home, like petroleum jelly. The pupils should be provided with clear instructions on how to run this practical activity.

ii) The use for petroleum jelly is to test the presence of dust particles from the wind.

iii) The activity requires the participation of all pupils. Make sure that your lesson is inclusive.

iv) The activity will be done outside of the classroom and the observations can be made some time after the start of the lesson.

v) Guide your class in the activity and guide them to answer the reflection question.

vi) During the activity, continue to facilitate how pupils are answering and note their progress. At the end of the lesson, provide summative questions and summarize the lesson learned.



Answer to the application activity 3.2

After hanging the paper in the tree for some hours, some dust particles will stick to the paper due to the presence of petroleum jelly. The conclusion is that the experiment confirms the presence of dust particles in the environment, and they are among the sources of air pollution.

3.3. Consequences of polluted air

a) Learning objectives

- Identify causes and dangers associated with air pollution

b) Teaching and learning resources

- Computers connected to the Internet, plastics or rubber materials like tyre or other kinds of plastics/rubber available for the activity 3.3, textbooks and charts on air pollution, charts on sources of consequences of pollution, photographs of polluted cities or towns, or drawings of consequences of air pollution in manila papers.
- The teacher may improvise appropriate teaching and learning resources based on the lesson context and locally available materials.

c) Activities' guidance and answers



Guidance to the activity 3.3

i. You may introduce the lesson using probing questions such as 'what do you think are the consequences of polluted air? Let them have a brainstorming session.

ii. Let pupils perform experiment in the activity 3.3, this activity needs your facilitation so that pupils could not get burnt.

iii. Let them hold group discussions at school and answer the questions asked in the activity 3.3.

iv. Let them write their findings down and correct them when they are back in the class.

v. Summarise the lesson by highlighting main points as pupils write summary notes.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 3.3

1. Smokes will spread in the whole place.
2. The smell is bad
3. Polluted air causes coughing, irritation of eyes, and throat leading to asthma attacks

Guidance to the application activity 3.3

i. This is the application activity 3.3, and it involves research, and it is a take-home activity.

ii. Make sure that the instructions are clear for all the pupils and make it inclusive.

iii. The first instruction is to visit a sector officer in charge of agriculture, but if the pupil lives far away from the sector, she/he may consult a secondary school teacher who teaches Chemistry.

iv. As a teacher, be flexible when more than one pupils want to visit the sector officer because this might encourage them to work as a group and that improves their communication, and collaboration and cooperation skills.

v. Let the pupils know the time for submission, and how they will present their findings.

vi. During the activity, keep facilitating pupils on how to conduct the activity and be flexible to assist them along the way of their research, and note their progress. At the end of the lesson, ask some questions on the performed activities or content learnt to assess the learning and summarise the content.

Answers to the application activity 3.3

Answers may vary from one pupil to another.

3.4. Protection of air against air pollutants

a) Learning objectives

- Explain how to protect air against air pollutants
- Develop positive attitude towards avoiding air pollution

b) Teaching and learning resources

- Computers connected to the Internet, textbooks and charts on air pollution, charts on control measures of pollution, photographs of clean and well-kept environment with many trees, or drawings of such an environment in manila papers.
- The teacher may improvise appropriate teaching and learning resources based on the lesson context and locally available materials.

c) Activities' guidance and answers

Guidance to the activity 3.4

i) You may introduce the lesson using probing questions such as 'how do you think we can reduce air pollutants in the atmosphere?

ii) Let them have a brainstorming session.

iii) Let pupils look at the picture and say what is going on in the pictures of the activity 3.4 pupil's book. For the pupils with visual impairment assist them or pair them with the advanced leaners to help them in the description of the pictures.

iv) Let them suggest how the activity in the picture is going to affect the air.

v) Guide the pupils to reach on the intended answers

vi) Build on the pupils' prior knowledge and introduce them to air pollution and its common pollutants.

vii) Emphasize the fact that 'PLANTING OF TREES' is the best way of ensuring clean air. Bring to their attention the fact that this is why the Rwanda government emphasizes on plating two trees every time one is cut.

viii) Summarise the lesson by highlighting main points as pupils write summary notes.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 3.4

1. Pupils in the picture are planting trees in their schools.
2. The activity in the picture is good to the environment.
3. The activity in the picture is good and very important because planting plenty of trees creates forest cover, which helps absorb carbon dioxide. Forests act as both water catchment areas and recycling points for gases.

Guidance to the application activity 3.4

i. This application activity might be done in groups or individually depending on your preferred choice.

ii. Let pupils look at the picture and say what is going on in the pictures of the application activity 3.4 pupil's book. For the pupils with visual impairment assist them or pair them with the advanced leaners to help them in the description of the pictures.

iii. Let them categorize the activities in the pictures between 'Environment friendly' and 'Environment harmful' to check their understanding of the learnt content.

iv. During the activity, keep on observing and understanding what pupils do and answer, and note their progress. At the end of the lesson, ask some questions on the performed activities or content learnt to assess the learning and summarise the content.

Answers to the application activity 3.4

1. Environment harmful (a), Environment friendly (b), Environment harmful (c), Environment friendly (d).
2. Some ways to reduce air pollution include:
   - Reducing the use of vehicle and use more public transport
   - Using solar panels for electricity.
   - Using electric car and motorbikes.
   - Setting up industries far away from residential areas.
   - Reducing the use of charcoal and firewood and use gas or biogas in cooking.

3.5. End unit assessment

I. Guidance to end unit assessment

1. This part provides the guidance and answers for end of unit assessment with cross reference to the textbook
2. Tell the pupils that the last part of the unit is the end of unit assessment comprising the project where they apply what they learnt and questions that check pupils' knowledge, understanding, skills, attitudes and values related to the whole content of the unit.
3. **For the project:**

- Request every pupil to do the project either at school or at home. As a teacher, you will decide if it is possible to do the project at school or home and give them guidance. They will use materials available in their community.
- Request pupils to submit their work, organize a class display table or mini exhibition and allow pupils explain how they made their objects and how it is useful.
- Reward the objects made or work of pupils by providing feedback and marks.
- Select some of good objects made by pupils and display them in class.

4. **For the quiz,** choose some questions for the quiz depending on available time for the assessment. The quiz must be done individually whereby pupils will submit their responses papers to be marked
5. As all questions may not be done during the period(s) allocated to the end of unit assessment, request them to answer the questions during their self-study time or as homework.
6. Mark their answers and keep records of every student's achievement.
7. Provide the remedial, consolidation and extended activities when necessary.

II. Answers to the end of end unit assessment

A. Knowledge and understanding

1. C. Harmful gases and dust in the air
2. C. Carbon dioxide
3. C. Volcanic eruption
4. Trees
5. Smoke
6. Global warming
7. Sulphur dioxide, nitrogen dioxide and carbon monoxide.
8. Burning charcoal produces carbon monoxide in a closed room due to limited supply of air. This can cause carbon monoxide poisoning.
9. Acid rains, destruction of atmosphere, global warming.
10. Smoking cigarettes in public places is disastrous because it causes respiratory diseases to the passive smokers (non- smokers) and it also contributes to air pollution.
11. (a) Global warming is when the earth gets hotter than it should be because of too much pollution in the air.

    (b) Planting trees improves forest cover which in the long run helps in cleaning up the surrounding air by using carbon dioxide during photosynthesis; and avoid mining and using a lot of vehicles that releases smokes and carbon dioxide.
12. Air pollution is severe in cities because this is where most industries are located, also there are always very many vehicles in the cities.
13. Vehicles release smokes like carbon dioxide which leads to global warming, their engines produce sound and causes noise pollution, we cut trees to build roads, and a lot of movement of vehicles cause a lot of dust in the atmosphere.

B. Skills

14. C. Use clean cooking methods
15. C. Reuse or recycle it
16. Filter
17. Solar
18. False

C. Attitudes and Values

19. True

20. False

Choose the correct answer

21. B. Ride a bike instead of using a car

22. Respect

3.6. Additional activities

A. Remedial Activities

1. Air pollution is _____
2. _____gases are examples of air pollutants.
3. State how air pollution affects plants.
4. Define air pollutant?
5. One method of controlling air pollution is to plant_____.

❖ **Answers to the remedial activities**

1. Contaminating air with air pollutants.
2. Carbon dioxide, carbon monoxide, Sulphur dioxide, nitrogen dioxide.
3. Air pollution brings acid rain which burns plants. Also smoke or dust particles covers leaves which interferes with their normal processes of surviving (Like receiving the sunlight for photosynthesis).
4. Air pollutants are substances that cause air pollution
5. Trees

B. Consolidation activities

- 1.Name one natural and one artificial source of air pollution.
- 2.Suggest different ways of safeguarding air against pollutants.

❖ **Answers to the Consolidation activities**

1. Natural dust from volcanic eruptions and that created by wind. Artificial – smoke from industries, fumes from vehicles, etc.

2. Ways of safeguarding air against pollutants:

1. Use less cars: Walk, ride the bike, or take the bus instead of going by car all the time . Fewer cars means less smoke in the air.

2. Plant more trees: Trees clean the air by taking in pollution and giving out fresh oxygen.More trees means cleaner air.
3. Use less energy: Turn off lights and gadgets when not in use. Power stations burn fuel to make electricity, so saving energy helps reduce pollution.
4. Don't burn rubbish: Burning rubbish creates dirty smoke. Recycle or compost instead.
5. Use safe products: Some sprays and cleaners have chemicals that pollute the air. Use natural or eco-friendly ones instead.
6. Factories need filters: Factories should use special filters to stop smoke and gases from going into the air.
7. Reduce waste: Recycling helps reduce pollution from landfills.

C. Extended activities

1. As a P6 pupil, use your critical thinking skills to explain the term "go green" as a solution to air pollution in our environment.
2. What differentiates a polluted and a non-polluted environment?
3. The consequences of not taking care of our environment are so severe. Discuss.

❖ **Answers to the extended activities**

1. The term 'go green' means reducing pollution by all possible ways, mostly by planting trees. When you reduce pollution, you help to conserve our environment.
2. Polluted is contaminated with pollutants, the other one is not.
3. Refer to consequences of air pollution sub-heading 3.3 as explained in pupil's book.

Unit 4

COWS AND GOATS MANAGEMENT



i. Key unit competence:

Pupils will be able to explain how to keep and care for cows and goats properly and show good habits towards animals living in homes and communities.

ii. Prerequisites

Pupils will learn better this unit if they remember various classes of animals that is mammals, birds, fishes, reptiles and amphibians in primary 4. Further, in primary 5, pupils were taught about chicken rearing and good practices that should be carried out in chicken farming. In this class, pupils will further be taught and practice goat and cattle farming. They will learn about proper housing for goats and cattle, goat and cattle breeds, how to feed goats and cattle and how to prevent goat and cattle diseases. Ask pupils some reflection questions or give them a task to revise on these content leanrt in Primary 4 and 5.

There is also need to sensitize pupils about the fact that the concepts in this unit can be applied in Agriculture under animal management. Let them know that when they continue and further their education in this area, they may become Agricultural officers, veterinary doctors or professional farmers.

Introductory activity and guidance

a) Guidance on the introductory activity

- This topic is about cattle and goat rearing.
- Ask pupils to look at the picture on the introductory activity and let them discuss what they see. N.B: For pupils will visual impairment pair them with advanced pupils to help describe the picture. Be inclusive and accommodate everyone.
- Let them brainstorm in five minutes to discover what is taking place in the picture of the introductory activity.
- Ask them questions under the picture and try to orient their answers to the right ones.
- The picture is associated with traditional cattle keeping whereby cattle are kept within a very small piece of land, which is not friendly to the environment.

- Remind pupils about the cross-cutting issue environment sustainability.
- What topics do they think this unit will include based on the picture?
- Give time for some brainstorming and after share the main sub-units.

b) Possible answers to the introductory activity

1. The cows are feeding on the pasture/grass (The cattle is grazing on the outside pasture).
2. The boy is herding his cows in the outside pasture.
3. It is not good. This practice goes against the modern way of farming which discourages practices like zero grazing because it is not environment friendly.

iii .Generic competences:

- Research and problem solving: Developed through field visits, conducting research on cattle and goat breeds, diseases, feeding practices, and applying appropriate animal management techniques.
- Critical thinking: Developed through comparing local and exotic breeds, selecting suitable animals for rearing, and analyzing appropriate methods for disease prevention and proper livestock management.
- Co-operation, interpersonal relations and life skills: Developed through group discussions, collaborative farm activities, and participation in school livestock projects while sharing responsibilities and ideas.

iv .Basic values:

- Hard work: Developed through active participation in feeding, cleaning, and caring for livestock, recognizing that successful animal farming requires commitment and diligence.
- Dignity and integrity: Developed through treating animals humanely, maintaining proper hygiene, and carrying out livestock management activities responsibly.
- Patriotism: Developed through appreciating the contribution of livestock farming to food security, household income, and Rwanda's socio-economic development.

v. Cross cutting issues to be addressed

1. Environment, climate change and sustainability

As farming or keeping of animals is being started, there are a lot of trees cut down to construct the animal houses or sheds. While cutting down trees and clearing the farm area we should be conscious not to destroy the environment. In fact, we should plant two trees for every one tree cut down.

2. Gender education

Inform pupils that both men as well as women can earn a living by becoming professional

farmers or practicing commercial farming. Farming is NOT a preserve of one gender!

3. Peace and values education

Caution pupils against using farm tools as weapons to hurt others. Stress the fact that people should live in peace and harmony in order to develop. Inform them that they should be willing all the time to accommodate views of others. Also, in case a child gets hurt accidentally others should provide first aid before taking him or her to the hospital. Also, inform them that as they share animals according to the government policy it should be peacefully and harmoniously done.

4. Financial education

Let the pupils understand that farming or keeping of farm animals is highly beneficial to the people and government of Rwanda. They can comfortably ache a living fromfarming in future therefore they should not look at it in a negative way.

5. Standardization culture

Encourage pupils to learn about new innovations in modern farming. They should migrate from traditional farming to modern farming methods. Also, caution them against using counterfeit materials and feeds when doing farming.

6. Inclusive learning

Allow all pupils to participate during class/group activities irrespective of their gender, physical disability or mental challenges. Also during pairing, bright pupils should be mixed with disadvantaged pupils.

vi. List of lessons

#	Lesson title	Learning objectives	Number of periods: 12
1	Characteristics of a good cowshed/ goat shelter	• List of a good cowshed/ goats shelter	2
2	Types of cattle breeds	• Identify the types and originality of cow breeds	1
3	Types of goat breeds	• Identify the types and originality of goat breeds	1
4	Characteristics of cattle breed to rear	• Select proper cattle to rear based on their characteristics	1
5	Characteristics of goat breed to rear	• Select proper goats to rear based on their characteristics	1

6	Proper feeding of cows and goats	• Identify elements of a good diet of cattle.	1
7	Cattle or goat health sanitation conditions	• Explain health sanitation conditions of cattle and goats.	1
8	Common diseases of cattle/goat	• Explain the most common cattle diseases	1
9	Prevention and control measures of cattle and goat diseases	• Explain the prevention and the treatment of cattle and goat diseases.	1
10	Importance of cattle/goat farming	• Explain the importance of cattle/goat farming.	1
11	End of unit assessment	• Assess the acquired competences • Demonstrate the achievement of the key unit competence	1

Guidance on different lessons

4.1. Characteristics of a good cowshed/ goat shelter

a) Learning objectives

- List the characteristics of a good cowshed/ goats shelter

b) Teaching and learning resources

- Animal shelters at school, at home or in a farm, photographs/ charts showing animal shelters, or videos on internet (If available).

c) Activities' guidance and answers

Guidance to the activity 4.1

i. Introduce the lesson using probing questions such as 'how do you think we should keep our domestic animals?

ii. Let pupils look at the picture and say what is going on in the pictures of the activity 4.1 in the pupil's book. For the pupils with visual impairment assist them or pair them with the advanced leaners to help them in the description of the pictures.

iii. Guide the pupils to reach on the intended answers

iv. Build on the pupils' prior knowledge and introduce them to the characteristics of a good cowshed or goat shelter.

v. Emphasize the fact that grazing is not the best way of rearing our cattle.

vi. Bring to their attention the fact that also the Rwandan government emphasizes on not living with animals inside the same house we live in.

vii. Summarise the lesson by highlighting the main points as pupils write summary notes.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 4.1

a. In picture A there is a cow, eating grass in a manger and there is water aside in a clean pot. In B, there are calves that are in the separate cowshed.

b. All the pictures show clean cowshed with roofs and fences.

c. Yes, there are in clean houses.

Guidance to the application activity 4.1

i) This application activity involves matching items composing a cowshed or goat shelter with their purposes.

ii) Guide the pupils in pairs to do the activity, and this will improve their communication, collaboration and cooperation skills.

iii) While pairing your pupils, make sure that they are paired according to their needs and their abilities so that they could help one another. N.B: Be gender responsive and inclusive in your pairing.

iv) Provide the answers to the table on how the pupils should have matched the items with their purposes.

v) During the activity, keep facilitating pupils on how to do and answer the questions, and note their progress. At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarise the content.

Answer to the application activity 4.1

Items	**Purposes**
Roof	Keeps animals from rain and sun.
Fence	Keeps harmful animals out.
Paved floor	Makes it easy to clean and cows cannot slip.
Clean pot	Holds clean water for cows and goats to drink.
Open building	Keeps the air inside the cowshed or goat shelter.

4.2. Types of cattle breeds

a) Learning objectives

- Identify the types and origin of cow breeds

b) Teaching and learning resources

- Photographs / charts showing various types of cows, textbooks, videos from the internet.
- The teacher may improvise appropriate teaching and learning resources based on the lesson context and locally available materials.

c) Activities' guidance and answers

Guidance to the activity 4.2

i. Introduce the lesson using probing questions such as 'Do you think all cows are the same?'

ii. Let pupils look at the pictures and say what is going on in the pictures of the activity 4.2 in the pupil's book. For the pupils with visual impairment assist them or pair them with the advanced leaners to help them in the description of the pictures.

iii. This is an individual task that requires pupils to reflect on the picture.

iv. Guide the pupils to reach on the intended answers. But do not allow chorus answers, let them answer one-by-one.

v. Emphasize the fact that in Rwanda we have different types of cows, and their origins are different.

vi. Build on the pupils' prior knowledge and introduce them to the types of cattle breeds.

vii. Summarise the lesson by highlighting the main points as pupils write summary notes

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 4.2

1. Different cows. Yes, there is.
2. Some cows are big others are small, others have big udders (big breasts), others have horns, and others have humps on their backs.

Guidance to the application activity 4.2

i. This application activity 4.2 aims to assess pupils' comprehension on the learnt content.

ii. Provide time to do the exercise individually and note down their progress. It might

also be provided as a homework and answers will be given in the next lesson.

iii. Make sure that the instructions on how to complete the activity are clear for every pupil.

iv. During the activity, keep facilitating pupils on how to do and answer the questions, and note their progress.

v. At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarise the content.

Answer to the application activity 4.2

1. Inyambo
2. (a) Guernsey, (d) Friesian, (e) Jersey
3. Indigenous cattle breeds are good for Rwanda's environment because they **can walk long distances** and **cannot be affected by Rwanda's climate**.

4.3. Types of goat breeds

a) Learning objectives

- Identify the types and origins of goat breeds

b) Teaching and learning resources

- Photographs / charts showing various types of goats, textbooks, videos from the internet.
- The teacher may improvise appropriate teaching and learning resources based on the lesson context and locally available materials.

c) Activities' guidance and answers

Guidance to the activity 4.3

i. Introduce the lesson using probing questions such as 'How many goats do have you at home? Are all the goats similar?'

ii. Let pupils look at the pictures and say what is going on in the pictures of the activity 4.3 in the pupil's book. For the pupils with visual impairment assist them or pair them with the advanced leaners to help them in the description of the pictures.

iii. This is an individual task that requires pupils to reflect on the picture.

iv. Guide the pupils to reach on the intended answers. But do not allow chorus answers, let them answer one-by-one.

v. Emphasize the fact that in Rwanda we have different types of goats, and their origins are different.

vi. Build on the pupils' prior knowledge and introduce them to the types of goat breeds.

vii. Summarise the lesson by highlighting the main points as pupils write summary notes.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 4.3

1. Different goats. Yes, there are some differences in the goats from the picture.
2. Some goats are big, others have many colors, others have horns, others have big udders (Big breasts).

Guidance to the application activity 4.3

i. This application activity 4.3 aims to assess pupils' comprehension on the learnt content.

ii. Provide time to do the exercise in groups and note down their progress.

iii. During the activity, keep facilitating pupils on how to do and answer the questions, and note their progress.

iv. Make sure that the instructions on how to complete the activity are clear for every group.

v. At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarise the content.

Answers to the application activity 4.3.

1.

Goat breeds	**Countries of origin**
Boer	South Africa
Galla	Kenya
Toggenburg	Switzerland
Alpine	France

2. Boer goats grow faster and are best to crossbreed.

3. Local goats are more resistant to diseases.

4.4. Characteristics of cattle breed to rear

a) Learning objectives

- Select proper cattle to rear based on their characteristics

b) Teaching and learning resources

- Photographs / charts showing cows for rearing, textbooks, and videos from the internet.
- The teacher may improvise appropriate teaching and learning resources based on the lesson context and locally available materials.

c) Activities' guidance and answers

Guidance to the activity 4.4.

i. Introduce the lesson by reminding pupils about the previous lesson. Ask them whether they remember the various breeds of cows. Let them name some of them.

ii. This task must be done by the whole class and it requires pupils to reflect on the picture.

iii. Guide the pupils to reach on the intended answers. But do not allow chorus answers, let them answer one-by-one.

iv. Build on the pupils' prior knowledge and introduce them to the characteristics of cattle breeds to rear.

v. At this point you can explain the various features to look for when choosing cattle to rear.

vi. Emphasize the need to identify high productivity breeds, both drought and diseases resistant breeds and those with good feeding habits.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 4.4.

1. The cow in (b) has a bigger udder than the cow in (a). The cow in (b) seem to have a lot of weight than the cow in (a).
2. The cow in (b) is the best because it gives more milk, and it can give more beef too.
3. Cow in b. **Remind them this from the previous lesson.**

Guidance to the application activity 4.4

i. This application activity 4.4 involves reading the scenario, and it enhances the reading skills of pupils.

ii. Allow pupils to read the text in groups and make sure that by forming groups you are **being gender sensitive and inclusive** for pupils with special needs.

iii. Give clear instructions that they should read the text and circle the correct characteristics that shows the best quality of breeds to rear.

iv. Make sure that everyone in the group is participating to ensure collaboration and cooperation.

v. During the activity, keep facilitating pupils on how to do and answer the questions, and note their progress.

vi. At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarise the content.

Answers to the application activity 4.4

1. Qualities: Climate adaptation, very big, and produces 17 littres of milk.
2. Yohana should buy the cow A because it has all the good qualities of a good cow to rear.

4.5. Characteristics of goat breed to rear

a) Learning objectives

- Select proper goats to rear based on their characteristics

b) Teaching and learning resources

Photographs / charts showing goats for rearing, textbooks, and videos from the internet.

c) Activities' guidance and answers

Guidance to the activity 4.5

i. Introduce the lesson by reminding pupils about the previous lesson. Ask them whether they remember the various breeds of goats. Let them name some of them.

ii. This is task to be done with the whole class that requires pupils to reflect on the pictures.

iii. Guide the pupils to reach on the intended answers. Let them answer one-by-one.

iv. Build on the pupils' prior knowledge and introduce them to the characteristics of goat breeds to rear.

v. Explain to pupils the various features to look for when choosing goats to rear.

vi. Emphasize the need to identify high productivity breeds in terms of giving meats, milk and producing more than one kids, both drought and diseases resistant breeds and those with good feeding habits.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 4.5.

1. A mother goat that has one kid.
2. A mother goat that has three kids.
3. The mother in B is very productive and is the best to rear.

Guidance to the application activity 4.5

i. This application activity 4.5 targets the critical thinking of pupils on the learnt content.

ii. Allow all pupils to participate and note their progress. Make sure you are being gender sensitive and inclusive for pupils with special needs.

iii. Make sure that everyone in the class is participating to ensure communication skills.

iv. During the activity, keep facilitating and provide the reason to the correct answer for those 'True' questions. You can do the same and explain the reason for the 'false' questions.

v. At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarise the content.

Answers to the application activity 4.5

1. False
2. False
3. True
4. True
5. False

4.6. Proper feeding of cows and goats

a) Learning objectives

- Identify elements of a good diet of cattle

b) Teaching and learning resources

Sample animal feeds such as napier, lucern and a variety of concentrates, photographs/charts showing various types of livestock feeds, textbooks and the internet.

c) Activities' guidance and answers

Guidance to the activity 4.6

i. Guide pupils in the revision of the previous lessons by asking some questions on types and characteristics of cattle and goats. This lesson promotes the **standardization culture** in buying and feeding animals.

ii. Ask pupils to look the pictures in the activity 4.6 and let them brainstorm for a few minutes.

iii. Guide their answers to the intended ones. All pupils should have equal chances of answering.

iv. Emphasize the fact that cattle and goats are fed depending on their type and the age. Also their status, for example, whether pregnant or not.

v. Also, emphasize the need to give balanced ration to cattle/goats. Balanced diet contains all body nutrients necessary for healthy growth of the animals.

vi. At this point, you can introduce the concept of proper feeding of your cattle or goats.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 4.6

1. Cows are feeding on mixed cereals
2. Goats are feeding on grass
3. Other examples of feedings are in the pupil's book under activity 4.6

Guidance to the application activity 4.6

i. This application activity 4.6 aims to assess pupils' comprehension on the learnt content.

ii. Provide clear instructions on how to complete the activity because this is an individual task, and it may serve as a homework.

iii. During the activity, keep facilitating pupils on how to do and answer the questions, and note their progress.

iv. At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarise the content.

Answers to the application activity 4.6

Food elements	Source
Minerals and vitamins	All types of salts
Fat and oils	Soybeans
Carbohydrates	Banana stem and potato tubers
Proteins	Legumes

4.7. Cattle and goat health sanitation conditions

a) Learning objectives

- Explain health sanitation conditions of cattle and goats.

b) Teaching and learning resources

Photographs/charts showing various types of livestock, textbooks and videos from the internet.

c) Learning activities

Guidance to the activity 4.7

i. Introduce the lesson using probing questions such as 'How do you make sure your goats and cows are clean? Do we clean them with water ?'

ii. Let pupils look at the picture and say what is going on in the picture of the activity 4.7 in the pupil's book. For the pupils with visual impairment assist them or pair them with the advanced leaners to help them in the description of the pictures.

iii. This is an individual task that requires pupils to reflect on the picture.

iv. Guide the pupils to reach on the intended answers. But do not allow chorus answers, let them answer one-by-one.

v. Build on the pupils' prior knowledge and introduce them to the cattle and goats' health sanitation conditions.

vi. Summarise the lesson by highlighting the main points as pupils write summary notes.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 4.7

1. Possible answers: people, goats, cows, bloom, etc.
2. People in the picture are cleaning the cowshed.
3. Check the content on 4.7 in the pupil's book.

Guidance to the application activity 4.7

i) This application activity 4.7 aims to assess pupils' comprehension on the learnt content.

ii) Provide time to do the exercise in individually, or give this activity as a takeaway homework.

iii) Give clear instructions on how to do the work and when to submit it.

iv) Note each pupil's progress after the completion of the activity.

v) At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarise the content.

Answers to the application activity 4.7

1. Cut bushes
2. Separate sick animals
3. Open house for fresh air
4. Clean the water pots
5. Ensure drainage

4.8. Common diseases of cattle and goat

a) Learning objectives

- Explain the most common cattle's diseases.

b) Teaching and learning resources

Photographs/charts on goats or cows that are sick, computers, and videos from internet.

c) Activities' guidance and answers

Guidance to the activity 4.8

iv) Introduce the lesson by reminding pupils about common diseases that they know or might have come across. They may name some diseases that affect human beings.

v) You can then ask them if they think the same diseases affect animals. The answer may be yes or no.

vi) Take advantage and ask pupils to look at the picture and say what is going on in the pictures of the activity 4.8 in the pupil's book. For the pupils with visual impairment assist them or pair them with the advanced leaners to help them in the description of the pictures.

vii) Guide the pupils to reach on the intended answers.

viii) Build on the pupils' prior knowledge and introduce them to the characteristics of a good cowshed or goat shelter.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 4.8

1. Picture (B), there is a cow which is sick and its skin is damaged.
2. Picture (A) is healthy and in good condition.
3. The cow in B is healthy.

Guidance to the application activity 4.8

i) This application activity 4.8 involves reading the text and it focuses on communication skills.

ii) Guide the pupils on pairs to do the activity, and this will improve their communication, collaboration and cooperation skills.

iii) While pairing your pupils, make sure that they are paired according to their needs and their abilities so that they could help one another. N.B: Be gender responsive and inclusive in your pairing.

iv) Provide the answers to the text provided.

v) During the activity, keep facilitating pupils on how to do and answer the questions, and note their progress. At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarise the content.

Answers to the application activity 4.8

1. The disease is anaplasmosis.
2. Symptoms are: Ticks on the skin, the cow was weak, and it has fever.

4.9. Prevention and control measures of cattle and oat diseases

a) Learning objectives

- Explain the prevention and the treatment measures of cattle and goat diseases.

b) Teaching and learning resources

- Pupil's textbook, photographs/ charts showing animal drugs or pesticides, or videos on internet of animals being treated (If available).

c) Activities' guidance and answers

Guidance to the activity 4.9

i. Introduce the lesson using probing questions such as 'What do we call people who treat animals? Have you ever seen any medicine for any animal (Let them name any medicine be it for a dog, cat, cow, goat, etc.)?

ii. Let pupils look at the pictures and say what is going on in the pictures of the activity 4.9 in the pupil's book. For the pupils with visual impairment assist them or pair them with the advanced leaners to help them in the description of the pictures.

iii. Guide the pupils to reach on the intended answers

iv. Build on the pupils' prior knowledge and introduce them to the prevention and control measures of cattle and goat diseases.

v. Emphasize a cross-cutting issues of standardization culture, and let them know that the only person who should treat an animal is a veterinary.

vi. Highlight that the government of Rwanda has put at least one veterinary in each sector so that the animal can receive proper and standard medication.

vii. Summarise the lesson by highlighting the main points as pupils write summary notes.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 4.9

1. In picture B there is a farmer who is spraying pesticide on a cow in its cowshed.
2. In picture A, there are two veterinaries who are giving a vaccine to a small goat kid.
3. For a cow or goat not to get sick, you can ensure that their food and shelter are always clean, and make sure that they receive proper vaccination when they are still young, etc.

Guidance to the application activity 4.9

i) This application activity involves matching actions with the right way of prevention of diseases.

ii) Guide the pupils in pairs to do the activity, and this will improve their communication, collaboration and cooperation skills. Alternatively, this activity can be provided to pupils as homework.

iii) While pairing your pupils, make sure that they are paired according to their needs and their abilities so that they could help one another. N.B: Be gender responsive and inclusive in your pairing.

iv) Provide the answers to the table on how the pupils should have matched the items with their purposes.

v) During the activity, keep facilitating pupils on how to do and answer the questions, and note their progress. At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarise the content.

Answer to the application activity 4.9

Actions	**Way of prevention**
Spraying cows to kill ticks	Spraying pesticides
Cleaning the food and water containers every day	Hygiene
Keeping a sick cow away from other healthy cows	Quarantines
Give medicine to a sick goat	Use of drugs
Calling veterinary officer to give an injection	Vaccination

4.10. Importance of cattle/goat farming

a) Learning objectives

- Explain the importance of cattle/goat farming
- Argue for the benefits of owning either a cow or a goat

b) Teaching and learning resources

- Cow or goat products such as milk, belts and handbags, meat, charts on the products above, or videos on how the products from cows and goats are made.
- Optional: Poster paper, crayons, marker pens, or paint, ICT resources for additional research.

c) Activities' guidance and answers

Guidance to the activity 4.10

i) Introduce the lesson by telling pupils that they will be trying to answer this science question: Should a villager purchase a goat or a cow?

ii) Divide the class into small groups of 3-4 pupils for this lesson.

iii) Explain to pupils they have been asked by a villager to help them make the decision to either buy 1 cow or 1 goat. Explain that the villager only has enough money to purchase one, not both. The pupils' job will be to help the villager decide which he should purchase.

iv) Ask pupils to predict the benefits/products they can get from cows, and the benefits/products they can get from goats.

v) Ask pupils to predict, which animal the villager should purchase: a cow or a goat? Chart the class results.

vi) Review the content in the pupil's book regarding the importance of cattle and goat farming (nutritional benefits, economic benefits, social benefits, and agricultural benefits). Allow pupils to take notes by filling in the following table:

Products from goats	Importance	Products from cows	Importance

Emphasize the fact that livestock farming can be a source of income in the future if pupils take it seriously.

Then , proceed as follows:

vii) Tell pupils they are going to work in small groups to create arguments for the villager about which type of animal he should purchase. Their arguments should include information they have learned in this lesson as well as from previous lessons.

viii) Allow groups to present their arguments to the "villager" represented by the "Teacher" or other pupils.

Answers to the activity 4.10

Predictions might include:

a) Benefits/products of cows: Milk, meats, horns for decoration, products from its skin (clothes, belts, shoes, bags), cheese, yogurt

b) Benefits/products of goats: milk, meats, products from its skin

c) Predictions of which the villager should purchase will vary

Note: Arguments do not have to be all the same and there is not a preference between a cow and goat. This application activity targets the critical thinking of pupils on the learnt content. Good arguments should include information pupils have learned from this Unit 4.

Guidance to the application activity 4.10

i. This application activity 4.10 targets the critical thinking of pupils on the learnt content.

ii. Allow all pupils to participate and note their progress. Make sure you are being gender sensitive and inclusive for pupils with special needs.

iii. Make sure that everyone in the class is participating to ensure communication skills.

iv. During the activity, keep facilitating and providing the reason to the correct answer (those 'True' questions). You can do the same and explain the reason for the 'false' answers.

v. At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarise the content.

Answers to the application activity 4.10

1. False
2. False
3. True
4. False
5. True
6.

Products from goats	Importance	Products from goats	Importance
Meat	Provide food/ protein	Milk	For drinking and making daily
Skin	Used to make leather items	Meat	Provide food/ protein
Milk	For drinking and making cheese	Skin	Used to make leather

4.11. End unit assessment

I. Guidance to end of unit assessment

1. This part provides the guidance and answers for end of unit assessment with cross reference to the textbook.
2. Tell the pupils that the last part of the unit is the end of unit assessment comprising the project where they apply what they learnt and questions that check pupils' knowledge, understanding, skills, attitudes and values related to the whole content of the unit.
3. **For the project:**

- Request every pupil to do the project either at school or at home. As a teacher, you will decide if it is possible to do the project at school or home and give them guidance. They will use materials available in their community.
- Request pupils to submit their work, organize a class display table or mini exhibition and allow pupils explain how they made their objects and how it is useful.
- Reward the objects made or work of pupils by providing feedback and marks.
- Select some of good objects made by pupils and display them in class.

4. **For the quiz,** choose some questions for the quiz depending on available time for the assessment. The quiz must be done individually whereby pupils will submit their responses on papers to be marked.

5. As all questions may not be done during the period(s) allocated to the end of unit assessment, request them to answer the questions during their self-study time or as homework.
6. Mark their answers and keep records of every student's achievement.
7. Provide the remedial, consolidation and extended activities when necessary.



II. Answers to the end of unit assessment

A. Knowledge and understanding

1. C. A house for cows
2. C. Zebu
3. B. Banana leaves and Napier grass
4. Goat shelter
5. Inyambo
6. Milk
7.
   a) False
   b) False
   c) True
   d) True
   e) False
8. Indigenous: Means local or native breeds; Exotic means imported or foreign breeds.
9. Good milk producer, good feeder and quality meat, resistant to climate, and resistant to sickness.
10. It reduces chances of the animals becoming sick thereby ensuring healthy farm.
11. It is an advantage to grow crops when you are a cattle or goat farmer because animal wastes can be used to produce manure which is then used to enrich soil for increased crop yields.
12. To reduce cases of malnutrition and improve nutrition in the country and to provide some source of family income.

B. Skills

13. C. Call a veterinary doctor
14. C. When the floor is dry and easy to clean
15. A. Fencing
    B. Brushing
    C. True
    D. For the answer, refer to the pupil's book.

Unit 5

WORD PROCESSING AND SPREADSHEET



i. Key unit competence:

Manipulate documents using a Word Processing program Perform mathematical calculations using a Spreadsheet program and format a spreadsheet document.

iii. Prerequisites

Pupils are familiar with writing because they learned units related to Word Processing in P4 and P5. They are also familiar with using a computer as they have been using it from P3

Introductory activity and guidance

i) Make pupils read the scenario in the introductory activity and answer the related questions under your guidance

ii) Let pupils tell the programs to use for the tasks stated in the introductory activity

iii) Guide the pupils so that they say the how to make the text bold, underlined and bold using the program they stated (Word)1) The program is Word. It can be also done in Excel.

B. Possible answers for introductory activity

1. the program is worg. it can be also done in Excel.
2. 2) He will use Excel
3. 3) He will use the font tools which are (for Bold) and U (for Underlined)

iii. Generic competencies:

- Critical thinking: Developed through selecting appropriate formatting tools, organizing data in spreadsheets, applying formulas and functions, and managing files and folders effectively.
- Research and problem solving: Developed through creating and organizing digital documents, solving practical data management tasks, and applying spreadsheet functions to perform calculations and present information accurately.
- Communication (Literacy and Numeracy): Developed through producing well-formatted documents, presenting information clearly using word processing tools, and organizing numerical data in spreadsheets using appropriate formulas and tables.

iv. Basic values:

- Dignity and integrity: Developed through responsible use of computers, careful management of digital files, and respect for ICT equipment and digital information.
- Self-reliance: Developed through independently creating, editing, formatting, saving, and retrieving documents and spreadsheets to accomplish assigned tasks.
- Commitment and resilience: Developed through practicing ICT skills repeatedly, correcting errors, and improving the quality and accuracy of digital documents and spreadsheet work.

v. Cross cutting issues to be addressed

1. Critical thinking:

Tell pupils that will writing they have to think deeply and make sure that what they write will be understood by the reader.

2. Peace and values education:

Encourage pupils to write text which promotes peace and positive values. They must be aware that they should avoid to write , that it is not good to write provocative text and using offensive language.

iv. List of The Lessons

#	Lesson title	Learning objectives	Number of periods: 12
1	Making text bold, italic and underlined	• To format text by applying bold, italic, and underline styles.	2
2	Changing font type, font size and font color	• To change the font type, font size and font color of text in a document	2
3	Opening and saving a workbook	• Open a new and existing workbook, and save their work using appropriate file names	2
4	Working with worksheets	• Manipulate worksheets by inserting, deleting, renaming new sheets • Modify a worksheet environment by changing columns and rows properties	2
5	Formatting a cell	• Change the appearance of a cell content	2
6	Mathematical operations	• Perform basic mathematical operations on cell content	1
7	End of unit assessment	• Assess the acquired competences • Demonstrate the achievement of the key unit competence	1

Guidance on different lessons

5.1. Making text bold, italic and underlined

a) Learning objectives

To format text by applying bold, italic and underline styles.

b) Teaching and learning resources

Word processing software, Sample text for formatting practice, Printed guides showing toolbar icons for formatting, Projector for demonstration and students' textbooks.

c) Activities' guidance and answers



Guidance on activity 5.1

i) Review previously learned skills to help pupils perform successfully in the current lesson.

ii) Introduce the purpose of text formatting (bold, italic, underline) in writing.

iii) Ask pupils to explain why sometimes we make words stand out in writing.

iv) Show examples from a textbook or written examples with bold, italic, or underlined text.

v) Explain the purpose of each these formats: Bold (to highlight important words or titles), Italic (for emphasis or names such as animals or books titles), Underline (for drawing attention to specific parts of text)

vi) Demonstrate and show to pupils how to apply bold, italic, and underline formatting in a document. Use the following steps.

vii)Open a word processing program and type a short sentence on the projector then demonstratehow to make text bold, make text italic, make text underlined.

Note: Ensure to develop the values and generic competencies related to this lesson.



Answers on activity 5. 1

1) Open a Word processing program and write the given text

2. Click on the tools **B**,U, *i* to make the text bold, underlined, or *italic.*



Guidance on application activity 5.1

Present the task clearly by writing it on the chalkboard or using a projector for better visibility.

Give pupils enough time to complete the activity at their own pace.

Circulate around the classroom to provide guidance and support as needed.

Promote active participation from all students throughout the lesson.

Offer extra support to pupils who may require additional help.

Answers on application activity 5.1

1. The tool **B** is used for making text bold, tool I is used for making text italic and tool U is used for underlining text
2. The written text can be made bold, italic, and underlined by using the the process shown in the student book

5.2. Changing font type, font size and font color

a) Learning objectives

To change the font type, font size and font color of text in a document

b) Teaching and learning resources

Computers with installed word processors, sample paragraphs for formatting, chart displaying different font type and sizes, printable quick reference cards showing formatting steps and students' textbooks

Guidance on activity 5.2

i) Begin by asking the pupils questions. For example, ask pupils if they have ever seen writing that looks beautiful or very big on a computer

ii) Show pupils printed or digital examples of texts with different fonts and sizes.

iii) Explain to pupils the importance of formatting the document. For example, Fonts help us express different moods or uses—like titles, headings, or body text.

iv) Show how to locate and use the font type and size tools by using a projector

v) Demonstrate how to Change the font type, Change the font size

vi) Show live examples of how the sentence changes with different settings.

vii) Encourage Pupils practice changing font type and size. For example ask pupils to type their name or a short sentence and change the font type and font size

viii) Ask pupils explain how to increase the font size and font type

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers on activity 5. 2

The questions 1 to 5 are practical. Pupils type the RAINBOW and change its font color, font size and type. They realize that when they change the font, the text appearance also changes.

Guidance on application activity 5.2

i) Make sure each pupil has access to a computer or digital device.

ii) Encourage pupils to apply the keyboard skills they have previously learned during practice sessions.

iii) Present the task clearly by writing it on the chalkboard or using a projector for better visibility.

iv) Give pupils sufficient time to complete the activity at their own pace.

v) Circulate around the classroom to provide guidance and support as needed.

vi) Promote active participation from all pupils throughout the lesson.

vii) Offer extra support to pupils who may require additional help.



Answers on application activity 5.2

The questions 1 to 4 are practical. Pupils write the proposed text and change its font color, font size and font type by following the instructions given in the questions. As they change the fonts, the text appearance also changes.

5.3. Opening and saving a workbook

a) Learning objectives

Identify the elements of a workbook environment and their uses

b) Teaching and learning resources

Computer with a Spreadsheet program, projector, wallchart, P6 textbooks, images of a Spreadsheet environment with labels.



Guidance on the activity 5.3

i) Introduce the lesson by asking questions that will make them discover that a program for doing calculation is Excel (Or Gnumeric Spreadsheet). Questions can be: what are the programs we find in a computer? Which one can be used for accounting?

ii) Ask pupils to open that program

iii) Show them the image (screenshot) in the activity 5.5. Ask them to identify the parts will labels on their Spreadsheet windows

iv) Let them explain how to open a spreadsheet program

v) Let them save that document after writing some text and closing it. After they open the document they have saved. When they fail to do what is requested, provide a demonstration on how to do it.

vi) Open a new Spreadsheet window again and guide pupils so that they try to give the names of the parts they had earlier identified. For each part they also give its use.

vii) When they can not name them or give their use, help them.

viii) Summarize the key features and functions of spreadsheets

Answers on activity 5. 3

Questions 1 and 2:

Pupils open Excel by clicking on the Start menu, searching for "Excel," and then clicking on the Excel icon to open it. Or they open Gnumeric by clicking on Applications then on Office then on Gnumeric Spreadsheet

3. Pupils write their names in the Excel workbook they have opened

4.Use the CTLR+S shortcut.

Guidance on the application activity 5.3

This application activity is practical and will be done in the computer laboratory. These are practical questions and verifying there answers requires to go to every pupils' computer.

i) Write the questions on the blackboard and explain what is going to be done

ii) Let pupils do the activity and move around to provide guidance to those having difficulties and see if they can do it well

iii) Call one pupil to come and demonstrate how to do the activity was done

iv) Give time to pupils who did not do it well to do it again

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the application activity 5.3

Questions 1, 2) Pupils open Excel (or Gnumeric) and do the activity and save it as p6 marks.

5.4. Working with worksheets

a) Learning objectives

- Manipulate worksheets by inserting, deleting, renaming new sheets
- Modify a worksheet environment by changing columns and rows properties

b) Teaching and learning resources

Computer with a Spreadsheet program, projector, wallchart, P6 textbooks,

Guidance on the activity 5.4

i) Begin with a brief discussion on why multiple worksheets might be useful in organizing different types of data. Also discuss why we need to resize rows, columns and renaming worksheets or deleting them.

ii) Ask pupils to do the questions of the activity 5.4 and if they can't do some of the questions tell them they will be able at the end of the lesson

iii) After giving to pupils the opportunity to do it, demonstrate by using a projector how to insert delete, rename worksheet; modifying columns, rows, cells. After demonstrating each activity, let pupils follow the demonstrated steps.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 5.4

1. We need many worksheets so that we may arrange our information in many areas (sheets) especially when the information is not in the same category
2. It helps when there is much information that can not fit in a cell, therefore the cell is resized so that the information is well visible

Question 3&4) These are practical questions and the answer can be checked by looking on pupils' work on their computers.

Guidance on the application activity 5.4

This is a practical activity that will be done in the computer laboratory.

i) Write the questions and let pupils practice using their computers

ii) Go to every pupil to see if they can do what is requested well and provide appropriate feedback and reinforcement

iii) As correction, call one pupil to project and show how he/she does it

iv) Give time to pupils so that they can do it again if they did not do it well before

Answers to the application activity 5.4

1) This is a practical activity. The result can be checked only using a computer.

5.5. Formatting a cell

a) Learning objectives

Change the appearance of a cell content

b) Teaching and learning resources

Computer with a Spreadsheet program installed, projector, wallchart, P6 textbooks, images showing cell content that was formatted.

Guidance on the activity 5.5

i) Start the lesson by asking what formatting is and what they can do if they are asked to format text (this was learned in Word or Abiword)

ii) Provide the questions in the activity 5.5. If pupils cannot do some of the questions tell them that after the lesson they will be able to do it.

iii) For every formatting activity start by making pupils try without your guidance then do a demonstration that they will follow to do formatting well. The topics to cover one by one are: Changing font type, font size, Inserting or removing cell borders, text font colour, text alignment.

iv) Move around to see how pupils are following your demonstration to do what you said

v) Do a quick summary after which pupils do again what they have learned.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 5.5

The questions a to g are practical and their answers can be checked only in a computer (not on paper). The steps on how to do different tasks are in the pupil's book in section 5.5 on formatting a cell.

Guidance on the application activity 5.5

The application activity will be done in the computer laboratory.

i) Write the questions, read and explain them to make sure pupils understand what they are going to do
ii) Let pupils do the practical activity and move around to facilitate and correct them
iii) For corrections a pupil comes and do a single question using your computer which is connected to the projector.
iv) Give another time to pupils so that they can correct themselves if they did not do well the questions

Answers to the application activity 5.5

The questions are practical and the result can be checked only on a computer

5.6. Mathematical operations

a) Learning objectives

Perform basic mathematical operations on cell content

b) Teaching and learning resources

Computer with a spreadsheet program installed, projector, wallchart, P6 textbooks, images illustrating use of the operations +, -, *, /

Guidance on the activity 5.6

i) Start the lesson by making pupils do the activity 5.8 in groups of 3 or 4. They calculate the totals for question 2 and realize how much time it takes to calculate that total
ii) Still in groups let pupils write the table in Excel (or Gnumeric Spreadsheet)
iii) Show them how to write a formula in Excel and let them do it in their table to calculate the total. They do it for all the rows.
iv) Make clear the difference between a formula and a function
v) Demonstrate the sum, subtraction, multiplication and division using the examples provided in the student book and individually pupils do the same
vi) With the list of 20 pupils that you have written in advance, call pupils so that they calculate totals using formulas

vii) Ask pupils to summarize the lesson by telling the formula to use for addition, subtraction, multiplication and division

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 5.6

Pupils make the given table in Excel (or Gnumeric Spreadsheet) and calculate the total manually then fill it

1. The result is this table:

No.	Item	Quantity in litres or Kg	Amount per item	Total amount
1	Milk	10	500	5,000
2	Juice	1	4500	4,500
3	Tomatoes	10	1000	10,000
4	Potatoes	20	450	9,000
5	Fruits	5	1800	9,000
6	Cabbage	3	600	1,800
7	Onions	2	800	1,600
Total				40,900

Yes. A formula can be used and give the same results quickly. Learners use Excel (or Gnumeric Spreadsheet) to calculate the totals.

For question 2 &3 the result can be checked only on a computer.

Guidance on the application activity 5.6

This is an activity that will be done in the computer laboratory

i) Write the questions and explain them before pupils start working on them

ii) Move around to facilitate and correct those who have finished

iii) Call pupils to correct using the teacher's computer which is connected to the projector so that other pupils see the answers

iv) Give time to pupils so that they can improve their work by referring on the corrections done.

Answers to the application activity 5.6

This is a practical activity for which answers are got after doing calculations using formula. These are the numbers after calculation:

1. Total amounts are respectively cells from top to bottom are: 42000, 6000; 2000
   The general total: 50,000
2. The remaining fees to be filled in the respective cells from top to bottom are: 73000, 17000, 77800
3. Sum function is a function that adds numbers in Excel

5.7. End of unit assessment

5.7.1. Project activity: My Report Card

Objective: Create a school report card using a word processing program (AbiWord or MS Word) with necessary formatting to make it look beautiful

- **Skills and competences to be developed:**
  - Communication in official language through the project presentation
  - Critical thinking: when pupils are arranging the words, rows and columns to make a school report
  - Creativity and innovation: when pupils apply different formatting styles to make the school report look more beautiful
- **Guidance on the project activity**
  i) This project activity will be done individually at school or at home
  ii) Start by recalling the information that they find on their school reports
  iii) Give the project title and let pupils identify what they will do and the project timeline
  iv) If there are some pupils that will do the project at school, set the time for using the school's computers with supervision
  v) Submission of the work: Pupils will submit by email by whatsapp. They may not know how to use an email but this becomes an occasion to search and know it

Let pupils present their work and let them be appreciated by their colleague who choose the best project.

- **Possible answers to the project activity**

Pupils make report card that looks like the one they receive from the school. They try to do more formatting to make the report look more beautiful.

5.7.2. Assessment questions

I. Guidance on end of unit assessment

The end unit assessment will be done in the computer laboratory. The questions assess their competences in manipulating spreadsheets and their feelings towards the use of spreadsheet programs.

i) Write the questions and read them while also explaining what is requested.
ii) Move around to help those that may be having difficulties
iii) Correct pupils who have finished and provide appropriate feedback and reinforcement.
iv) Correct all the questions collectively. For the questions 1 to 3 listen to many answers to know the pupils feelings towards the use of Spreadsheets.

II. Answers to end unit assessment

- **Knowledge and understanding**

1) a) B 2)d) Home 3)c) A group of worksheets

4) c) Right click → Rename 5) c) Puts it in the middle

6) a) It makes important text stand out

7) It can help people manage money by helping them do calculations rapidly. For example calculations that would take hours can be done in minutes. It also help them arrange data well.

8) It makes student marks fair because there are no errors of miscalculations that often happen.

- **Skills**

9) a, b) The result will look like this:

No	Last name	First	Class	Cell	Sector
3	UWIMANA	Tedy	P6	Kinunga	Remera
4	KALISA	Joe	P6	Gasozi	Remera
5	INEZA	Sandra	P6	Gasozi	Remera
6	KAMANZI	Steven	P6	Kinunga	Kabeza
7	UWASE	Chantal	P6	Gashaka	Kabeza
8	HABIMANA	Joseph	P6	Gashaka	Remera

c) The font size is changed by selecting the table and choosing 14 in the font size box 

d) Save the work by clicking on the save icon , choosing where to save and writing the new name.

- **Attitudes and values**

10) The positive feeling is that pupils like using formula in spreadsheets because it makes the calculations easy and quick

5.8. Additional activities

Remedial Activities

1. What program is used for Word processing?
2. What happens if you close a document without saving it?
3. What is a folder on a computer?
4. What happens when you delete a folder?
5. What is the use of the Title Bar?
6. What is the name of where you type the text.

Answers to remedial activities

1. AbiWord or Microsoft Word
2. A warning message appears asking if you want to save the document before closing.
3. A folder is a digital space used to organize files, like a cupboard for storing books.
4. The folder is moved to the Trash or Bin.
5. It shows the document name and the application title.
6. In the Document Area.

Consolidation activities

1. Why is saving your work important?
2. What happens when you modify a document but don't save changes?
3. Why do people create folders on a computer?
4. How can the Ruler help when typing a document?
5. Make a table containing a list of the food items that were bought at your home.
   a) Use Excel to calculate the total for each item
   b) Calculate the general total

Answers to the consolidation activities

1. To avoid losing your progress and be able to access your document later.
2. The changes will be lost when the document is closed.
3. To organize and store files well, just like putting papers in different drawers.
4. It helps adjust margins and organize text alignment.
5. Pupils make a table containing all food items that were bought. They calculate the totals and general total using formula

Extended activities

1. Why do we have to set the correct date and time on the laptop?
2. What happens if you disconnect from all networks?
3. Why is organizing files in folders important for efficient computer use?
4. Why is the Scroll Bar essential in word processing software?
5. Why is text formatting necessary in professional documents presentation?
6. Make a table containing a list of the food items that were bought at your home.
   a) Use Excel to calculate the total for each item
   b) Calculate the general total
   c) Your family wants to reduce the budget by 70%. Use formula to calculate the budget for each item and the general total

Answers to the extended activities

1. To ensure accuracy for saved files, scheduled tasks, and system operations.
2. Applications that require internet access, like browsers and email, won't function.
3. Organizing files in folders helps users locate documents quickly.
4. It allows users to navigate through long documents efficiently without losing track of their place.
5. Proper formatting improves readability, emphasizes key points and enhances the document's presentation.
6. Pupils make a table containing all food items that were bought. They calculate the totals and general total using formula and do it again by applying a reduction of 70%.

Unit 6

INTRODUCTION TO EMERGING TECHNOLOGIES



i. Key unit competence:

Pupils will be able to:

- Determine the characteristics, and practical applications of various emerging technologies
- Construct a machine learning model using available machine learning platforms

ii. Prerequisites

To effectively learn about emerging technologies, pupils need:

- Understand how everyday technologies like mobile phones, television, cameras, smart watches, computers, and tablets are used at home and school.
- Be able to use search engines to do research

Introductory activity and guidance

A. Guidance on the introductory activity

- As a teacher, start with brainstorming sessions with the pupils and ask pupils to list technologies they use daily such as smartphones, television, tablets, computers, smart watches. Write answers on the board and let pupils categorize them as communication, entertainment, home appliances.
- Clarify unfamiliar terms to pupils such as Virtual Reality (VR) and highlight real-world applications.
- Provide context to pupils, show pupils that Robots were first used in factories in the 1960s, but VR became popular after 2010.

B. Possible answers to the introductory activity

1. I use a mobile phone, radio, TV, and sometimes a computer at school
2. I have seen a robot that helps deliver medicine in hospitals and a drone that takes pictures from the sky.
3. Matching device with their function

a)

Devices	Functions
Smartwatch	Shows digital health information
Robot	Helps in factories
VR glasses	Let's you see a place as if you are there
Drone	Flies and delivers things

b) Robot, Drone, VR glasses, and Smartwatch are all examples of new (emerging) technologies. You can mention any one as the answer, for example the drone is a new technology.

iii. Generic competencies:

- Critical thinking: Developed through analyzing community problems, comparing different emerging technologies, and selecting appropriate technologies to solve real-life challenges.
- Research and problem solving: Developed through researching applications of emerging technologies, exploring their uses in different sectors, and proposing innovative technological solutions to community needs.
- Communication (Literacy and Numeracy): Developed through group discussions, presentations, demonstrations, and explaining how emerging technologies improve services in education, health, transportation, and other sectors.
- Creativity and innovation: Developed through designing and presenting original technology-based solutions to everyday problems and imagining new ways technology can improve people's lives.

iv. Basic values:

- Self-reliance: Developed through encouraging learners to explore and use technology independently to solve problems and improve everyday activities.
- Commitment and resilience: Developed through adapting to technological changes, continuously learning about new technologies, and embracing innovation responsibly.
- Patriotism: Developed through appreciating how emerging technologies, such as drones and Artificial Intelligence, contribute to Rwanda's development in areas such as healthcare, education, agriculture, and public services.

v. Cross cutting issues to be addressed

1. Environmental Sustainability:

Discuss how technologies like AI can help in protecting the environment by predicting the weather and taking measures in advance. Here emphasize that it is good to protect the environment and more by using new technologies.

2. Inclusive education:

Make pupils know that there are technologies that support people with disabilities, such as AI-powered speech recognition for individuals with speech impairments or wearable devices enhancing mobility.

3. Financial education:

Make learners understand the economic benefits of using emerging technologies whereby a task that would be done by humans and require too much financial means can be done using technologies like robots and therefore reduce the cost. They should therefore embrace these technologies.

vi. List of lessons

No	Lesson title	Learning Objectives	Number of periods: 6
1	Use of technology in our lives	Identify common emerging technologies used at home, school, and in the community	1
2	Types of Emerging Technologies	Describe how each type of emerging technologies can improve services in different areas	1
3	AI as an emerging technology	Describe AI in respect to the characteristics of emerging technologies.	1
4	Machine Learning	Describe AI in respect to the characteristics of emerging technologies.	1
5	Working with Machine Learning	Use machine learning platforms to train a model that can be used in recognizing items	2
6	End of unit assessment	• Assess the acquired competences • Demonstrate the achievement of the key unit competence	1

Guidance on different lessons

6.1. Use of technology in our lives

a) Learning objectives

Identify common emerging technologies used at home, school, and in the community

b) Teaching and learning resources

Projector, Visual Aids such as smartphone, washing machine, fan, laptop, microwave, real objects such as a calculator, remote control.

c) Activities guidance and answers

Guidance to the activity 6.1

i) Introduce the scenario by displaying it on a projector.

ii) Ask pupils if they have ever used technology like Mutesi does?

iii) Divide the class into small groups and assign them the activity.

iv) Each groups share responses with the whole class.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 6.1

1. b) Phone. Her phone rang to wake her up
2. a) Her phone informed her it might rain.
3. b) Projector and computer.
4. c) School computers.
5. c) She saw her score immediately and knew her mistakes.

Guidance on the application activity 6.1

This application activity will be done in groups.

i) Before starting the application activity make sure that all concepts are understood especially those that are asked in the application.

ii) Explain that pupils will now do an activity to share their ideas and opinions about the new technologies in our daily life.

iii) Move around to provide support to groups that may not understand well the task

iv) Let some groups presents their work. As they present, summarize some points on the blackboard.

Answers to the application activity 6.1

1. Other ICT devices that can be used in teaching and learning are : tablets, smartboards, radios, televisions, printers, digital cameras, mobile phones
2. A chatbot is a computer program that talks with people using a voice or text messages.
3. Chatbot are found in websites and apps for banks and other institutions that give service to people.
4. - Learning would be slow because teacher would use papers only and write all notes on the blackboard Slow
   - Banking would be difficult because people would need to go to the bank to get any service
   - Communicating to people would take longer,

   etc

6.2. Types of Emerging Technologies

a) Learning objectives

Describe how each type of emerging technologies can improve services in different areas

b) Teaching and learning resources

- Computer, projector and Pupil's book
- Posters or digital slides showing Types of emerging technologies, AI, IoT, Virtual Reality (VR), Augmented Reality (AR)
- Robotics kits, 3D Printing.

Where the real objects stated above are not available, use images or videos.

c) Activities' guidance and answers

Guidance to the activity 6.2

i) Start by showing pictures or short video clips of different technologies in action such as a robot in a factory, a drone in flight, upils using VR glasses, traffic lights controlling cars

ii) Ask pupils to answer the questions of the activity 6.2.

iii) Ask pupils to map emerging technologies to where they are used and what they are used for

iv) For every new technology given and its use, complement what the pupils say about it.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 6.2

1. A robot is a machine that can do tasks automatically.
2. A robot can help in doing tasks that would be done by humans. For example in the story, the robot was used in a factory to pack products.
3. A drone is a flying machine that does not need a pilot to fly. It can be used to deliver items.
4. - Good things: Makes work easier and faster, helps in emergencies (like drones delivering blood), makes learning more fun (like VR trips).
   - Bad things: Some emerging technologies can be expensive, they may not work without electricity or internet, they can replace some human jobs and some AI search tools like CHATGPT can be used by pupils in cheating.

Guidance on the application activity 6.2

- Begin with a short video for demonstration using a projector or pictures of people using VR headsets, robots in factories, smart assistants (like Siri or Alexa)
- Ask pupils to recall what they saw in the Mugisha story
- Divide pupils into pairs or small groups and let them discuss each question together.
- Let groups present their findings

Answers to the application activity 6.2

1. When you use VR glasses, you feel like you are inside a different place. It creates a 3D virtual world that looks real.
2. Places like: Nyungwe National Park or Lake Kivu.
3. Robots can work faster, don't get tired, and can do dangerous or repetitive jobs. This helps industries save time, reduce mistakes and keep people safe.
4. Yes, technologies like sensors, solar panels, or smart irrigation can help save energy, reduce pollution and use resources well.
5. Pupils give new technologies they would invent such as robot to clean water.
6. Advantages:
   They can answer questions quickly
   They help with spelling, reading and information searching
   Make learning fun and interactive

Disadvantages:

Pupils may become too dependent on them

They need internet and electricity to work

They don't replace real teachers or deep understanding

6.3. AI as an emerging technology

a) Learning objectives

- Describe AI in respect to the characteristics of emerging technologies

b) Teaching and learning resources

P6 student book especially the section 6.3, and section 6.1 on characteristics of AI, internet connectivity for doing research on characteristics of AI.

c) Activities' guidance and answers

Guidance to the activity 6.3

- Start the lesson by reading the scenario of the activity 6.3 and guide learners so that they answer the questions. Also guide them through questions and answers so that they discover the title of the lesson.
- Make learners recall the characteristics of emerging technologies as seen in lesson "6.1. Use of technology in our lives".
- Now let learners explain how AI as an emerging technology has all the characteristics they have recalled.
- Conclude the lesson by making learners discover ethics principles in AI that must be respected by users and inventors.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 6.3

1) Use of AI in Learning

2) YouTube suggested History videos because the phone had information about his mother, who is a History teacher and uses her phone on youtube.

3) A telephone can predict words by using Artificial Intelligence (AI). It studies the words you usually type and uses installed AI technology to predict the next word.

4) Other examples of AI tools include Google Assistant, Alexa, ChatGPT and Cortana.

5) i) Siri gets the voice by: b) Using the microphone

ii) Siri understands the voice by: a) Using AI

Guidance on the application activity 6.3

Guidance

The application activity 6.3 will be done individually in the classroom.

- Write the questions of the application activity 6.3 or display them using a projector then choose one pupil to read them and make sure learners understand them.
- Move among the students to correct them while you also explain why some of their answers are wrong.
- Let the class correct collectively all the questions. Make sure they explain why they gave those answers.



Answers to the application activity 6.3

1. a) False b) True c) True
2. b) Data

Problem 1: AI makes wrong decision due to incorrect data

Solution: Make sure AI is trained with correct, accurate, and reliable data.

Problem 2: People losing their jobs.

Solution: Train people with new skills so they can work with AI instead of being replaced by it.

6.4. Machine learning

a) Learning objectives

Explain the key components of Machine Learning and how they work together.

b) Teaching and learning resources

P6 Student Book (specifically the section 6.4), machine learning diagram, video illustrating machine learning, computer and projector for displaying a video.

c) Activities' guidance and answers



Guidance on the activity 6.4

- Start the lesson with the activity6.3 that pupils will do individually.
- Using questions and answers, and basing on the answers of the activity 6.4 make pupils discover the lesson title.
- Through questions and answers let the pupils explain what is machine learning.
- Show the diagram of machine learning and let learners guess what it means. Alternatively, display a video explaining how machine learning works.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 6.4

1. c) Learning in class, at home and in the class
2. c) Learning from data such as pictures
3. Example 1. Computer that recognizes cats and dogs: You show the computer many pictures of dogs and cats. For each picture, you tell the computer whether it is a dog or a cat. Later, if you show new pictures of a dog or a cat, the computer can identify them correctly because it has learned from the data.

   Example 2. Smart voice assistant (like Siri or Alexa): It learns from how people speak and respond. After some time, it can understand speech, suggest answers, and perform tasks like setting alarms or playing music.

d) Guidance and answers on the application activity 6.4

Guidance

- Briefly do a synthesis of the lesson by explaining the components of machine learning and how they work together.
- Display or write the questions and emphasize that students will attempt the activity individually.
- As pupils do the application activity move around to correct those that have finished.
- Let pupils write the correct answers on the blackboard and compare their answers with the correct ones.
- Engage the class in discussion and reasoning focusing on why the computer cannot learn by just reading names and how predictions depend on datasets and algorithms.

Answers to the application activity 6.4

1. By watching many picture of dogs and cats
2. a) Dataset b) Intelligent
3. Dataset, learning algorithm, prediction.

6.5. Working with Machine Learning

a) Learning objectives

Use machine learning platforms to train a model that can be used in recognizing items

b) Teaching and learning resources

The P6 student book, pictures of avocadoes and mangoes, computers with internet connectivity to be used in practicing training models using appropriate platforms, projector to be used for demonstrating the steps.

c) Activities' guidance and answers



Guidance on the activity 6.5

- Start the lesson with the activity 6.5 and from the answers let learners discover the lesson title.
- Open the platform machinelearningforkids.co.uk and use a projector to demonstrate how to train a model that will recognize mangoes and avocadoes. Images of mangoes and will have been prepared in advance.
- As you demonstrate the steps, let learners follow the steps you have demonstrated.
- Test the model until the accuracy of the model is improved near 100%
- Let learners summarize the steps about how to train a machine learning model.

Note: Ensure to develop the values and generic competencies related to this lesson.



Answers to the activity 6.5

1) No, a mango and an avocado do not have the same shape. A computer learns to distinguish between them by looking at their outlines and features.

2) The computer said it is an avocado because it was trained using images of an avocado and a mango so that it can recognize them.

3) c) It will guess well because it will learn more.

When the computer is trained with many images of mangoes and avocados, it will recognize them more accurately.

d) Guidance and answers on the application activity 6.5

Guidance

This application activity will be done in groups and in the computer laboratory. It is based on practical activity to create a machine learning model that recognize specified items.

- Put pupils in groups. Every group will do all the two questions. Explain the questions so that they are understandable for everyone.
- Let groups present their projects/model and let pupils appreciate those projects.
- Check if the projects presented have an accuracy above 90%. If not the model can not be considered as a success.
- Let the groups with the best projects demonstrate how to do it.

Answers to the application activity 6.5

(1&2) Open the machinelearningforkids.co.uk platform and follow the steps. Upload the images you have prepared in advance:

- For the model to recognize images of dogs and cats, upload those images
- For the project of a model tha recognize numbers, you will upload numbers from 1 to 10. Every number will have its own box and name.

Now train each model so that they can do their tasks to recognize images.

6.6. End of unit assessment

I. Guidance to end unit assessment

A. Guidance on end of unit assessment

i) This test checks how much pupils understand new technology. Prepare the classroom atmosphere for the assessment.

ii) Read the questions for the class and make sure that every question is understandable

iii) Let pupils do individually the questions and move around to facilitate those that may be having difficulties

iv) Correct every single pupil and record the performance. As you correct, provide positive reinforcement and advice where necessary

v) Call pupils to provide the answers for every single question

II. Answers to end unit assessment

Knowledge and understanding

1. Emerging technology is a new technology that is still growing and improving. It is used to solve problems and make life easier in smart ways.
2. c) Robot that gives medicine
3. c) It is new and still growing
4. b) Transporting blood to rural hospitals
5. b) AI (Artificial Intelligence)

Skills

6. I could use cardboard for the body, bottle caps for wheels, sticks or straws for arms, and wires from old electronics to connect parts. If I had a small motor or battery, the robot could move or pick up small things.
7. c) Train

Attitudes and values

8. Zipline drones have improved healthcare in Rwanda by delivering blood and medicine to remote hospitals and health centers. This saves lives, especially when roads are damaged or far away.

9.

10. An AI tool that recognizes animals can help a farmer with many animals such as cows, goats, hens by helping him/her count them quickly using a camera. This saves time and errors that happen when sorting animals and counting them.

6.7. Additional activities

Remedial Activities

1. Match the Technology to Its use. Match the emerging technology on the left with what it is used for on the right.

Technology	Use
A. Drone	1. Delivers blood to hospitals
B. Virtual Reality	2. Feels like you're visiting a new place
C. Robot	3. Packs items in a factory
D. Smart Assistant	4. Helps answer questions using AI

2. Complete the sentences using, robot, AI, traffic lights, drone, learning

A _____ can fly and deliver blood to remote areas.

_____ help control cars and people at busy roads.

A _____ works in a factory to pack products.

_____ helps doctors and teachers make better decisions.

VR can make _____ more fun and interesting

3. Read each statement and write True or False.

i) Robots can get tired like humans.

ii) AI can help teachers and pupils.

iii) VR is used to make food.

iv) Drones can help in emergencies.

v) Emerging technologies are old and no longer used.

Answers to remedial activities

1. A – 1, B – 2, C – 3, D – 4
2. i.Drone ii.Traffic lights iii. Robot iv. AI v. Learning
3. i.False ii.True iii. False iv. True v. False

Consolidation activities

1. Write a short story about how emerging technology solved a real-life problems
2. In groups, create a short role-play/skit showing how one emerging technology (e.g., robot, drone, VR) helps people in real life.
3. How can drones help in rural areas?
4. Give one way AR can help pupils in class.

Answers to the consolidation activities

1. In a rural hospital in Rwanda, the roads were flooded and no car could bring blood for a patient. A Zipline drone flew over the river and delivered the blood quickly, saving the patient's life.
2. The role play can be like: A nurse calls for blood delivery. A Zipline drone flies in and drops the blood pack, everyone claps, and the patient is saved.
3. By delivering medicine or blood where roads are bad
4. It shows 3D images to help them understand topics better.

Extended activities

1. Do a research on how Rwanda uses at least two emerging technologies.
2. How can Artificial Intelligence (AI) help farmers in Rwanda?
3. List 3 ways that Virtual Reality can improve education.
4. What are some challenges people may face when using new emerging technology? Suggest solutions
5. Why is it important to learn about emerging technologies at a young age?
6. What rules should we follow to use emerging technology safely and responsibly?

Answers to the extended activities

1. In Rwanda, drones are used to deliver blood and medicine. They fly over rivers and hills to reach hospitals faster. This has saved many lives.
2. AI can help farmers know the best time to plant crops by checking the weather. It can also help detect crop diseases early and suggest solutions so farmers can have better harvests.
3. It helps pupils visit places like museums or forests without leaving school.

   It makes learning fun and helps pupils understand difficult topics better.

   It allows pupils to practice skills like science experiments safely.
4. Challenges: No electricity, not enough internet, high cost

   Solutions: Use solar power, build more internet centers, share technology in schools and libraries.
5. Because it helps children prepare for the future, solve problems using smart tools and

create new ideas that can help their communities grow.

6. Don't share personal information online.
   Use devices with permission.
   Take breaks and don't stare at screens for too long.
   Be kind and respectful when using technology to talk to others.

Unit 7

PROGRAMMING FOR CHILDREN



i. Key unit competence:

Pupils will be able to use scratch to design shapes, do calculations, and create simple animations with sprites and backgrounds.

ii. Prerequisites

Pupils are already familiar with programming because they learned the Unit "Programming for Children" in Primary Four and Primary Five in which they tackled Turtle Art and Scratch. This will serve as a foundation for this unit, which is like a continuation of the same units learn in previous years.

Introductory activity and guidance

A. Guidance on the introductory activity

i) Read aloud the story "Grandma's Morning Robot" with expression.

ii) You can have a few pupils take turns reading parts of it.

iii) Support pupils who struggle with reading by pairing them up or reading the story again slowly.

iv) Guide the class to complete the comprehension questions related to the story.

v) Pupils can:

- Answer them individually on paper.
- Work in pairs or groups to discuss answers.
- Share their answers with the class.

vi) Link this story to upcoming lessons on Scratch or Turtle Blocks so students see the purpose of programming.

vii) Highlight key points:

- The robot follows only exact instructions.
- Programming is giving a list of steps to a computer or robot.
- Blocks are used to build programs (just like Scratch or Turtle Blocks).

viii) Ask pupils to write steps for a different routine (like taking bath or cooking porridge).

ix) Encourage **creativity** when they share routines and programming examples.

B. Possible answers to the introductory activity

1. She received a breakfast robot.
2. The robot only follows instructions exactly as they are given.
3. Because Grandma forgot to tell it to pick up the bread first, so the robot didn't do anything.
4. She used blocks like move forward, turn left, pick up, wait, and say "Breakfast is ready".
5. For example for brushing my teeth we use these steps: I get my toothbrush, put toothpaste on it, brush my teeth, rinse my mouth, and wash the brush.
6. Because robots and computers don't know what to do unless you tell them step-by-step in the correct order.

iii. Generic competences:

- Creativity and innovation: Developed through designing animations, games, interactive stories, and drawings using Scratch and Turtle Art to express ideas and solve simple problems creatively.
- Critical Thinking: Developed through arranging programming blocks logically, debugging simple programs, drawing geometric shapes, and applying mathematical operations using Scratch.
- •Research and Problem Solving: Developed through exploring programming features, experimenting with different commands, and creating projects that address real-life situations and learning tasks.

iv. Basic values:

- Self-reliance: Developed through independently creating, saving, editing, and improving programming projects using Scratch and Turtle Art.
- Commitment and resilience: Developed through practicing programming skills, correcting errors (debugging), and improving projects through continuous testing and refinement.
- Hard work: Developed through active participation in programming activities, completing creative projects, and applying effort to produce quality animations, games, and drawings.

v. Cross cutting issues to be addressed

1. **Gender education:** Encourage both boys and girls to participate in programming tasks equally. Explain that programming is for both girls and boys
2. **Environmental Education:** Discuss how 3D shapes (like cubes/cylinders) are used in packaging, and reflect on how packaging affects waste and recycling.
3. **Financial education:** Discuss how using computerized systems help in reducing the cost of doing tasks. For manual systems where workers use papers, the tasks become more difficult and takes a lot of time.

vi. List of lessons

No	Lesson title	Learning Objectives	Number of periods:14
1	Drawing Polygons	• Explain the steps to save, open, delete and rename a project • Identify turtle art instruction to draw cylinder, cube, cuboids • Use Turtle Blocks to draw regular polygons by combining movement and turning blocks.	2
2	Drawing Equilateral and Right-Angled Triangles	• Use Turtle Blocks to draw both an equilateral triangle and a right-angled triangle. • Apply correct turning angles in Turtle Blocks to accurately draw triangle shapes.	2
3	Drawing 3D shapes in Turtle Art	• Use Turtle Arts to draw simple 3D-like shapes by combining lines and angles to create depth. • Construct and produce different geometric shapes using turtle art instructions	2
4	Using Scratch in Mathematical operations	• Create a simple Scratch program that applies a mathematics idea • State how to compute the sum, difference, product, quotient or average in Turtle Art • Compute and perform different calculations in Scratch	2
5	Drawing shapes in Scratch	• Use Motion, Turn and Pen blocks in Scratch to draw simple geometric figures • Explain how angles and steps are used in Scratch to form different shapes.	2
6	Using Variables in Scratch	• Create and use variables in Scratch to store and change data such as keeping track of scores	1
7	Making interactive stories in Scratch	• Create an interactive story having a beginning, middle and end by using Scratch blocks	1

8	Creating a Simple Game in Scratch	• Use control, sensing, and motion blocks to build a simple game. • Use variables to track score or time. • Show creativity for designing and creating more projects reflecting the real life experience • Appreciate the way of expressing the ideas through projects, animations • Be proud to arrange commands and produce animations	1
9	End of unit assessment	• Assess the acquired competences • Demonstrate the achievement of the key unit competence	1

Guidance on different lessons

7.1. Drawing Polygons

a) Learning objectives

Use Turtle Blocks to draw regular polygons by combining movement and turning blocks

b) Teaching and learning resources

XO Laptops with Turtle Blocks or a Windows computer in which Turtle Art was installed (by using Sugarizer setup), projector for doing demonstrations, Whiteboard/Chalkboard for drawing shapes and explaining concepts, Polygon reference chart (with names and pictures), pupils' books.

c) Activities guidance and answers

Guidance to the activity 7.1.1

i) This activity reviews the properties of a square and how to create it using Turtle Blocks.
ii) Display the questions clearly on the chalkboard or using a projector.
iii) Read through each question with the class, ensuring understanding before they begin.
iv) Instruct pupils to copy all the questions into their notebooks.
v) Have the pupils complete the exercise individually and quietly.
vi) Walk around and give individual guidance where needed.
vii) Discuss the answers right answers for the questions.

Answers to the activity 7.1.1

1. A square has 4 sides.
2. Yes, all sides of a square are equal in length.
3. A square has 4 corners (or angles).
4. Each angle is 90 degrees.
5. You will see a straight line that is 100 steps long.
6. It turns the turtle 90 degrees to the right (clockwise)
7.
   - Use forward 100 to draw one side.
   - Then use right 90 to turn the turtle.
   - Repeat this 4 times to draw all 4 sides and complete the square.

Guidance on the application activity 7.1.1

i) Ask the pupils to write the question in their notebooks and begin the activity individually.

ii) As they work, walk around the classroom to observe and assess their responses.

iii) Once everyone has finished, invite one or more students to share their answers with the whole class.

iv) Encourage the rest of the pupils to compare their answers with those presented.

v) Finally, have the pupils write down the key points in their notebooks.

Answers to the application activity 7.1.1

1. Programming is the process of giving instructions to a computer to make it perform tasks or solve problems.
2. a. False

   b. False. The angle to use for a pentagon is 72 degrees
3. This is Open-ended question: expected output depends on the shape the student chooses, such as a triangle, square, or hexagon.

7.2. Drawing Equilateral and Right-Angled Triangles

a) Learning objectives

- Use Turtle Blocks to draw both an equilateral triangle and a right-angled triangle.
- Apply correct turning angles in Turtle Blocks to accurately draw triangle shapes.

b) Teaching and learning resources

- XO Laptops with Turtle Blocks installed or any computer with Turtle Blocks software
- Projector or teacher laptop to demonstrate steps and examples
- Rulers and protractors to help pupils understand triangle sides and angles before drawing them digitally.
- Step-by-step instruction sheet guide for drawing each type of triangle using Turtle Blocks.

c) Activities' guidance and answers

Guidance to the activity 7.1.2

i) Instruct learners to draw the three types of triangles in their exercise books and guide them so that they answer the other questions

ii) Guide learners so that they explain the properties of each triangle

iii) Do a demonstration of how to draw those triangles in Turtle Art or let learners do it if they can.

iv) Guide learners so that they give a summary of how to draw those triangles in Turtle Art

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 7.1.2

1. Draw or let pupils draw the three triangles on the chalkboard.
2. Equilateral triangle has all sides equal.
3. The right angled triangle has one angle with 90^0
4. To draw a triangle use the Forwardd block and turn block.

Guidance on the application activity 7.1.2

i) Write or display the activity instructions clearly on the board or projector for all pupils to see.

ii) Ask pupils to turn on their laptops and open Turtle Blocks.

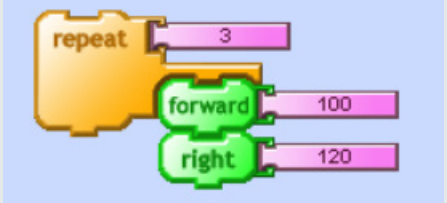
iii) Explain the activity step by step, making sure everyone understands what they are expected to do.

iv) Move around the classroom to support pupils.

v) Select a few pupils to present their Turtle Block projects using the projector or show their screens to the class.

vi) Invite pupils to describe their process, including what changes they made to create triangles of different sizes and how they discovered the angles and steps to draw a right-angled triangle.

Answers to the application activity 7.1.2

1) Blocks for an equilateral triangle	2) Blocks for an Equilateral triangle with a different size	3) Blocks for a Right angled triangle
		

7.3. Drawing 3D shapes

a) Learning objectives

Use Turtle Arts to draw simple 3D-like shapes by combining lines and angles to create depth

b) Teaching and learning resources

Computer and projector, XO laptops or any computer with Turtle Art, Pre-made Turtle Art files (sample code for cubes, pyramids, 3D boxes), Geometry shape posters or flashcards, physical 3D shape models

c) Activities' guidance and answers



Guidance to the activity 7.1.3

i) Write the questions and read them aloud and explain them
ii) Let pupils draw the shapes while you move around to help those that are having difficulties
iii) Invite 2 to 3 pupils to share their answers or drawings.
iv) Summarize key points on the properties of 3D shapes
v) Invite pupils to try drawing a cube using Turtle Art
vi) With your guidance, learners summarize how to draw a 3D shape in Turtle Art.

Note: Ensure to develop the values and generic competencies related to this lesson.



Answers to the activity 7.1.3

1. The three solid shapes: Cube, Cylinder and Sphere
2. - The first shape is flat and has only 2 dimensions, it's a 2D shape (e.g., square).

   - The second shape looks like it has depth and shows more than one face, it's a 3D solid shape (e.g., cube).
3. Using Turtle Art, learners draw a rectangle.



Guidance on the application activity 7.1.3

i) Ask pupils to open Turtle Blocks on their laptops and give a quick recap of the cube they drew before.
ii) Allow them to work individually and provide guidance to pupils facing difficulties
iii) Correct learners, provide appropriate feedback and record the marks
iv) Invite a few pupils to display their Turtle Blocks on the projector.

Answers to the application activity 7.1.3

1) The properties of a cube are: It is a 3D solid shape. It has 6 square faces, 12 edges, 8 corners (vertices). All edges are equal in length and the angles are right angles (90°).
2) A smaller cube is to be drawn by changing the distance in the **Forward** block
3) The properties of a cylinder: A cylinder is a 3D solid shape with these properties: It has 2 flat circular faces (top and bottom), it has 1 curved surface connecting the circles, it has no corners (vertices) and have two circular faces which are equal and parallel.
4) Let pupils use the blocks given to draw a cylinder

7.4. Use Scratch to perform mathematical operations

a) Learning objectives

Create a simple Scratch program that applies a mathematics idea

b) Teaching and learning resources

XO Laptops, Projector, Sample Scratch projects showing math concepts, Printed activity worksheets (Step-by-step guides or challenges for pupils to follow during the lesson) and P6 Student book

c) Activities' guidance and answers

Guidance on the activity 7.2

i) Start with a quick discussion on the operators they know and how to use them
ii) Guide the pupils so that they open Scratch and identify the different operators
iii) Let the pupils use the blocks in the student book to see how blocks can be assembled to do addition, subtraction, multiplication and division
iv) Guide pupils so that they give a summary on how to use operators in Scratch

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 7.2

1. The 4 examples of arithmetic operators are: Addition, subtraction, multiplication and division.
2. Comparison operators are used to check if one number is bigger, smaller, or the same as another.
3. The And, Or, Not operators are known as logical operators. Logical operators help computers decide based on rules. Those rules are:
   - AND: Everything must be true.
   - OR: At least one must be true.

4.

- NOT: It changes true to false and false to true.

5. It is a practical activity. The answer should be verified using a computer.



Guidance on the application activity 7.2

i) Ask pupils to turn on their laptops and open Scratch.

ii) Explain the activity, making sure everyone understands what they are expected to do.

iii) Move around the classroom to support pupils.

iv) Select a few pupils to present their work using the projector or show their screens to the class.

v) Invite other pupils to describe their projects.



Answers to the application activity 7.2

1. The Scratch program to add two numbers will be like the one presented in the Pupil's book in section 7.2. This answer should be verified using a computer.
2. For calculating the area, modify the program to add two numbers by multiplication whereby the Length will be multiplied with the height. This answer should be verified using a computer.

7.5. Drawing shapes in Scratch

a) Learning objectives

- Use Motion, Turn and Pen blocks in Scratch to draw simple geometric figures
- Explain how angles and steps are used in Scratch to form different shapes

b) Teaching and learning resources

Computers with Scratch, Projector or screen for demonstration and printed papers showing different shapes

c) Activities' guidance and answers



Guidance on the activity 7.3

Ask pupils to turn on their laptops and open Scratch.

Explain the activity making sure everyone understands what they are expected to do.

Move around the classroom to support the pupils.

Select a few pupils to present their work using the projector or show their screens to the class.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 7.3

The questions 1 to 5 are practical. Let the pupils open Scratch and practice using the different pen blocks to change the pen color and pen size.

Guidance on the application activity 7.3

Let pupils open Scratch and start a new project.

They can delete the default sprite and choose a new one (or keep the scratch Cat).

Allow pupils to begin drawing a pentagon by following the given instructions.

Move around the classroom to support pupils.

Let the pupils present their answers and appreciate the different answers.

Answers to the application activity 7.3

1. A regular pentagon has 5 equal sides. In Scratch, use a loop to repeat these steps 5 times:
   - Repeat (5)
   - move (100) steps
   - turn clockwise (72) degrees
2. Use this block (set pen color to...) before drawing to change the pen's color.
3. Add a wait (0.5) seconds block after each move to see the shape draw slowly, step-by-step.
4. Changing the angle changes the shape.
   - If the angle is too small, the shape curves in on itself.
   - If the angle is too large, it may not form a closed shape.
   - **Example**: Turning 60° would draw a hexagon, not a pentagon.
5. Use the same repeat and turn pattern, but adjust the angle and number of repeats. A hexagon has 6 sides.
6. A If you turn **60°** each time and keep rotating in the same direction, you will get an **hexagon.**

7.6. Using variables in Scratch

a) Learning objectives

Create and use variables in Scratch to store and change data such as keeping track of scores

b) Teaching and learning resources

Computers in which scratch program is installed, computer with a projector or

interactive whiteboard, sample project illustrating the use of variables.

c) Activities' guidance and answers

Guidance on the activity 7.4

i) Display the result (output) of a sample project on using variables and guide learners so that they discuss on the project demonstrated. Your guidance must be in such a way that they discover the use of variables

ii) Display or write the questions of the activity 7.4 and guide the pupils so that they answer them

iii) Demonstrate how to create and use variables and let pupils follow your demonstration. Note that before doing the demonstration you must give the pupils the opportunity to show what they know.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 7.4

1. A variable is used to keep the score.
2. We use the block: **change [score v] by (1)**
3. We use the block: **set [score v] to (0)**

Guidance on the application activity 7.4

i) Write the questions of the application activity 7.4 or display them using a projector and explain the questions to make sure that every pupil understands what is expected.

ii) Move around to facilitate the pupils or correct those that have finished. Also provide appropriate feedback

Answers to the application activity 7.4

1. It resets the variable so the score or time starts fresh each time the game begins.
2. Go to the **"Variables" tab**, click **"Make a Variable"**, give it a name, choose **"For all sprites"**, then click **OK**.
3. Use the **Show variable [name]** block to display the variable value during the game.
4. Variables can be used to track a player's score and measure how much time is left in a game.

7.7. Making an interactive story in Scratch

a) Learning objectives

Create an interactive story having a beginning, middle and end by using Scratch blocks

b) Teaching and learning resources

Computers in which Scratch program is installed, computer and a projector, video for demonstrating what is interactive story

c) Activities' guidance and answers

Guidance on the activity 7.5

i) Start the lesson by showing a video of an interactive story and let them say the features

ii) Guide the pupils answer the questions of the activity 7.5

iii) Display the short interactive story which is in the section 7.5 and let pupils analyse its blocks one by one

iv) Explain the blocks that should be used in an interactive story and give an example of a list of blocks that the pupils will assemble to make an interactive story

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 7.5

1. Use the "**say [Hello!] for [2] seconds**" block from the **Looks** category.
2. You start a story using the "**when green flag clicked**" block from the ***Events*** category. It
3. We change the background to show different places or times in the story. It helps the viewer understand where the action is happening.
4. You can add sounds, music, costumes, animations, and dialogues.

Guidance on the application activity 7.5

i) These are practical questions which are asked in a theory way. Pupils will do the questions on their computers then explain what they have done.

ii) The teacher writes the questions on the blackboard and reads each question aloud to ensure all pupils understand the task.

iii) Pair the pupils so that they do the activity and move around to make sure that every pair understands the task

iv) Let sample pairs present their work which is then appreciated by colleagues. Let the class choose the best story of all

Answers to the application activity 7.5

1. Use the "**when this sprite clicked**" block from the Events category and connect it to a "**say [Hello!] for [2] seconds**" block from the **Looks** category.
2. Use three different backdrops to show each part of the story. You can switch between them with the "switch backdrop to [scene]" block from the **Looks** category.
3. Use a block like "**when flag clicked**" or "**when [dog] touches [edge or object]**", then add the "start sound [bark]" block from the Sound category.
4. Use the "**glide [1] secs to x: [ ] y: [ ]**" block from the **Motion** category to move smoothly or use "**go to [x: y:]**" to jump to a new place.

7.8. Creating a simple game in Scratch

a) Learning objectives

Use control, sensing, and motion blocks to build a simple game.

Use variables to track score or time.

b) Teaching and learning resources

Projector and a computer for doing demonstrations, Computers in which Scratch program is installed, sample game project for demonstration

c) Activities' guidance and answers



Guidance on the activity 7.6

i) Using a projector, show a sample game, play it and show its code

ii) By referring to the shown game and its blocks let the pupils answer the questions of the activity 7.6

iii) Go through the steps of making a game and make sure that every step is understandable

iv) Let pupils use their computers to assemble the blocks that are in the section 7.6 and see the result. Let them explain.

v) Together with the pupils give the common blocks that can be used to make a game and explain how those blocks are used.

Note: Ensure to develop the values and generic competencies related to this lesson.



Answers to the activity 7.6

1. We use the "when green flag clicked" block.
2. The "if key (right arrow) pressed then change x by 10" block.
3. The target goes to the top again, and the score goes up by 1.
4. We put the score code in the target sprite.



Guidance on the application activity 7.6

i) This application activity is to apply the theory knowledge and as a teacher, you will continue to help them. This activity will be done in 20 minutes.

ii) Write the question of the application activity 7.6 and explain what is expected

iii) Put pupils in groups so that they do the activity. Guide them so that they set the rules for the game they are going to make

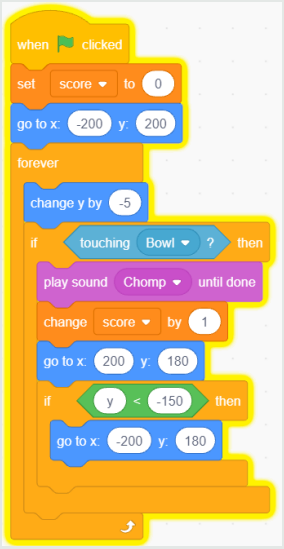
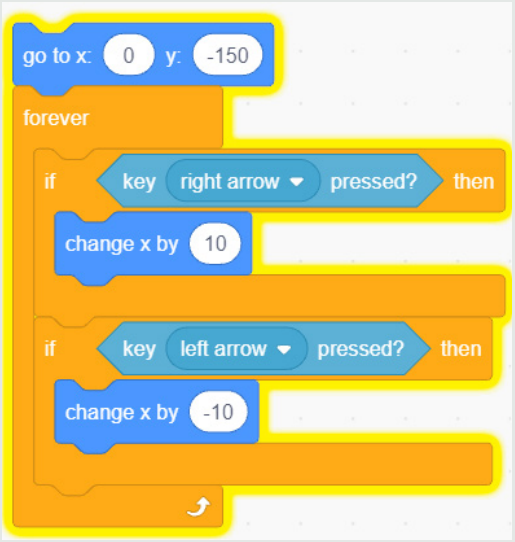
iv) Guide them so that they write down the blocks they are going to use

v) Let the groups assemble the blocks while you move around to provide support

vi) Let sample groups present their games and appreciate the work done

vii) Let every group try the blocks presented to see the result. As they do so they also improve the game by adding other features.

Answers to the application activity 7.6

Blocks for the apple sprite	Blocks for the bowl sprite
	

Note that this game can be improved by making the apple change its next position when it falls again so that the player continues to move the bowl.

The game in which the falling apple is received by a bowl. For every case when the apple is successfully received, the score increases by 1. The gave has two sprites (apple and bowl) and a variable must be created to keep trach of the score.The answer should be verified using a computer.

7.9. End of unit assessment

I. Guidance on end of unit assessment

i) Write the assessment and clearly explain what is going to be done ensuring that pupils understand what is expected before they begin.

ii) While learners work, move around the room to supervise and provide clarifications if needed

iii) Once completed, collect and mark the assessments, then provide feedback during the next lesson.

iv) Make sure all pupils have answered every question in the End of Unit Assessment as well as any additional questions you assign.

v) Correct all pupils and record their marks.

vi) Let pupils present the correct answers to the class.

II. Answers to the end of unit assessment

- **Knowledge and Understanding**

1) b) 900 2) b) Say "Hello" 3) a) False

- **Skills**

4) Polygons are drawn by using the Forward and Right (or Left) blocks. In the Forward block. specify the number of spaces while in the Right block we specify the turn angle.
5) It will make a right angled triangle.
6) It will draw an Octagon.

- **Attitudes and values**

7) If a program is not work well look if the blocks are well arranged and if there is no mistake. Ask for help or do research on the internet. It is not good to give up because trying is important in learning.
8) Programming is fun and helps in thinking better. It also helps in learning how to solve problems step by step. It initiates young pupils so that they become strong programmers in the future.
9) Yes, coding helps in many ways. People use it to make websites and apps.

7.10. Additional activities

Remedial Activities

1. How many sides and angles does a triangle have?
2. Name three examples of regular polygons.
3. What is the angle you need to turn to draw a square?
4. What is a **background** in Scratch?
5. How do you **change the background** in a Scratch project?
6. What does the clear block do?
7. What shape is formed when we repeat move and turn 4 times with a 90° turn?

Answers to remedial activities

1. A triangle has 3 sides and 3 angles.
2. Three examples of regular polygons: Equilateral triangle, Square and Regular hexagon
3. You need to turn 90 degrees to draw a square.

4. A background is the scene or setting that appears behind the sprites in a Scratch project.
5. You can change the background by clicking on the "**Choose a Backdrop**" button in the bottom-right corner and selecting a new one from the library.
6. The **Clear** block erases all drawings made with the pen on the stage.
7. A square is formed.

Consolidation activities

1. Which block do you use to make the turtle move forward?
2. What does the **Repeat** block do?
3. Create a Scratch program that draw a Square
4. What happens if you forget to use the pen down block?
5. What are **costumes** in Scratch, and how are they used in animations?
6. What blocks would you use to make a sprite move and wave at the same time?
7. What happens when you use the **next costume** block without a **wait** block?

Answers to the consolidation activities

1. You use the "forward" block to make the turtle move forward.
2. The repeat block runs the same set of instructions a certain number of times.
3. Use the Move and Turn 90° blocks four times to draw a square
4. If you forget the pen down block, the turtle or sprite will move but it won't draw anything on the screen.
5. Costumes are different looks or appearances of a sprite. In animations, changing costumes quickly can make the sprite look like it's moving or doing an action.
6. You can use the "move" block and the "next costume" block together in a repeat block, and add a wait block for smooth animation.
7. The sprite changes costumes too quickly, and the animation will look too fast.

Extended activities

1. You want to draw a hexagon. What angle will the turtle turn each time?
2. Use Scratch to draw a triangle. Which blocks did you use?
3. Can you draw a shape with more than 6 sides? What angle did you use?
4. Can you draw a right-angled triangle using only forward and turn blocks?
5. Can you make a sprite walk across **different backgrounds**? How?
6. Which block is used to **ask a question and get an answer** from the user?
7. Can you describe the steps to draw a square using Scratch?

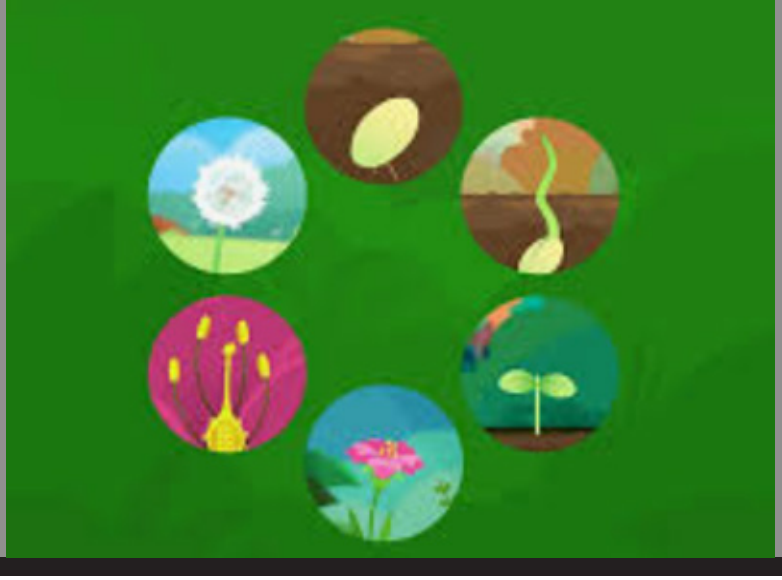
Answers to the extended activities

1. The turtle should turn 60 degrees each time.
2. I used pen down, move, and turn 120 degrees blocks, repeated 3 times.

3. Yes, for example, an octagon has 8 sides. The turtle should turn 45 degrees each time.
4. Yes, but you need to use a 90-degree turn for the right angle and adjust the other two angles so they add up to 90 degrees.
5. Yes, Use the next backdrop block to change scenes and use move and next costume blocks to make the sprite walk.
6. The **ask [ ] and wait** block is used to ask a question and get an answer.
7. The blocks to use to draw a square are: Pen down, Repeat 4 times: Move forward and Turn 90 degrees.

Unit 8

PLANTS REPRODUCTION



i. Key unit competence:

Pupils will be able to identify and explain plant reproduction, recognize its different types, and demonstrate care for plants.

ii. Prerequisites

This unit is about reproduction in plants. Remember, pupils have come across plants before. For example, in primary 4, they learnt about types of plants and the process of germination. In primary 5, they were introduced to importance of plants and trees. Therefore, link what the pupils already know with what they will learn here. Also, it will help if you bring to the awareness of pupils the fact that this topic is related to reproduction in plants in biology and vegetative reproduction in Agriculture. It is also good to tell the pupils that this topic is used a lot in Agriculture. If they study more, they can become plant experts or botanists. Provide some questions regarding these Prerequisitess in order to assess their progress. This will help you know what to teach more and which pupils need extra help.

Introductory activity and guidance

A. Guidance on the introductory activity

- As a way of introducing the unit, ask the pupils to carefully observe the picture on the introductory activity of unit 8 in the pupil's book.
- Let pupils comment on the picture.
- Let the pupils answer to the provided questions.

B. Possible answers to the introductory activity

1. The picture A is about creating awareness on how to maximize land usage in the reproduction of plants and how plant reproduction can be useful when taken care of.
2. In picture B, the crops dried and did not yield any production.
3. They most likely used wrong planting materials or methods and did not take good care of their crops. They should choose good planting materials and take good care of them.

iii. Generic competences:

- Critical thinking: Developed through comparing sexual and asexual reproduction, classifying different methods of plant reproduction, and explaining their importance in plant growth and survival.
- Research and problem solving: Developed through observing flowers, conducting practical activities on plant propagation, carrying out research, and investigating different methods of plant reproduction.
- Co-operation, interpersonal relations and life skills: Developed through group discussions, practical investigations, flower dissection, and collaborative planting and propagation activities.

iv. Basic values:

- Hard work: Developed through active participation in practical activities such as flower dissection, grafting, layering, and stem cutting, while caring for plants.
- Dignity and integrity: Developed through handling plants and practical materials responsibly and showing commitment to protecting and caring for living organisms.
- Commitment and resilience: Developed through practicing different plant propagation techniques patiently, learning from observations, and improving practical skills in plant production.

v. Cross cutting issues

1. **Financial education**: Bring to the attention of pupils the fact that they should practice crop production so that they improve food security in the country.
2. **Environmental conservation and sustainability:** Caution pupils on the practice of picking up flowers from plants everywhere. This will make them interfere with plant reproduction
3. **Gender education:** Let pupils know that people of all gender can take care of plants and pursue a career in plant management.
4. **Inclusive education:** All pupils should be encouraged to participate during lessons and practical. Special arrangement should be made to take care of pupils with special needs. For example, provide brail for blind pupils, large print text for those with sight problems and allocate physically challenged pupils to others to assist them during field trips and practical activities. Further, this category should be given tasks that they can manage during the practical sessions.

vi. List of lessons

#	Lesson title	Learning objectives	Number of periods: 8
1	Parts of a complete flower	• Identify the reproductive parts of a flower. • Draw and label a flower.	2
2	Types of plant reproduction	• Define and explore the types of plant reproduction. • Take interest in growing plants at home and at school	2
3	Sexual reproduction in flowering plants	• Explain the processes of sexual reproduction of flowering plants • Identify plants that reproduce sexually or asexually by looking at them	1
4	Asexual reproduction in plants	• Explain the process of asexual reproduction of flowering plants • Recognize just by looking at plants and flowers thosereproducing either sexually or asexually.	1
5	Reasons for plants reproduction	• Explain reasons of plant reproduction • Value both individual and group work in protecting plants and nature.	1
6	End of unit assessment	• Assess the acquired competences • Demonstrate the achievement of the key unit competence	1

Guidance on different lessons

8.1. Identification of parts of a complete flower

a) Learning objectives

- Identify the reproductive parts of a flower
- Draw and label a flower.

b) Teaching and learning resources

Textbooks, charts on parts of a flower, computers connected to the internet, different flowers collected from the field, razor-blade or scalpel, hand lens, well-painted diagram on parts of a flower on manila paper for use if you lack charts.

c) Learning activities

Guidance on the activity 8.1

i. You may begin this lesson by asking pupils whether they recall the parts of a plant they learnt about in primary 3.

ii. Obviously, one of the parts that pupils will mention is the flower. Build on this and introduce activity 8.1.

iii. Take pupils outside the class to collect flowers. Let them come back to class with them, observe the collected flower, and dissect them. Remember to caution pupils against destroying plants as they collect flowers. This is not good for the environment.

iv. Let the pupils observe the dissected flowers using naked eyes, then hand lens. Ask them what they can see.

v. Provide pupils with charts showing parts of a flower. Let them compare what they notice on the dissected flowers and what they see on these charts.

vi. Guide pupils to draw the parts of a flower in their notebooks, they will label them later.

vii. At this point, you can highlight the main points emphasizing on parts of a flower and their functions.

viii. Let pupils know that the parts of the flower can be put into two distinct groups, these are especially male and female parts.

ix. Wind up the lesson by giving pupils a task of collecting flowers and identifying the male and female parts and drawing them in their notebooks as a further activity.

x. Summarize the lesson by highlighting the key points as pupils write down short notes.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 8.1

1. Going out
2. Collecting flowers
3. Dissection of flowers
4. Not all flowering plants produce complete flowers with both male and female organs. Some flowering plants have only male flowers, while others have only female flowers

Guidance to the application activity 8.1

i. This application activity 8.1 involves filling the name in the table and it focuses on communication skills.

ii. Guide the pupils in pairs to do the activity, and this will improve their communication, collaboration and cooperation skills.

iii. While pairing your pupils, make sure that they are paired according to their needs and their abilities so that they could help one another. N.B: Be gender responsive and inclusive in your pairing.

iv. Provide the answers to the table provided.

v. During the activity, keep facilitating pupils on how to do and answer the questions, and note their progress. At the end of the lesson, ask again some questions on the performed activities or content learnt to assess the learning, and summarize the content.

Answers to the application activity 8.1

Main Part	Role	Name of the part
External part	Joins the flower to the rest of the plant	Stalk
External part	Looks like leaves and protect the flower	Sepals
External part	Bright, colorful parts that attract insects	Petals
Male part	The whole male reproductive part of a flower	Stamen
Male part	Contain pollen grains	Anther
Male part	Holds the anther	Filament
Female part	The whole female reproductive part of a flower	Pistil
Female part	Receives pollen grains	Stigma
Female part	Connects stigma to ovary	Style
Female part	Contains ovules (Female reproductive cells)	Ovary

8.2 Types of plant reproduction

a) Learning objectives

Define and explore the types of plant reproduction.

b) Teaching and learning resources

Textbooks, chart or photographs, computers connected to the internet, plant parts such as sugar cane cuttings, seeds and root tubers.

c) Guidance and answers to the activity 8.2

Guidance on the activity 8.2

i. Introduce the lesson by reminding pupils about the previous lesson. Ask them whether they remember different parts of a flower. Let them name some of them.

ii. Let the pupils observe the picture. This is task to be done with the whole class that requires pupils to reflect on the picture.

iii. Guide the pupils to reach the intended answers, but do not allow chorus answers; let them answer one-by-one.

iv. Build on the pupils' prior knowledge and introduce them to definition and types of plant reproduction.

v. At this point you can explain what plant reproduction is and its types.

vi. Emphasize on the need of knowing different methods of plant reproduction, because the economy of Rwanda is based on agriculture and most of agriculture methods are going to be useful to pupils in the future.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 8.2

1. Children have sand in the sky-blue backets.
2. A standing boy has a watering can in his hand.
3. The sitting children in the picture are trying to plant some seedlings (using the method that will be learnt in the next lesson: seed germination).

Guidance to the application activity 8.2

i. This application activity 8.2 targets the critical thinking of pupils on the learnt content.

ii. Allow all pupils to participate and you will note their progress. Make sure you are being gender sensitive and inclusive for pupils with special needs.

iii. Make sure that everyone in the class is participating to ensure communication skills.

iv. During the activity, keep facilitating and provide the reason to the correct answer for those 'True' statements. You can do the same and explain the reason for the 'false' statements.

v. At the end of the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarize the content.

Answers to the application activity 8.2

1. True
2. True
3. False
4. True
5. False

8.3 Sexual reproduction in flowering plants

a) Learning objectives

- Explain the processes of sexual reproduction in flowering plants
- Model seed dispersal for plants

b) Teaching and learning resources

- Textbooks, charts on sexual reproduction in flowering plants, computers connected to the internet, a well-painted diagrams of processes such as fertilization and pollination and parts of a seed on manila papers for use if you lack charts.
- Flowers collected from outside.
- Learning resources to build the "seeds" (thick paper, cotton fibres, wires).
- Scissors, rulers.

c) Guidance and answers to the activity 8.3

Guidance to the activity 8.3

i) Introduce the lesson by telling pupils that they will be investigating to answer this science question: How does the shape and size of a seed affect how far it can travel?

ii) Let pupils observe the picture. This task is to be done with the whole class.

iii) Guide the pupils to discuss the questions in this activity 8.3, as they sit at their desks. Allow pupils to share their findings in a discussion forum in class.

iv) Take advantage of this and explain what pollination is and the two types of pollination: self and cross pollination.

v) At this point, you can introduce the **three agents of pollination: animals (birds and insects), wind, and water.** Explain how each of these occurs. Refer back to the original picture. Ask pupils to determine which agent of pollination is at work there.

Then, proceed as follows:

vi) Remind pupils they are investigating the science question: How does the shape and size of a seed affect how far it can travel?

vii) Take pupils outside to investigate the plants around the school campus. Have pupils collect any seeds they can find.

viii) Back in the classroom, have pupils observe the seeds they found and look for similarities and differences. Have pupils record the similarities and differences in their exercise books.

ix) Explain to pupils that they will use their knowledge and observations to design a seed that travels the furthest in the wind.

x) Divide the class into pairs to carry out this task.

xi) Show pupils the learning resources they can use for this task.

xii) Tell pupils to draw their individual design for their seed in their exercise books before discussing with their partner.

xiii) Allow pupils to share their individual ideas with their partner afterwards.

xiv) In partners, they will decide which design to build together. It is okay to combine ideas as well. The pairs' solutions do not need to be the same. Accept all innovations.

xv) Give groups the learning resources to build their designs.

xvi) To test their seeds, have pupils either: drop their seeds from a tall height, launch their seeds into the wind outside, or if there is not enough wind outside, the teacher could use a fan or have pupils make papers fans to create wind. Have pupils take notes of which seeds traveled the furthest.

xvii) Have pupils compare their seeds to other groups. Look for similarities and differences.

xviii) Have pupils group their created seeds into the three agents of pollination. Have pupils explain their reasonings for groupings.

xix) Discuss which seeds might be the most successful in a windy area. Which seeds might be the most successful in a calm area?

xx) Restate the science question that was introduced at the beginning of the lesson: How does the shape and size of a seed affect how far it can travel? Have pupils answer the science question in their own words. Encourage them to use information from the lesson.

Answers to the activity 8.3

1. Pupils might see: Different flowers in the garden, a flying bee with somethings on its legs (pollens), two similar flowers, etc.
2. The bee has picked them from a flower of a certain plant.
3. The bee might be taking them to another flower of the same kind (Occurrence of pollination) or different kind (No pollination)

Note: During discussions, pupils will provide different answers depending on their observations and findings of similarities and differences.

Guidance to the activity 8.3.4

i. Guide pupils to carry out this activity in groups. The composition of the groups should be informed by size and ability of pupils. Note: The results will be gotten like after 7 days.

ii. At the end of the experiment, pupils should observe that germination has occurred. Let them say or find out what it is and share with other class members.

iii. You can then clarify that seeds germinate into new plants after dispersal.

iv. At this point, you can introduce the fact that certain conditions are necessary for germination to occur.

v. Wrap up the lesson by highlighting the main conditions necessary for germination as pupils take summary notes.

Answers to the activity 8.3.4

Pupils will provide different answers depending on their observations. Use their findings to identify the ways in which plant reproduce. For the answer refer to the content pupil's book under sub-topic 8.3.4.

Guidance to the application activity 8.3

i. This application activity 8.3 aims to assess pupils' comprehension on the learnt content.

ii. Provide time to do the exercise individually and note down their progress. It might also be provided as a homework and answers will be given in the next lesson.

iii. Make sure that the instructions on how to complete the activity are clear for every pupil.

iv. During the activity, keep facilitating pupils on how to do and answer the questions, and note their progress.

v. At the end the lesson, ask again some questions of activity 8.3 and application activitty 8.3 to assess the learning and summarize the content.

Answers to the application activity 8.3

1. The flower is a reproductive organ of a plant. It helps to produce new plants.
2. For Q2-Q5, refer to the content pupil's book under the sub-topic 8.3.1, 8.3.2, and 8.3.3

8.4. Asexual reproduction methods

a) Learning objectives

- Explain the process of asexual reproduction of flowering plants

- Recognize just by looking at plants and flowers those reproducing either sexually or asexually.

b) Teaching and learning resources

- Textbooks, charts on asexual reproduction in flowering plants, computers connected to the internet, different vegetative parts collected from the field, a well-painted diagrams of processes such as grafting, cuttings, budding, layering among others on manila papers for use if you lack charts.

c) Guidance and answers to activity 8.4

Guidance on the activity 8.4

i. You may introduce the lesson using probing questions such as 'how do you think plants reproduce without the meeting of male and female seeds?

ii. Let them have a brainstorming session.

iii. Let pupils look at the picture and say what is going on in the pictures of the activity 8.4 pupil's book. For the pupils with visual impairment assist them or pair them with the advanced leaners to help them in the description of the pictures.

iv. Guide the pupils to reach on the intended answers.

v. Build on the pupils' prior knowledge and introduce them to asexual reproduction method.

vi. Summarise the lesson by highlighting main points as pupils write summary notes.

Answers to the activity 8.4

1. Pupil might see: Sugarcane cut into pieces, a man digging holes, a man in the field, etc.
2. A man putting the sugarcane pieces in the dug holes.
3. Some pupils might answer 'yes', other 'no'. But the right answer is 'yes', they will grow.

Guidance on the application activity 8.4

i. Guide pupils to carry out this activity individually. Note: This activity should be done after this sub-topic as a project work that will benefit pupils at their homes.

ii. At the end of the project, pupils should observe that grafting has occurred. Let them find out what happens to their trees and share with other class members at the end of the term.

iii. You can then clarify that grafting helps to improve the quality of a produce of certain plant.

iv. At this point, you can introduce the fact that certain conditions are necessary for grating to occur. A scion and a root stock should be united and grown together but they should be of the same type.

v. Wrap up the lesson by highlighting the main conditions necessary for grafting as pupils take summary notes.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the application activity 8.4

Pupils will provide different answers depending on their observations. Use their findings to identify the importance of the plant's reproduction. Refer to the next sub-topic 8.4 in the pupil's book.

8.5. Importance of plant reproduction

a) Learning objectives

Explain reasons of plant reproduction

b) Teaching and learning resources

Charts, textbooks on reproduction in flowering plants, computers connected to the internet for videos.

c) Guidance and answers to the activity 8.5

Guidance on the activity 8.5

i. You may introduce the lesson using probing questions such as 'From what just finished learning on the reproduction of plants, do you see any benefit?

ii. Let pupils look at the picture and say what is going on in the pictures of the activity 8.5 pupil's book. For the pupils with visual impairment assist them or pair them with the advanced leaners to help them in the description of the pictures.

iii. Guide the pupils to reach on the intended answers.

iv. Guide them to discover the importance of plant reproduction as listed in their pupil's books.

v. Summarise the lesson by highlighting main points as pupils write summary notes.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 8.5

1. We see new young plants growing. Reproduction allows plants to make more plants and grow in different places.
2. We see an old tree and a young sprout. Reproduction allows plants to live for a long time because new plants grow to keep the plant family alive when old plants die.
3. We see harvested crops and food. Reproduction is important to provide food (fruits, seeds, and vegetables); without it, we would not have food to eat.
4. Plants produce seeds and fruits that are used for eating and making medicines.

Guidance to the application activity 8.5

i. This application activity 8.5 targets the critical thinking of pupils on the learnt content.

ii. Allow all pupils to participate and note their progress. Make sure you are being gender sensitive and inclusive for pupils with special needs.

iii. Make sure that everyone in the class is participating to ensure communication skills.

iv. During the activity, keep facilitating and provide the reason to the correct answer for those 'True' questions. You can do the same and explain the reason for the 'false' questions.

v. At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarize the content.

Answers to the application activity 8.5

1. True
2. False
3. True
4. False
5. True

8.6. End unit assessment

I. Guidance to end unit assessment

21. This part provides the guidance and answers for end of unit assessment with cross reference to the textbook.

22. Tell the pupils that the last part of the unit is the end of unit assessment comprising the project where they apply what they learnt and questions that check pupils' knowledge, understanding, skills, attitudes and values related to the whole content of the unit.

23. For the project:

- Request every pupil to do the project either at school or at home. As a teacher, you will decide if it is possible to do the project at school or home and give them guidance. They will use materials available in their community.
- Request pupils to submit their work, organize a class display table or mini exhibition and allow pupils explain how they made their objects and how it is useful.
- Reward the objects made or work of pupils by providing feedback and marks.
- Select some of good objects made by pupils and display them in class.

24. For the quiz, choose some questions for the quiz depending on available time for the assessment. The quiz must be done individually whereby pupils will submit their responses on papers to be marked.

25. As all questions may not be done during the period(s) allocated to the end of unit

assessment, request them to answer the questions during their self-study time or as homework.

26. Mark their answers and keep records of every student's achievement.
27. Provide the remedial, consolidation and extended activities when necessary.

II. Answers to the end of unit assessment

A. Knowledge and understanding

1. C. To help in reproduction
2. D. Flower
3. b) Anther
4. b) Stigma
5. pollination
6. Self-pollination
7. Seeds
8. Germination
9. Pollination is the transfer of pollen grains from anther to stigma of a flower. Whereas fertilization is the fusion of male and female gametes to form a zygote
10. Refer to the figures in the pupil's book in the content under subheading 8.1 (b and c).
11. Stem
12. (i) Use of suckers (ii) Use of cuttings.
13. Bees help in pollination.
14. Refer to the figure in the pupil's book, in the content under subheading 8.1.
15.

	Part	Function
a)	Petal	Attracts insects and birds
b)	Sepal	Protects young flower
c)	Stigma	Receives pollen grains during pollination
d)	Style	Holds stigma in position
e)	Ovary	Contains ovules
f)	Anther	Produces pollen grains

B. Skills

Choose the correct answer

16. a) Pencil and paper
17. b) Sugarcane

Unit 9

SUSTAINABLE WASTE MANAGEMENT



i. Key unit competence:

Pupils will be able to explain and sort different types of waste, use ways to manage waste safely, and show care for the environment by handling waste properly.

ii. Prerequisites

This is a relatively new topic to pupils. However, it is closely related to what pupils learnt in primary 4 about the environment and in primary 5 under water and water pollution. You are therefore expected to show pupils how to manage wastes diseases hence helps to reduce health related costs. You may take advantage of that and introduce the concept of proper waste management as a way of keeping our environment clean. You may also bring to the attention of pupils the fact that the concepts in this unit can be applied in environmental management in social studies, environmental chemistry and Agriculture.

Further, pupils should be made aware of the fact that when they continue with their education in this area, they may become environmentalists. Ask a few questions to pupils on the Prerequisites to assess their prior knowledge; this will orient your methodologies in this unit.

Introductory activity and guidance

A. Guidance on the introductory activity

- Ask pupils to form groups and let them observe the picture on the introductory activity of unit 10, in the pupil's book.
- Let pupils describe deeply what the picture is showing. Ask pupils to explain what is happening in the picture using the questions that follow.
- Ask pupils if there is a better way in which they can manage the wastes in the picture.

B. Possible answers to the introductory activity

1. Responses from pupils might vary depending on what they see.
2. The boy's foot is injured by a broken glass bottle.
3. The accident was caused by wastes that were not properly managed.

iii. Generic competences:

- Critical thinking: Developed through classifying waste according to its decomposition and harmfulness, analyzing sources of waste, and selecting appropriate waste management methods.
- Research and problem solving: Developed through field observations, investigating waste management practices in the community, and proposing practical solutions such as composting, recycling, and safe disposal.
- Co-operation, interpersonal relations and life skills: Developed through group discussions, waste collection activities, preparation of waste classification charts, and collaborative environmental conservation projects.

iv. Basic values:

- Dignity and integrity: Developed through handling waste responsibly, following safety procedures, and maintaining cleanliness in the school and community.
- Hard work: Developed through active participation in waste collection, sorting, compost preparation, and environmental cleaning activities.
- Patriotism: Developed through appreciating the importance of keeping Rwanda clean and contributing to environmental protection through proper waste management practices.

v. Cross cutting issues

1. Standardization culture

Emphasize the need to use certified equipment and machines which do not pollute the environment when dealing with wastes. Caution pupils against using counterfeits which may lead to accidents or injury of users or frequent breakdowns.

2. Inclusive learning

All pupils should be encouraged to participate during lessons and practical. Special arrangement should be made to take care of pupils with special needs. For example, provide brail for blind pupils, large print text for those with sight problems and allocate physically challenged pupils to others to assist them during field trips and practical activities. Further, this category should be given tasks that they can manage during the practical sessions.

3. Financial education

Pupils should be made aware of the fact that a clean environment ensures healthy living devoid of related costs. Also, buying quality equipment will lead to saving money in the long run as they stay for long and costs on repairs are reduced.

4. Environment, climate change and sustainability

This is a topic that directly affects the environment. Emphasize the need to reduce wastes in our environment and other forms of pollution in general. Caution pupils against activities that may cause environmental pollution in general.

5. Peace and values education

People should learn to live in harmony and they should be ready to help other people, especially those who are affected by natural disasters like floods or landslides.

6. Gender education

Both boys and girls should participate in conserving our environment. It is not a preserve of the boy or girl gender. Also, emphasize to pupils that anybody irrespective of their gender can pursue a carrier in environmental health. Give examples of role models who are successful in the area where the pupils come from.

iv. List of lessons

#	Lesson title	Learning objectives	Number of periods: 8
1	Classification of wastes	• To differentiate between biodegradable and non-biodegradable wastes. • Name different sources of waste • Show responsibility by separating hazardous, organic, and recyclable waste.	**1**
2	Sources of waste: Municipal, medical and agricultural	• Identify different sources of wastes	**1**
3	Sources of waste: Automobiles and construction	• Identify different sources of wastes	**1**
4	Sources of waste: Electronics and industrial	• Identify different sources of wastes	**1**
5	Waste management techniques	• Cite, explain and practice • waste management techniques. • Investigate different ways people manage waste in their community • Practice basic waste management methods like sorting and safe disposal	**2**
6	End of unit assessment	• Assess the acquired competences • Demonstrate the achievement of the key unit competence	**2**

Guidance on different lessons

9.1. Classification of wastes

a) Learning objectives

- To differentiate between biodegradable and non-biodegradable wastes

b) Teaching and learning resources

- Computers connected to the Internet for videos, textbooks, and charts on waste management.
- The teacher may improvise by using locally available waste materials (e.g., food remains, leaves, plastics, and bottles) for pupils to observe and classify wastes.

c)Activities' guidance and answers

Guidance on the activity 9.1

i) This lesson will involve more discussions on the **environment sustainability**.

ii) Ask pupils probing questions such as: Have you ever heard of the word 'waste? What does it mean?

iii) Guide pupils to discover the definition of waste, which is 'any unwanted material in the environment.

iv) At this point, guide pupils to go through the pictures then answer the questions associated with them.

v) Let pupils give the answers to the questions as you correct them. Examples of categories of wastes are biodegradable and non-biodegradable wastes. You can also mention that wastes are further divided into groups of their decomposition and harmfulness.

vi) Explain the above kinds of wastes as pupils write summary notes. Refer to pupil's book.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to activity 9.1

1. Picture A shows wastes that can be decomposed.
2. Picture B shows wastes that cannot be decomposed.
3. Yes, there is a difference according to their decomposition.
4. Refer to the summary of unit 9.1.

Guidance to the application activity 9.1

i) This activity involves research work in groups and discussions.

ii) Remind pupils that it is important to always separate wastes into biodegradable/non-biodegradable, harmful, etc.

iii) Let pupils take a walk around the school compound. Let them identify the types of wastes that they can see.

iv) Guide pupils to group the wastes as shown in the table of the application activity 9.1. Let them compare their work to those of other class members.

v) Guide pupils to find out the advantages and disadvantages of the various types of wastes.

vi) Encourage them to always practice this when they come across wastes!

vii) At the end of the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarise the content.

Answers to application activity 9.1

1. Doing the activity.
2. Doing the activity.
3.

Biodegradable wastes	Non-biodegradable wastes	Harmful wastes
Compost to grow crops, clean organic wastes to feed animals, kitchen garbage, animal dung, vegetable remains, ...	Plastics, polythene bags and glass	Chemical wastes (Antifreeze, Brake fluid, Car wax, Diesel fuel, Gasoline. Fuel oil, Kerosene, gas ,.....)

4. Refer to the pupil's book sub-heading 9.1
5. Refer to the pupil's book sub-heading 9.1

9.2. Sources of wastes

a) Learning objectives

- To differentiate between biodegradable and non-biodegradable wastes

b) Teaching and learning resources

- Computers connected to the Internet for videos, textbooks, and charts on waste management.
- The teacher may improvise by using examples of waste sources found in the local community, such as homes, markets, schools, construction sites, and industries.

c) Activities' guidance and answers

Guidance on the activity 9.2

i) This lesson still talks about the environment sustainability, and you might need to use the internet to demonstrate possible sources of wastes.

ii) Let pupils think about the activities that take place in various places mentioned in the activity 9.2. They should say some other sources of wastes based on the hints given in this activity.

iii) You may also guide the pupils to make a field study and observe physically the sources of wastes.

iv) Let pupils give the answers to the questions as you correct them.

v) Explain different wastes as pupils write summary notes.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to activity 9.2

1. There is a machine that removing waste on construction site.
2. The common thing is that all the places shown in the pictures are sources of wastes.
3. Picture A shows a construction site with its wastes, picture B shows the wastes in the industry, then picture C shows waste from the market.

Guidance to the application activity 9.2

i) This activity involves working in groups, and it may serve as a homework.

ii) Guide pupils to brainstorm on the activities that take place in the places mentioned in the application activity 9.2.

iii) Let pupils discuss answers in their groups. Let them identify the activities that they were able to find.

iv) Let them compare their work to those of other class members.

v) Guide pupils to find out the advantages and disadvantages of the various types of wastes.

vi) Encourage them to always aim to have as minimal wastes from their homes as possible to take care of their environment.

vii) At the conclusion of the lesson, pose additional questions regarding the activities conducted or the material learned to evaluate the understanding and summarize the content.

Answers to application activity 9.2

1. *a.*

i. Farms: Crop cultivation, harvesting, and animal rearing (Agricultural activities).

ii. Garage: Vehicle reparations and maintenance (Automobile activities).

iii. Industries: Production of goods, food processing (Industrial activities).

iv. Medical centres: Treatments, laboratory services (Biomedical activities).

b.

i. Farms: Crops remains, animal remains, etc.

ii. Garage: Old tyres and grease containers

iii. Industries: Dirty waters and chemicals, metal scraps, plastic scraps, etc.

iv. Medical centres: plastics gloves, syringes, etc.

2. Urban wastes, construction wastes, and electronic wastes.

- Urban wastes: Food remains
- Construction wastes: Broken bricks, metals , pieces of woods, rubble from stones
- Electronic wastes: damaged computers, TVs, radios etc.

9.3. Waste management techniques

a) Learning objectives

- To cite, explain and practice waste management techniques.
- Go through the engineering design process to design a solution with recycled materials to build something useful for the classroom or school campus

b) Teaching and learning resources

- Scissors, scotch or glue.
- Learning resources to create with (e.g., wood, cloth, paper, plastic, foil, cardboard, wires, strings etc.).

c) Activities' guidance and answers

Guidance to the activity 9.3

i) Introduce the lesson by telling pupils the problem as stated in pupil's book.

ii) Let pupils look at the pictures on page 119 of the pupil book. Ask pupils to describe what they see in the pictures. For pupils with visual impairment assist them or pair them with the advanced learners to help them in the description of the pictures.

iii) Build on the pupil's prior knowledge, introduce them to the appropriate waste management techniques. Highlight that the government of Rwanda (and our school campus) has put effort on collecting wastes from each house and should pay for the waste disposal each month to dump all the garbage in one place (landfill). Tell pupils that a better option than the landfill would be to re-use or recycle the materials into new useful objects.

iv) Explain to pupils that their useful object will need to pass the following tests:

a) Must be useful in the classroom or for the school campus

b) Must fit on the pupil desks

c) Must use at least 2 recycled waste materials for a new purpose

v) Ask pupils to think about items that might be useful in the classroom or the school campus. Some ideas might be garden planters, waste bins for each classroom, recycling bins for each classroom, a box or bin to put their pens and pencils, a box or bin to hold the teacher's chalk and duster. Create a class list of potential objects.

vi) Divide the class into small groups of 3-4 pupils to carry out this task.

vii) Tell pupils to draw their individual design in their exercise books before discussing with others. Encourage them to think about the criteria (passing the tests) as they design their solution.

viii) Allow pupils to share their individual ideas with their group afterwards.

ix) As a group, they will decide which design to build together. It is okay to combine ideas as well. The groups' solutions do not need to be the same. Accept all innovations.

x) Give groups the learning resources to build their designs. Or ask pupils to bring learning resources they find around campus or at home.

xi) As groups finish building, request pupils to submit their work. Organize a class display table or mini exhibition and allow pupils to test their objects by asking themselves or others to demonstrate the following:

a) Does the object solve a problem in the classroom or the school campus?

b) Does the object fit on the pupil desk?

c) Does the object use at least 2 recycled materials for a new purpose?

Then, proceed as follows:

xii) Have pupils explain their design.

xiii) Prompt pupils to talk about the following:

a) How did the group decide which object to design

b) How they decided which recycled learning resources to use

c) How the object is useful to the classroom or school

Answers to the activity 9.3

- The solutions do not need to be the same design or use the same learning resources. Accept all innovations.
- Possible answers in discussions:
  - We decided to design *a* ____ because ____.
  - We used the following recycled learning resources ____ because ____.
  - The object is useful because ____.

Guidance to the application activity 9.3

i) This application activity involves fill in the table waste management methods actions with the correct examples.

ii) Guide the pupils in pairs to do the activity, and this will improve their communication, collaboration and cooperation skills. Alternatively, this activity can be provided to pupils as homework.

iii) While pairing your pupils, make sure that they are paired according to their needs and their abilities so that they could help one another. N.B: Be gender responsive and inclusive in your pairing.

iv) Provide the answers to the table on how the pupils should have filled in each waste management technique.

v) Throughout the activity, continue to guide pupils on how to respond to the questions and monitor their progress. At the conclusion of the lesson, pose additional questions regarding the activities conducted or the material learned to evaluate the learning outcomes and summarize the content.

Answer to the application activity 9.3

Waste management technique	**Example**
Reuse	Old plastic bottle used to grow flowers
Composting	Making manure from food wastes
Landfilling	Burying garbage on a large land
Recycle	Changing the plastics into chairs
Reduce	Using a few plastics containers
Incinerating	Burning medical wastes

9.4. End unit assessment

I. Guidance to end unit assessment

1. This part provides the guidance and answers for end of unit assessment with cross reference to the textbook.
2. Tell the pupils that the last part of the unit is the end of unit assessment comprising the project where they apply what they learnt and questions that check pupils' knowledge, understanding, skills, attitudes and values related to the whole content of the unit.
3. **For the project:**

- Request every pupil to do the project either at school or at home. As a teacher, you will decide if it is possible to do the project at school or home and give them guidance. They will use materials available in their community.
- Request pupils to submit their work, organize a class display table or mini exhibition and allow pupils explain how they made their objects and how it is useful.
- Reward the objects made or work of pupils by providing feedback and marks.
- Select some of good objects made by pupils and display them in class.

4. **For the quiz,** choose some questions for the quiz depending on available time for the assessment. The quiz must be done individually whereby pupils will submit their responses on papers to be marked.
5. As all questions may not be done during the period(s) allocated to the end of unit assessment, request them to answer the questions during their self-study time or as homework.
6. Mark their answers and keep records of every student's achievement.
7. Provide the remedial, consolidation and extended activities when necessary.

II. Answers to the end of unit assessment

A. Knowledge and understanding

1. b) Biodegradable waste
2. c) Expired medicines
3. b) To collect, treat, and dispose of waste safely
4. Non-biodegradable
5. Composting
6. Hazardous
7. Non-biodegradable waste cannot decompose naturally (for example polythene and plastics, while degradable waste can decompose (for example, kitchen wastes).
8. Recycling is the process of taking old or used materials and turning them into new products, then use it again;

   Re-using means using something again instead of throwing it away. This helps reduce waste and save money.
9. By putting in place strict legislations that hinder environmental degradation and educating its citizens on the importance of environmental conservation.
10. Recycle, Re-use, Reduce.
11.

Type of waste	Example	How to manage
Industrial	Smoke	Fitting chimneys and catalytic converters
Medical	Syringes and needles	Incineration
Electronic	Old computer parts	Recycling
Agricultural	Chemical containers	Re-using

B. Skills

12. b) Sorting waste into different groups
13. b) Group wastes into biodegradable and non-biodegradable
14. Recycling
15. Safe disposal
16. True
17. False
18.
    - **Biodegradable:** Orange peel, paper
    - **Non-biodegradable:** Plastic bottle
    - **Hazardous:** Battery, broken glass

C. Attitudes and values

19.a) To keep the environment and people safe

20.b) Being responsible and separating wastes

21.Dangerous

22.Responsibility

23.True

24.False

9.5. Additional activities

a) Remedial activities

1. Waste is _____

2. Give 3 examples of non-biodegradable wastes.

3. How would you manage harmful wastes?

❖ **Answers to the remedial activities**

1. Unwanted materials in the environment.

2. Polythene, pipes and plastics.

3. Burning in an incinerator.

b) Consolidation activities

1. What is waste?

2. (a) Mention the two main categories of wastes.

   (b) What is the difference between them?

3. Which other categories of wastes do you know?

4.What sources of wastes do you know of?

5.What is the difference between biomedical wastes and electronic wastes?

6.What sources of wastes do you know of?

7.What is the difference between construction wastes and automobile wastes?

8. Give four ways of disposing of wastes safely.

❖ **Answers to the consolidation activities**

1. Any unwanted material in the environment)

2. (a) Biodegradable and non-biodegradable wastes

(b) Biodegradable – can decompose naturally, non-biodegradable wastes – cannot decompose naturally

3. Municipal, industrial construction,
4. agricultural, biomedical, electronic and automobile wastes.
5. Biomedical from hospitals and health centres; electronic – from electronic equipment.
6. Municipal, industrial construction, agricultural, biomedical, electronic and automobile wastes.
7. Construction wastes from construction sites; automobile wastes – garages.
8. Safe collection, processing, transportation and finally disposal.

C. Extended activities for gifted and talented pupils

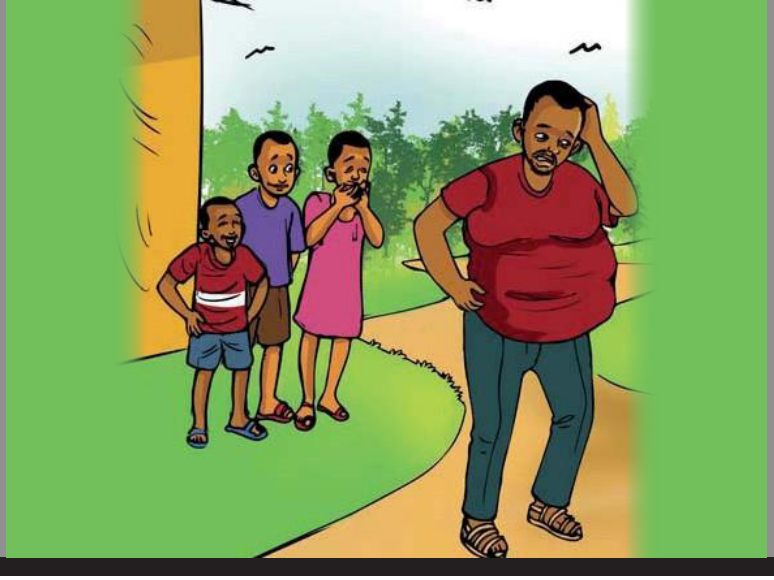
1. What are radioactive wastes.
2. What is the other name of biodegradable wastes?
3. Describe the process of managing wastes professionally.

❖ **Answers to the extended activities**

1. Wastes that emit radiations.
2. Organic wastes
3. Collecting, processing, transporting and finally disposing in a landfill

Unit 10

HUMAN CIRCULATORY SYSTEM



i. Key unit Competence:

Pupils will be able to identify the parts of the circulatory system, explain how it works, and demonstrate healthy habits to keep it functioning well

ii. Prerequisites

This unit builds on pupils' prior knowledge of the human body. In the lower primary classes, pupils were introduced to basic concepts about the human body. In Primary 4, they explored more advanced topics such as the sense organs and the human skeleton. In Primary 5, they furthered this understanding by learning about the digestive and reproductive systems. In this class, they will expand their knowledge by studying the circulatory and respiratory systems. As a teacher, you are encouraged to help pupils make connections between this content and the circulatory system in biology, which they will encounter in greater detail at higher levels of learning. Emphasize that at this stage, the pupils are expected to understand only the basic information, while deeper understanding of how these systems function will be developed in high school and beyond.

Introductory activity and guidance

A. Guidance on the introductory activity

i) Prepare the pupils by displaying the picture of the human circulatory system clearly to the whole class (e.g. using a chart, projector, or textbook)

ii) Activate prior knowledge by asking pupils to recall what they already know about the human body, especially any previous lessons on body systems like the digestive or reproductive systems.

iii) Guide observation by encouraging pupils to closely look at the red and blue lines (arteries and veins) in the image, prompting them to describe what they see using their own words.

iv) Ask open-ended questions such as those in the activity (e.g., "What do you think moves around inside your body to keep you alive?") to spark curiosity and critical thinking.

v) Encourage group discussion by letting pupils share their ideas in pairs or small groups

before responding in front of the class, promoting communication and collaboration skills.

vi) Connect to real life by explaining how the circulatory system helps transport nutrients, oxygen, and waste in the body, and why it's important for their health and survival.

vii) Assess learning informally by listening to pupil responses, asking follow-up questions, and clarifying any misconceptions while encouraging all pupils to participate.

B. Possible answers to the introductory activity

1. I see a human body with red and blue lines going all around it. There is a big red part in the chest, it looks like a heart.
2. The red fibers look like blood vessels. They carry blood around the body.
3. Blood moves around the body to keep us alive. It carries oxygen and food to all parts.
4. After digestion, the food goes into the blood, which then moves through blood vessels to all parts of the body.

iii. Generic competences:

- Critical thinking: Developed through explaining the functions of the circulatory system, interpreting blood circulation processes, analyzing common circulatory diseases, and relating healthy habits to the proper functioning of the system.
- Research and problem solving: Developed through conducting research, investigating circulatory health, interpreting blood pressure measurements, and proposing ways to prevent circulatory diseases.
- Communication (Literacy and Numeracy): Developed through presenting research findings, explaining the functions of the circulatory system and its organs, interpreting blood pressure results, and communicating healthy lifestyle practices.

iv. Basic values:

- Dignity and integrity: Developed through taking personal responsibility for maintaining good circulatory health by practicing hygiene, healthy eating, regular exercise, and routine health check-ups.
- Self-reliance: Developed through adopting healthy habits, monitoring personal health, and making informed decisions to prevent circulatory diseases.
- Commitment and resilience: Developed through consistently practicing healthy lifestyles, following preventive measures, and encouraging others to maintain good cardiovascular health.

v. Cross cutting issues

1. Standardization culture:

Bring to the attention of pupils the need to seek medical healthcare in standard and quality hospitals whenever they have problems with their circulatory system.

2. Financial education:

Emphasize the fact that pupils should practice good hygiene to avoid conditions and disease of the circulatory system in order to avoid spending money on treatment.

3. Gender education:

Emphasize to pupils that anybody irrespective of their gender can pursue a carrier in medicine. Give examples of role models who are successful cardiologists of specialist doctors in the area where the learners come from

4. Inclusive education:

All pupils should be encouraged to participate during lessons and practical. Special arrangement should be done to take care of learners with special needs. For example, provide braille for blind pupils, large print text for those with sight problems and allocate physically challenged pupils to others to assist them during field trips and practical activities. Further, this category should be given tasks that they can manage during the practical sessions.

vi. List of lessons

#	Lesson title	Learning objectives	Number of periods: 16
1	**Main function of circulatory system**	• Identify the main function of the circulatory system. • Describe how the circulatory system works in transporting blood.	2
2	**Organs of the circulatory system**	• Draw and name the key organs of the circulatory system	2
3	**Structure of the human heart**	• State the main parts of the heart • Draw and label the basic structure of the heart.	2
4	**Blood vessels**	• Identify types of blood vessels • Differentiate blood vessels based on function and structure	2

5	**The process of blood Circulation**	• Outline how blood flows through the heart and body. • Develop the habit of regularly checking blood pressure.	2
6	**Components of blood**	• List components of blood • Match each component with its function.	2
7	**Caring for our bodies and health of our circulatory system**	• Identify ways to keep the circulatory system healthy. • Take responsibility to keep their circulatory system healthy. • Practice good hygiene to keep the circulatory system healthy.	1
8	**Common diseases of circulatory system, symptoms, control, and prevention**	• Mention common circulatory system diseases. • Describe symptoms and prevention methods of circulatory diseases. • Be aware of diseases that affect the circulatory system.	1
9	**Blood pressure measurement**	• Observe and describe how blood pressure is measured using a device.	1
10	**End of unit assessment**	• Assess the acquired competences • Demonstrate the achievement of the key unit competence	1

Guidance on different lessons

10.1 Main function of circulatory system

a) Learning objectives

- Identify the main function of the circulatory system.
- Describe how the circulatory system works in transporting blood.

b) Teaching and learning resources

- Wall chart or poster showing the human circulatory system
- Flashcards with key terms
- Text books

IMPROVISATION: You may come up with your own charts drawn on manila papers in case your school does not have the circulatory system charts.

c) Guidance and answers to the activity

Guidance on the activity 10.1

This activity is meant to help pupils develop a clear understanding of the functions of the human circulatory system. You will proceed as follows:

i) Start by telling pupils how they will do research to find out the functions of the human circulatory system and write their findings.

ii) Then have a short discussion to remind pupils what they already know about the circulatory system.

iii) Divide them into pairs or small groups to encourage collaboration and sharing of ideas.

iv) Offer textbooks, printed articles, or visual aids. If available, let pupils use digital resources under supervision.

v) Walk around and support groups by asking guiding questions (e.g., "What does the blood do in the body?").

vi) Instruct pupils to write short, clear points from what they find rather than copying large chunks of text.

vii) Give each group time to share their findings with the class in their own words to build confidence and communication skills.

viii) Correct misconceptions, highlight correct answers, and encourage pupils to ask questions for clarification.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 10.1

1.&2. For the answers of questions in this activity, see the content under sub-heading 10.1 (Main function of human circulatory system) in the Pupil's book.

Guidance on the application activity 10.1

This activity is meant to assess pupils' understanding of the main functions of the human circulatory system. You will proceed as follows:

i) Briefly review the key functions of the circulatory system before starting the activity to ensure pupils are ready.

ii) Make sure pupils understand how multiple-choice questions work, they must choose only one correct answer for each.

iii) Allow pupils to complete the quiz individually or in pairs for peer support, depending on their ability levels.

iv) After the quiz, go through each question together, allowing pupils to explain their choices and correct any misunderstandings.

v) Summarize the main points again and encourage pupils to relate the functions of the circulatory system to real-life examples (e.g., breathing, eating, getting sick, or growing).

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the application activity 10.1

1. B
2. C
3. C
4. D
5. C

10.2. Organs of the circulatory system

a) Learning objective

-Name the key organs of the circulatory system

b) Teaching and learning resources

- Labelled diagrams of the human heart
- Textbooks
- Improvisation: You may come up with your own chart of structure of the heart drawn on manila papers in case your school does not have one. You can also organize for pupils to model a heart using clay or plasticine.

c) Guidance and answers to the activity 10.2

Guidance on the activity 10.2

This activity is meant to help pupils recognize and name the key organs involved in blood circulation. You will therefore proceed as follows;

i) Print or draw clear picture cards of the heart, blood vessels, and blood, along with matching word cards.

ii) Ask the pupils easy, relatable questions (e.g. Have you felt your heartbeat? to activate prior knowledge.

iii) Show the picture and word cards and ask pupils to come up and match each picture with the correct word.

iv) Ask pupils what they think each part does. Allow 1–2 pupils to share their ideas to spark discussion.

v) End the activity with a short explanation of each organ's role and how they work together in the circulatory system. Explain that they are the main organs of the circulatory system and that they work together to move blood around the body.

Answers to the activity 10.2

1. Human circulatory system. The red and blue lines represent blood circulating in the body
2. The red and blue lines represent blood circulating in the body. Red lines represent the oxygenated (clean) blood; Blue lines represent the deoxygenated (dirty) blood.
3. It is important for blood to move around the body because it carries oxygen and food to all parts of the body. Blood also takes away waste that the body does not need. This helps the body stay alive, strong, and healthy.
4. After digestion, the food is broken into small parts in the stomach and intestines. These small parts go into the blood. The blood carries the food to all parts of the body through blood vessels. This gives the body energy and nutrients to work and grow.

Teacher's wrap-up or Simple Explanation:

The heart is a strong muscle that pumps blood. Blood vessels are like roads that carry the blood all around the body. Blood moves through these vessels, bringing oxygen and food to every part. Together, they make up the circulatory system.

Guidance on the application activity 10.2

This activity is meant to help pupils apply their understanding of the circulatory system by analysing a real-life health scenario. You will therefore proceed as follows:

i) Read the scenario aloud or display it on the board. Make sure all pupils understand Amina's condition and that it relates to the circulatory system.

ii) Tell pupils the goal is to apply what they've learned about the heart, blood vessels, and blood to solve a real-life problem.

iii) Give pupils individual copies or write the multiple-choice questions on the board. Allow them to answer individually or in pairs.

iv) Go through each question with the class. Let pupils explain why they chose their answers, then guide them to the correct responses.

v) Summarize by linking Amina's symptoms to how each organ in the circulatory system works. Emphasize the importance of a healthy heart, blood, and blood vessels.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the application activity 10.2

1. b) Heart
2. c) To pump blood to all parts of the body
3. b) Blood is not reaching her hands and feet quickly
4. c) Blood vessels
5. b) Oxygen and food

10.3. Structure of the heart

a) Learning objectives

- State the main parts of the heart
- Draw and label the basic structure of the heart.

b) Teaching and learning resources

- Cross-section diagram or 3D model of the heart
- Clay or paper to make heart models

IMPROVISATION: You may come up with your own chart of the structure of a heart drawn on manila papers.

c) Guidance and answers to the activity 10.3

Guidance on the activity 10.3

This activity is meant to help pupils recall what they already know about the function of the heart. You will therefore proceed as follows:

i) Begin by telling pupils that they will explore how the heart is built and why it's important. Explain that you will ask a few questions to see what they already know.

ii) Read each of the four prompting questions slowly. Give pupils time to think and respond, either by raising hands or discussing briefly in pairs.

iii) Accept all reasonable answers. Praise effort and curiosity, even if the answers are not fully correct.

iv) After each question, add a simple explanation or correct information to build on their ideas (refer to the answers provided).

v) After all the four questions, explain that they will now learn more details about the heart and label its parts in their notebooks.

Answers to the activity 10.3

1. Four chambers: right atrium, right ventricle, left atrium, left ventricle.
2. Red = oxygen-rich blood. Blue = oxygen-poor blood.
3. Arrows show blood flow: body → right heart → lungs → left heart → body.
4. Pulmonary vein brings oxygenated blood from lungs to left atrium of the heart.
5. Pulmonary artery carries deoxygenated blood from right ventricle to lungs.

Guidance on the application activity 10.3

This activity is aimed at helping pupils visually understand and reinforce their knowledge of the human heart's structure. You will therefore proceed as follows;

i) Briefly go over the heart's parts (chambers, valves, septum) and how blood flows through them using a diagram or model before pupils begin the activity.

ii) Ask pupils to draw a simple, neat heart diagram in their notebooks, large enough to label clearly. Guide them to label the four chambers accurately.

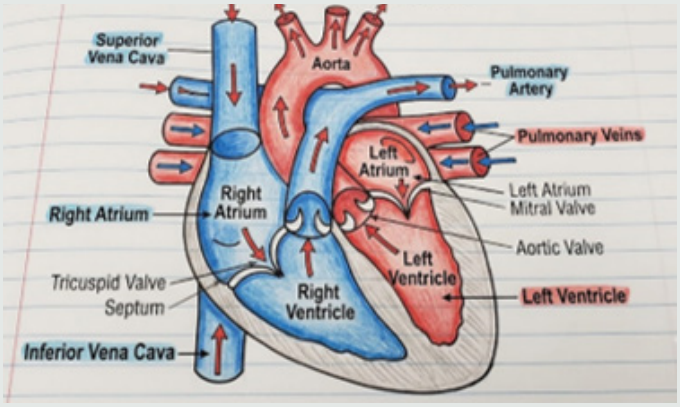
iii) Instruct pupils to use arrows to show the flow of blood through the heart, from the body to the heart, to the lungs, and back to the body.

iv) Remind pupils to use blue for deoxygenated blood (right side of the heart) and red for oxygenated blood (left side). Ensure they colour the appropriate chambers and vessels correctly.

v) Walk around as pupils' work, helping those who need support. When done, choose a few pupils to present or explain their drawing to reinforce learning.

Answers to the application activity 10.3

Structure of the human heart



10.4. Blood vessels

a) Learning objectives

- Identify types of blood vessels

-Differentiate blood vessels based on function and structure

b) Teaching and learning resources

- Charts comparing arteries, veins, and capillaries

- Straws or coloured strings to represent different blood vessels

c) Guidance and answers to the activity

Guidance on the activity 10.4

This activity is aimed at helping pupils recognize the names and roles of different blood vessels. You will proceed as follows;

i) Begin by telling pupils they will explore the three main types of blood vessels in the body: arteries, veins, and capillaries.

ii) Read each of the four prompting questions aloud one by one. Give pupils time to think or discuss in pairs before sharing answers.

iii) Accept all responses without judgment. Prompt pupils to explain their reasoning where possible, even if the answer is incorrect.

iv) Show the structure diagrams (Figs. 10.3, 10.4, 10.5) after each question to help pupils visualize and understand the correct answers.

v) After discussing all questions, briefly explain the role of each blood vessel and how they work together to transport blood throughout the body.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 10.4

1. Artery
2. Vein
3. Vein

Guidance on the application activity 10.4

This activity is meant to help pupils recognize and identify the major blood vessels involved in human blood circulation. You will proceed as follows;

i) Begin by showing the picture of human blood circulation to the pupils. Briefly explain that the picture illustrates how blood flows through the body and the role of different blood vessels.

ii) Encourage pupils to carefully observe the labelled blood vessels in the picture. Ask them to note the key features of each vessel's placement and direction of blood flow.

iii) Ask guiding questions like, "Where do you think blood is coming from and going to in each part of the circulation?" or "What do you already know about these blood vessels?"

iv) Give pupils time to individually name the blood vessels labelled a, b, and c. Walk around to check their understanding and offer hints if necessary.

v) Once pupils have made their guesses, discuss the correct answers as a class. Explain the role of each blood vessel and its part in the circulatory system.

Answers to the application activity 10.4

1. a) Veins
2. b) Capillaries
3. c) Arteries

10.5. The process of blood circulation

a) Learning objective

- Outline how blood flows through the heart and body.
- Build a model of the heart to simulate the pumping of the heart.

b) Teaching and learning resources

- 3 plastic water bottles
- Punch one hole in the first bottle cap big enough to fit a straw through, and a second hole that is smaller. 
- Punch two holes in the second bottle cap, big enough for a straw to fit through each hole. 
- 4 straws or clear tubing
- Bend the straws so that they can reach between the water bottles. You can use scotch to join the straws together to make them fit. 
- Optional: food coloring
- Optional: modeling clay or scotch

c) Guidance and answers to the activity 10.5

Guidance on the activity 10.5

This lesson is aimed at assessing and activating pupil's prior knowledge about how blood moves through the body. Pupils should work in small groups to carry out this task. If small groups are not possible, this task can be done as a pupil-led demonstration.

i) Tell pupils that they will do an investigation to answer this science question: How does the heart pump blood into the lungs?

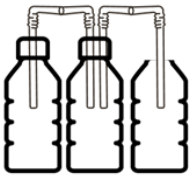
ii) Instruct pupils to feel for their heartbeat on their wrist or neck. Have them count how many heartbeats they feel for 10 seconds. Multiply their count by 6 to get their resting heartbeats per minute. Instruct pupils to record their resting heartbeats in their exercise books.

iii) Instruct pupils to jump or run in place for 10 seconds and then repeat step 2. Ask pupils if their heartbeats per minute have changed.

iv) Divide the class into small groups of 3-4 pupils to carry out this next part of the investigation.

v) Place three bottles on the pupil desks. Fill the first two bottles 80% with water. Leave the third bottle empty.

vi) Place the cap with one straw hole and one small hole on the first bottle. On the middle bottle, place the cap with two straw holes. Leave the third bottle without a cap.



vii) Carefully slide the straws through the bottle caps. If needed, place scotch or modeling clay around the straw bases to make an airtight seal.

viii) Instruct pupils to use their model.

a) Squeeze the middle bottle only. Observe what happens.

b) Squeeze the middle bottle while pinching the straw between the first and middle bottles. Observe what happens.

c) Keeping the middle bottle "squeezed", now pinch the straw between the middle and third bottles. Release the middle bottle and observe what happens.

d) If the "blood" (water) gets too low in any one bottle, reset the bottles so the first two bottles have 80% water and the third bottle is empty.

ix) Discuss with pupils:

a) Which parts of the heart were they modeling?

The parts of the heart and how blood flows through the heart to the lungs and the body. Display or draw a simple diagram of the heart and circulation path on the board to help pupils understand the four chambers of the heart.

b) How does blood pumping constantly keep organs functioning?

Pupils may use pages 132-133 of the pupil's book for reference.

Then, proceed as follows:

x) Explain to learners they will use what they've learned to answer the questions and think about the investigation they just completed.

a) How does this model relate to when we were counting our heartbeats?

b) What do you think would happen if we pinched both of the straws closed?

c) What would happen if one of the straws was halfway closed?

d) Why does the heart have four parts and two sides (left and right)?

xi) Restate the science question that was introduced at the beginning of the lesson: How does the heart pump blood into the lungs? Have pupils answer the science question in their own words. Encourage them to use information from the lesson.

Answers to the activity 10.5

- Possible answers in first discussions:

a) First bottle: Atrium of the heart

Middle bottle: Ventricle of the heart

Third bottle: Lungs of the body

Straws: Veins

Fingers: Valves of the heart

b) See pupil's book pages 132-133

- Possible answers in second discussions:

a) The model relates to our heartbeats because it shows how each beat is our heart pumping blood from the atrium to the ventricle and to the lungs.

b) If we pinched both straws closed, our heart could not beat.

c) If a straw was halfway closed, it would not beat as well, and less "blood" would travel to the next place.

d) The heart has four chambers to keep oxygen-rich and oxygen-poor blood separate. The right side of the heart sends deoxygenated blood to the lungs to get oxygen. The left side of the heart sends oxygenated blood to the rest of the body.

- Possible written response to science question:

The heart pumps blood to the lungs by taking in deoxygenated blood from the body, pumping it through the right side (atrium) to the lungs. The lungs return oxygenated blood to the heart's left side (ventricle) which then pumps it to the rest of the body.

Guidance on the application activity 10.5

This activity is meant to enhance pupils' ability to describe and visualize the process of blood circulation using diagrams and flowcharts. You will proceed as follows;

i) Explain to pupils that they will use what they've learned to answer questions and complete a diagram and a table about how the circulatory system works.

ii) Go through each question slowly. Allow pupils time to think, discuss in pairs, or write answers before moving to the next one, especially for the flow diagram.

iii) Display or draw a simple diagram of the heart and circulation path on the board to help pupils complete the flow diagram and understand the four chambers of the heart.

iv) Ensure pupils understand how to complete the table, whether it's comparing features, identifying parts, or functions. Clarify column headings if needed.

v) After pupils complete their work, allow them to exchange books and check each other's answers. Discuss the correct responses together as a class to reinforce understanding.

Answers to the application activity 10.5

1. The circulatory system is made of:

- The heart (pumps blood)
- Blood vessels (tubes that carry blood around the body: arteries, veins, and capillaries)
- Blood (the liquid that moves through the vessels, carrying oxygen and nutrients)

2. Deoxygenated blood comes from the body into the right atrium.

- The right atrium sends the blood to the right ventricle.
- The right ventricle sends the blood to the lungs to get oxygen.
- Oxygenated blood returns from the lungs to the left atrium.
- The left atrium sends the oxygen-rich blood to the left ventricle.
- The left ventricle pumps the blood to the rest of the body.
- Blood returns to the heart, and the cycle starts again.

Answer:

- The heart has 4 chambers to keep oxygen-rich and oxygen-poor blood separate.
- The right side of the heart sends deoxygenated blood to the lungs to get oxygen.
- The left side sends oxygenated blood to the rest of the body.

3.

Feature	**Arteries**	**Veins**
Direction of blood	Carries blood away from the heart	Carries blood towards the heart
Type of blood	Oxygenated blood (except in lungs)	Deoxygenated blood (except in lungs)
Pressure	High pressure	Low pressure
Walls	Thick walls	Thinner walls
Examples	Aorta, pulmonary artery	Vena cava, pulmonary vein

- Arteries carry oxygen-rich blood from the heart and have thick walls because the blood moves fast and under high pressure.
- Veins carry oxygen-poor blood back to the heart, have thinner walls, and have valves to stop blood from flowing backwards.

10.6. Components of human blood

a) Learning objectives

- List components of blood.
- Match each component with its function.

b) Teaching and learning resources

- Flashcards with blood components (red cells, white cells, etc.)
- Food colouring and water in jars to represent blood components
- The teacher may improvise by using simple locally available materials (e.g., coloured water, drawings, or labelled containers) to represent blood components if charts or flashcards are not available.

Guidance on the activity 10.6

This activity is meant to assess and activate pupils' prior knowledge about blood and its components, you will proceed as follows;

i) Arrange pupils in pairs or small groups to encourage participation and discussion. Ensure the board or chart paper is ready for listing responses.

ii) Begin with a simple question like, "What do you think blood does in your body?" to spark curiosity before giving instructions for the Think-Pair-Share activity.

iii) Give pupils 1 minute to think individually, 2–3 minutes to discuss in pairs, and then ask several pairs to share their thoughts with the class.

iv) Present the five questions one by one. Encourage open responses and write pupils' ideas on the board, grouping similar answers together.

v) After collecting responses, briefly explain the correct answers and connect pupils' ideas to the components of blood that they will study in more detail.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 10.6

a. Three colors: yellow, white, and red.

b. Yellow = plasma, white = white blood cells & platelets, red = red blood cells.

c. Yes, I had a cut, and blood came out. After some time, the bleeding stopped and a scab formed. The blood formed a clot and the wound began to heal.

d. White blood cells help the body fight diseases.

Guidance on the application activity 10.6

This activity is meant to assess and reinforce pupils' understanding of the four main components of blood. You will therefore proceed as follows;

i) Begin by briefly reviewing the four main components of blood and their functions to refresh pupils' memory before the activity.

ii) Go through each part of the activity (matching, fill-in-the-blanks, and short answers), making sure pupils understand what is expected in each section.

iii) Ask pupils to complete the activity individually to assess their personal understanding, then allow peer discussion afterward if needed.

iv) Walk around the classroom as pupils' work, offering help and clarifying any questions or misunderstandings about the blood components.

v) Go over the answers with the whole class, correcting misconceptions and reinforcing the correct functions of each blood component. Use this time for additional teaching if needed.

Answers to the application activity 10.6

Part1.

a) Plasma	b Transports nutrients and waste
b) Red blood cells	c Carries oxygen to body organs
c) White blood cells	d Protects the body from infections
d) Platelets	a Helps stop bleeding by forming clots

Part 2.

a) hemoglobin.

b) immunity.

c) wounds.

d) Plasma

e) oxygen

Part 3: Short Answer Questions

a) Plasma is important because it carries nutrients, hormones, proteins, and waste products throughout the body.

b) Without platelets, the blood wouldn't clot, and we could bleed a lot even from small injuries. Wounds would take longer to heal.

c) White blood cells fight infections and destroy harmful germs that enter the body, helping to keep us from getting sick.

10.7. Caring for our bodies and health of our circulation system.

a) Learning objectives

- Identify ways to keep the circulatory system healthy.

b) Teaching and learning resources

- Posters on healthy habits (diet, exercise, avoiding smoking)
- Role-play cards for pupils to act out healthy and unhealthy behaviours

c) Guidance and answers to the activity 10.7

Guidance on the activity 10.7

This activity is aimed at helping pupils identify healthy and unhealthy habits that affect the circulatory system. You will therefore proceed as follows:

i) Begin by asking pupils simple, relatable questions like: "What do you do to stay healthy?" or "Do you know how to take care of your heart or blood?" This will engage pupils and prepare them for the scenario.

ii) Tell the story of John and his cousin slowly and with feeling, to help pupils imagine the situation. You may ask a confident pupil to read it, or act it out briefly for more engagement.

iii) Write the 5 questions on the board. Organize pupils into pairs or small groups and let them discuss the questions together. Encourage all pupils to contribute.

iv) Ask each group to share their answers. As pupils respond, write correct points under the correct headings (heart, blood, blood vessels). Use this moment to correct misconceptions and praise thoughtful answers.

v) Summarize the discussion by linking it to the importance of the circulatory system and healthy habits. Clearly state what pupils will explore next (e.g., the components of blood, functions, and how to care for them).

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 10.7

1. He eats fruits and vegetables (a balanced diet).
   - He drinks clean water regularly.
   - He plays football (regular physical exercise).
   - He avoids harmful substances like cigarettes.
2. He smokes cigarettes.
   - He encourages others (John) to try smoking, which is dangerous.
   - He may not be exercising or eating healthy.
3. Smoking can damage the heart and blood vessels.
   - It reduces the amount of oxygen the blood carries.
   - It increases the risk of heart diseases.
   - It can block blood vessels, making it hard for blood to flow.
4. Sports help the heart beat stronger and pump blood well.
   - Exercise improves blood flow and strengthens the blood vessels.
   - It helps maintain a healthy weight and reduces stress.
   - It keeps the body active and reduces the chance of disease.
5. I eat healthy food.
   - I drink a lot of clean water.
   - I play games and do sports.
   - I avoid bad habits like smoking.
   - I rest and sleep well.
   - I try to stay calm and manage stress.

Guidance on the application activity 10.7

This activity is meant to encourage pupils to observe and reflect on real-life behaviours in their community that impact the health of the circulatory system, you will proceed as follows;

i) Explain the purpose of the activity to the pupils. Ensure they understand they are looking for behaviours in the community that could impact the heart, blood, and blood vessels (both positive and negative).

ii) Provide clear instructions on what to look for, including healthy habits (e.g., exercising, eating healthy) and harmful behaviours (e.g., smoking, drinking alcohol).

iii) Ensure safety: Remind pupils to be safe while observing in the community. They should stay in groups and observe without disturbing people or putting themselves in harm's way.

iv) Encourage respectful observation: Pupils should not make judgments or interrupt people but simply observe and take notes.

v) Give pupils a list of guiding questions (like the ones provided) to help them stay focused. Ask them to make brief notes or sketches of what they see during their community walk.

vi) After pupils return to class, organize group presentations where each group shares their observations. Guide them to discuss what they observed and make connections to what they've learned about the circulatory system.

vii) Encourage pupils to think critically about how they could encourage healthier behaviours in the community.

viii) Have pupils reflect on how they could promote good habits in their families and communities. Ask them to propose practical solutions (e.g., spreading awareness of healthy eating or exercise) that can help improve the health of the circulatory system in their community.



Answers to the application activity 10.7

ii. Physical exercise: Some people were seen jogging, walking, or riding bikes, which is good for heart health and improves blood circulation.

- Community sports: Groups of children were playing games like football or volleyball, promoting exercise and heart health.

ii. Smoking: Many people in the community were seen smoking, which is harmful to the heart and blood vessels.

- Drinking alcohol: Some adults were seen drinking alcohol in public, which can increase the risk of heart diseases and damage the liver.
- Unhealthy eating: People, especially in busy areas, were often eating junk food (e.g., chips, soda), which can lead to obesity and high blood pressure.
- Lack of exercise: Most people were sitting or not engaging in physical activities. There were few people walking or exercising

iii. Healthy eating habits: People were eating fruits, vegetables, and other healthy food choices, especially in markets or gardens.

- Drinking water: People were carrying or drinking clean water, which is important for maintaining healthy blood and hydration.

Note: Ensure to develop the values and generic competencies related to this lesson.

10.8. Common diseases of the circulatory system, symptoms, control, and prevention

a) Learning objectives

- Mention common circulatory system diseases.
- Describe symptoms and prevention methods of circulatory diseases.

b) Teaching and learning resources

- Charts showing symptoms, causes, prevention
- Visit or talk from a local health worker (if possible)
- Text books

c) Guidance and answers to the activity

Guidance on the activity 10.8

This activity is aimed at helping pupils recall or identify common circulatory diseases they may have heard of or seen in real life. You will proceed as follows;

i) Start the lesson by narrating a short story to capture pupils' attention and make the topic meaningful.

ii) Write the questions on the board and read them aloud.

iii) Ask pupils to share what they already know or have heard. Encourage participation from both boys and girls, and validate all answers.

iv) Allow them to discuss the questions in groups to promote peer learning and critical thinking.

v) Move around the class to listen, guide, and support pupils' discussions.

vi) Invite each group to present one or two key points they discussed.

vii) Write correct or important ideas on the board under headings like "Causes," "Symptoms," and "Prevention."

viii) Summarize what the pupils shared and connect their ideas to the diseases of the circulatory system.

ix) Clearly explain that they will now learn about specific conditions like high blood pressure, heart attack, and stroke, their symptoms, causes, and prevention.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 10.8

1. Yes, I have heard about them.
   - High blood pressure is when blood moves through the body with too much force.
   - A heart attack happens when the heart does not get enough oxygen.
   - A stroke is when blood stops flowing to the brain, which can make someone unable to talk or move.

2.Chest pain or tightness
   - Sweating a lot without doing exercise
   - Feeling very weak or tired suddenly
   - Having trouble breathing
   - Dizziness or fainting

3. Eating too much salty or fatty food
   - Not exercising or being lazy
   - Smoking cigarettes
   - Drinking alcohol
   - Stress and anger
   - Not getting medical checkups

4. Eating fruits and vegetables
   - Drinking enough clean water
   - Doing physical activities like running or playing sports
   - Avoiding smoking and alcohol
   - Going for regular health checkups
   - Managing stress by resting or talking to someone

5. Call for help or take them to the hospital immediately
   - Tell an adult or neighbour
   - Try to keep the person calm and help them sit or lie down
   - Do not give them food or water if they cannot speak
   - Stay with them until help comes

Guidance on the application activity 10.8

This activity is meant to assess and reinforce pupils' understanding of common diseases of the circulatory system. You will proceed as follows;

i. Begin by informing pupils that the activity is meant to help them apply and assess what they have learned about circulatory system diseases such as high blood pressure, heart attack, and stroke.

ii. Emphasize that the goal is not just to get correct answers, but to understand the health concepts.

iii. Read each question aloud slowly and clearly, especially for pupils with reading challenges.

iv. Allow pupils to answer the questions individually to promote critical thinking and personal responsibility.

v. Give them a reasonable time limit (e.g. 10–15 minutes) to complete the quiz.

vi. After pupils have finished, go over each question and ask for volunteers to share their answers and explain their choices.

vii. Clarify the correct answers and correct any misunderstandings to reinforce learning.

viii. Use pupils' responses to identify areas where additional teaching is needed.

ix. Provide encouragement and offer short remedial explanations or activities to pupils who struggled with certain questions.

Answers to the application activity 10.8

1. B
2. C
3. C
4. C
5. C
6. B
7. C

10.9. Blood pressure measurement

a) Learning objective

- Observe and describe how blood pressure is measured using a device.

b) Teaching and learning resources

- Blood pressure monitor (digital or manual, for demonstration)
- Chart showing normal and abnormal pressure ranges
- Text books

c) Activities' guidance and answers

Guidance to the activity 10.9

This activity is meant to help pupils understand the purpose of measuring blood pressure. You will proceed as follows;

i) Display the picture clearly where all pupils can see it (e.g. on a projector, printed copies, or drawn on the board).

ii) Ask pupils to observe silently for 1–2 minutes, then ask one of the pupils to read the guiding questions aloud.

iii) Invite pupils to give their opinions without fear of being wrong.

iv) Accept all answers and use follow-up questions to guide thinking, such as: "Have you seen this done before at a hospital or clinic?".

v) Explain that the nurse is checking the boy's blood pressure using an instrument called a sphygmomanometer or blood pressure meter.

vi) Connect this to the topic of circulatory system health and how we check for diseases like high blood pressure.

vii) Let pupils work in small groups to discuss what they saw in the picture and answer the questions together.

viii) Introduce or review content on how blood pressure is measured, why it matters, and how to prevent high blood pressure.

ix) Reinforce vocabulary like; nurse, blood pressure, sphygmomanometer, healthy habits

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 10.9

1. A nurse and a young boy (or patient).
    - The nurse is wearing a uniform or has medical tools, and the boy is sitting calmly.
2. The nurse is checking or measuring the boy's blood pressure.
    - The boy is getting a health check-up.
    - It looks like a visit to a clinic or health center.
3. Yes, it is called a sphygmomanometer (or a blood pressure meter).
    - It is used to measure blood pressure, how fast or hard blood flows in the body.

Guidance on the application activity 10.9

This activity is meant to enable pupils observe and practice how to measure blood pressure using a blood pressure meter. You will proceed as follows;

i) Begin by telling pupils they will learn how to measure blood pressure using a real tool from the science kit.

ii) Show how to use the blood pressure meter on a volunteer while explaining each step clearly and simply.

iii) Teach pupils what the numbers mean (e.g., normal is around 120/80 mmHg) and why measuring blood pressure is important.

iv) Organize pupils into small groups and supervise them as they take turns measuring each other's blood pressure.

v) Guide pupils to share what they observed and learned and connect the activity to real-life health and disease prevention.

N.B: If the science kit at your school has no blood pressure meter, you can borrow one from the nearest health center or take the pupils out on a hospital visit to observe a health worker measure blood pressure as they explain to them all the steps.

Answers to the application activity 10.9

1. The teacher/pupil selects a calm and seated volunteer.
   - The cuff is placed correctly on the upper arm.
   - The machine is turned on and a reading is taken.
   - Example reading shown on the screen: 118/76 mmHg
2. Systolic pressure (upper number): 118 mmHg
   - Diastolic pressure (lower number): 76 mmHg
   - Pulse rate: 75 beats per minute (example)
   - Measurement is written down on the board or worksheet.
3. Yes, the blood pressure reading is normal.
   - A healthy or normal blood pressure reading is usually between 90/60 mmHg and 120/80 mmHg.
   - This means the volunteer's heart is pumping blood at a safe and healthy pressure.
   - It shows that the person is likely not at risk of high blood pressure at the moment.

10.10. End of unit assessment

I. Guidance on End unit assessment

1. This part provides the guidance and answers for end of unit assessment with cross reference to the textbook.
2. Tell the pupils that the last part of the unit is the end of unit assessment comprising the project where they apply what they learnt and questions that check pupils' knowledge, understanding, skills, attitudes and values related to the whole content of the unit.
3. **For the project:**
   - Request every pupil to do the project either at school or at home. As a teacher, you will decide if it is possible to do the project at school or home and give them guidance. They will use materials available in their community.
   - Request pupils to submit their work, organize a class display table or mini exhibition and allow pupils explain how they made their objects and how it is useful.
   - Reward the objects made or work of pupils by providing feedback and marks.
   - Select some of good objects made by pupils and display them in class.
4. **For the quiz**, choose some questions for the quiz depending on available time for the assessment. The quiz must be done individually whereby pupils will submit their responses on papers to be marked.
5. As all questions may not be done during the period(s) allocated to the end of unit assessment, request them to answer the questions during their self-study time or as homework.

6. Mark their answers and keep records of every student's achievement.
7. Provide the remedial, consolidation and extended activities when necessary.

II. Answers to the end of unit assessment

A. Knowledge and understanding

1. For the answer, refer to the diagram in the student book. This should be a labelled diagram showing the heart, lungs, arteries, veins, with arrows indicating oxygenated and deoxygenated blood movement for the diagram, refer to the picture under the activity 10.3 in pupil's book.
2. Arteries, Veins.
3. Valves.
4. Aorta, Vena cava.
5. High pressure, Low pressure.
6. To fight infections, the clotting process.
7. Oxyhemoglobin (or attached to hemoglobin)
8. 1. A. Blood from the lungs
   2. Pulmonary artery
   3. A.Carries deoxygenated blood
   4. Vena cava
9. b) To transport blood, oxygen, and nutrients
10. c) Carbon dioxide
11. c) Protect the body from infections
12. Aorta
13. Vena cava
14. Valves

B. Skills

15. b) Arteries
16. b) Veins
17. Oxygenated, red
18. Deoxygenated, blue
19. True

C. Attitudes and Values

20. True
21. False
22. True
23. False

10.11. Additional activities

A. Remedial Activities

Questions

1. What organ pumps blood in the body?
2. Name one type of blood vessel.
3. What part of blood carries oxygen?
4. What do we call the red fluid found in the body?
5. Which side of the heart receives deoxygenated blood?
6. Which type of blood cell helps fight infections?
7. What is the name of the main artery in the body?
8. What is the function of valves in veins?

Answers to remedial activities

1. Heart
2. Artery / Vein / Capillary
3. Red blood cells
4. Blood
5. Right side
6. White blood cells
7. Aorta
8. To prevent blood from flowing backward

B. Consolidation activities

1. Name the four chambers of the heart.
2. What is the role of the septum in the heart?
3. What carries blood away from the heart?
4. What happens to blood when it reaches the lungs?
5. What is the liquid part of blood called?
6. Why is regular exercise good for the circulatory system?
7. Which blood vessel allows exchange of nutrients and oxygen?
8. Write one difference between arteries and veins.

Answers to the consolidation activities

1. Right atrium, right ventricle, left atrium, left ventricle
2. It separates the right and left sides of the heart to prevent mixing of oxygenated and deoxygenated blood
3. Arteries
4. It receives oxygen and releases carbon dioxide
5. Plasma
6. It strengthens the heart and improves blood circulation
7. Capillaries
8. Arteries carry blood away from the heart under high pressure; veins carry blood to the heart and have valves.

C. Extended activities

1. Explain why the left ventricle has a thicker wall than the right ventricle.
2. What would happen if the blood had no platelets?
3. How does smoking affect the circulatory system?
4. Suggest two ways to prevent high blood pressure.
5. Compare the structure of an artery and a vein.
6. Why is it important to check blood pressure regularly?
7. Create a flow diagram showing the circulation of blood in the body.
8. John's cousin smokes and often feels tired. Explain how this might be related to his heart health.

Answers to the extended activities

1. Because it needs to pump blood to the entire body, requiring more force.
2. Wounds would not clot quickly, leading to excessive bleeding.
3. It damages blood vessels and increases the risk of heart disease.
4. Avoid salty foods and exercise regularly.
5. Arteries have thick walls and no valves; veins have thinner walls and valves.
6. To detect any problems early and avoid complications like heart attacks.
7. Vena cava → right atrium → right ventricle → lungs → left atrium → left ventricle → aorta → body
8. Smoking reduces oxygen in the blood and damages heart vessels, causing fatigue and poor heart function.

Unit 11

HUMAN RESPIRATORY SYSTEM



i. Key unit Competence:

Pupils will be able to identify the parts of the respiratory system, explain how it works, and demonstrate healthy habits to keep it functioning well.

ii. Prerequisites

Before, the pupils learned about the circulatory system. The also learned about the reproductive and digestive systems. Now, this is a new lesson about the respiratory system. It is also one of the important body systems, just like the circulatory system.

Build on the knowledge of leaners in these areas regarding systems to introduce the respiratory system, its function, its hygiene and the diseases that affect it. Also, during the lessons, strive to bring to the awareness of pupils the fact that this topic is related to respiratory system in biology that they will study at secondary level of learning. Let them understand that at this level, they may only need the basic information otherwise, details of functioning of the respiratory system will be learnt in high school and even better, when later in life, they study medicine and become chest specialists.

Introductory activity

A. Guidance on the introductory activity

- This activity involves reading and the comprehension of the case study. Pupils will be able to find the answers to the reflection questions from the text.
- Ask a pupil to read the case study carefully, and this must be engaging, and you have to make sure that all the pupils are following the one who is reading.
- Pupils will take turns in reading, and it is the role of the teacher to make sure that all pupils are accommodated in regard to their special educational needs. If needed, those who are advanced might need to read slowly for other to catch up.
- After reading, let the pupils answer the questions of reflection and they will have to practice what was done in the scenario.
- Now introduce the respiratory system from the prior knowledge of pupils and from their answers in the activity.
- Guide pupils in the sub-headings that follow.

B. Possible answers to the introductory activity

1. Nose and lungs.
2. Tired, weak, and dizzy. Because there were in a small room and the windows and door were closed.
3. Opening the windows and door, to let the fresh air in the room.
4. Our respiratory system needs fresh air to function well.

iii. Generic competences:

- Critical thinking: Developed through identifying and explaining the functions of the respiratory system, interpreting the mechanism of breathing, analyzing causes and prevention of respiratory diseases, and relating healthy practices to the proper functioning of the respiratory system.
- Research and problem solving: Developed through conducting research on respiratory diseases, investigating the effects of smoking and air pollution, performing practical activities, and suggesting solutions for maintaining respiratory health.
- Communication (Literacy and Numeracy): Developed through presenting research findings, explaining the parts and functions of the respiratory system, discussing health practices, and sharing ideas during group activities and demonstrations.

iv. Basic values:

- Dignity and integrity: Developed through taking personal responsibility for maintaining respiratory health by avoiding harmful practices such as smoking, observing hygiene, and promoting healthy behaviours.
- Self-reliance: Developed through making informed decisions about personal health, applying preventive measures against respiratory diseases, and practicing healthy habits to protect the respiratory system.
- Commitment and resilience: Developed through consistently practicing healthy behaviours, following safety measures during practical activities, and encouraging others to prevent respiratory problems.

v. Cross cutting issues

1. **Standardization culture:** Bring to the attention of pupils the need to seek medical healthcare in standard and quality hospitals whenever they have problems with their respiratory system.
2. **Financial education:** Emphasize the fact that pupils should practice good hygiene to avoid conditions and disease of the respiratory system in order to avoid spending money on treatment.
3. **Gender education:** Emphasize to pupils that anybody irrespective of their gender can pursue a carrier in medicine. Give examples of role models who are successful chest specialists in the area where the learners come from.
4. **Inclusive education:** All pupils should be encouraged to participate during lessons and

practical. Special arrangement should be made to take care of pupils with special needs. For example, provide brail for blind pupils, large print text for those with sight problems and allocate physically challenged pupils to others to assist them during field trips and practical activities. Further, this category should be given tasks that they can manage during the practical sessions.

vi. List of lessons

#	Lesson title	Learning objectives	Number of periods: 8
1	Organs of the human respiratory system and their main functions	• Explain the main function of human respiratory system. • Describe the human respiratory System. • Draw and label the parts of the respiratory system.	2
2	Mechanism of respiration (breathing)	• To explain the process of respiration. • To describe the mechanism of respiration.	2
3	Good health practices and behaviors	• Explain the care of the human respiratory system. • Practice ways to take care of their respiratory system • Show care for the hygiene of the respiratory system. • Develop the habit of regular health check-ups	1
4	Diseases of respiratory system	• Identify main diseases of the respiratory system and state their causes, signs & symptoms and how to prevent them. • Be aware of diseases that affect the respiratory system. • Encourage others to avoid smoking.	1
5	Suffocation	• To define suffocation, state its causes and do first aid to a suffocation victim	1
6	End of unit assessment	• Assess the acquired competences • Demonstrate the achievement of the key unit competence	1

Guidance on different lessons

11.1. Organs of the human respiratory system and their main function

a) Learning objectives

- Explain the main function of human respiratory system.
- Describe the human respiratory system.

b) Teaching and learning resources

- Charts on parts of the human respiratory system and textbooks.

c) Guidance and answers to the activity 11.1



Guidance to the activity 11.1

i. You may begin the lesson by asking pupils probing questions such as: have you ever closed your mouth and nose at the same time for more than one minute? How did you feel?

ii. Guide pupils to conduct the activity 11.1. N.B: Provide assistance to pupils with special needs and disabilities.

iii. Let pupils do the activity and guide them to answer the questions.

iv. At this point, you can identify the organs of the respiratory system and ask their main functions.

v. Wrap up the section as you provide a summary of notes to take.

Note: Ensure to develop the values and generic competencies related to this lesson.



Answers to the activity 11.1

a) They will start to feel uncomfortable like feeling dizzy, heart might beat quickly. because of lack of oxygen.

b) The length might differ, but mostly it is between 60-90 secs.

c) We need oxygen to breath and this is taken via the nose.



Guidance on the application activity 11.1

i. This application activity boosts the critical thinking of pupils and this activity might be given as a homework.

ii. Let pupils reflect on the learnt content and answer the questions.

iii. Provide assistance to pupils with visual impairment on the picture that is under this activity.

iv. During the activity, keep facilitating pupils on how to do and answer the questions, and note their progress.

v. At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarize the content.

Answers to the application activity 11.1

1. Parts:
   a. Trachea
   b. Nose
   c. Lungs
   d. Diaphragm
   e. Bronchiole

2. Refer to the fig 11.1 (Parts of the human respiratory system) under the sub-heading 11.1 (Organs of the human respiratory system and their main functions).

11.2. Mechanism of respiration (breathing)

a) Learning objectives

- To explain the process of respiration.
- To describe the mechanism of respiration.

b) Teaching and learning resources

Items like bell jar, balloons, rubber sheet, stoppers and Y- shaped glass tube, charts on breathing mechanism and textbooks.

Improvisation: You may come up with your own breathing model and use it during the practical activity.

c) Guidance and answers to the activity 11.2

Guidance to the activity 11.2

i. This lesson will involve a practical activity. You will therefore organize the class as need arises during the lesson.
ii. Introduce the activity 11.2 on making a breathing model then demonstrating how it works.
iii. Ask pupils to compare this to the working of the breathing system in human beings.
iv. Let again the pupils assemble the items in this activity 11.2 as shown by the figure under Activity 11.2 in their book.
v. Let pupils observe and state what happens.
vi. Ask pupils what the various items in the model above correspond to and the actions.
vii. Guide the pupils to summarize what the various items and actions correspond to in the respiratory system.
viii. At this point, you can guide pupils to discover the mechanism of breathing in and out. Refer to pupil's book.

ix. Guide pupils to come up with a table on what happens to every organ in the respiratory system during breathing in and out.

x. Let pupils come up with summary notes on breathing mechanism. Assess their work then correct them accordingly.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to activities 11.2

2. The parts represent:

a) Bell jar Chest cavity (rib cage)

b) Balloons Lungs

c) Rubber sheet Diaphragm

d) Y-shaped tube Trachea and bronchi

3. Pull down the rubber sheet:

The balloons inflate.

4. Release the rubber sheet:

The balloons deflate.

5. Observation:

The balloons inflate and deflate repeatedly.

6. Relation to breathing:

Pulling the rubber sheet down is like breathing in (lungs fill with air).

Releasing it is like breathing out (lungs empty of air).

Guidance to the application activity 11.2

i. This application activity 11.2 involves multiple choice questions to choose the correct answer on the mechanisms of respiration and it focuses on communication skills.

ii. Guide the pupils in pairs to do the activity, and this will improve their communication, collaboration and cooperation skills.

iii. While pairing your pupils, make sure that they are paired according to their needs and their abilities so that they could help one another. N.B: Be gender responsive and inclusive in your pairing.

iv. Provide the answers to the provided application activity.

v. During the activity, keep facilitating pupils on how to do and answer the questions, and note their progress. At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarise the content.

Answers to the application activity 11.2

1. b
2. b
3. c
4. b
5. c

11.3. Good health practices and behaviours for the respiratory system

a) Learning objectives

- To explain the care of the human respiratory system.

b) Teaching and learning resources

- Transparent plastic bottle with cap or bell jar, cigarette, two balloons, rubber stopper, Y-shaped glass tube, rubber sheet tissue paper, biro pen casing, matchbox, string, reference materials from the pupil's textbook.
- Improvise the smoking effect model using available materials if needed.

c) **Guidance and answers to the activity 11.3**

Guidance to the activity 11.3

i) Introduce pupils to the science question they are going to be investigating: What are the effects of poor behaviors on our lungs?

ii) This lesson will involve a practical activity and group work. You will therefore organize the class as need arises during the lesson.

iii) Bring textbooks for reference in class.

iv) Guide pupils through the procedures given in activity 11.3 of the textbook.

v) Ask probing questions along the way. For example:

a) Why should the bottle be filled halfway with water? (Answer: to allow atmospheric pressure to push the smoke into the bottle).

b) What is the use of the tissue paper? (Answer: To show the effect of smoking on the lungs).

vi) Let pupils discuss the results of the experiment in their groups. They should write summary notes and nominate a group leader to present on their behalf.

vii) Highlight other points about safety of the respiratory system as listed in pupil's book in the content summary under sub-heading 11.3.

Then, proceed as follows:

viii) Explain to pupils that they have learned about what happens to lungs if you are smoking in the first investigation, and ways to take care of your respiratory system. In this investigation, they are going to determine how healthy their own lungs are.

ix) Divide the class into pairs.

x) Have pupils make the following table in their exercise books to collect data.

	Prediction	**Actual Count**
While sitting		
While exercising		

xi) Explain to pupils that they will be trying to count as high as they can, in a single breath, at a normal talking speed. They will do this while sitting, and again after exercising.

xii) Have pupils predict how high they can count in a single breath. They should record it on their data table.

xiii) Pupils will take turns sitting at their desk, inhaling one big breath, then as they exhale, count as high as they can before they run out of breath. They should record their number in their data table.

xiv) Then, have pupils run in place for 20 seconds. They should immediately repeat step (vi) to find their exercise lung capacity.

xv) Partners switch roles. The recording partner should do the activity while the other pupil records.

xvi) Explain to pupils that a healthy lung capacity for a child is counts above 20 while sitting. While exercising, it should be above 10.

xvii) Discuss with pupils:

a) What might it mean if the pupils' sitting score was under 20?

b) What might a person's score be if they were a smoker? An alcoholic?

c) What might a person's score be if they were eating foods rich in vitamins?

xviii) Restate the science question that was introduced at the beginning of the lesson: What are the effects of poor behaviors on our lungs? Have pupils answer the science question in their own words. Encourage them to use information from the lesson.

Answers to the activity 11.3

- Summarize the practical activity by highlighting the main aim of the experiment which is to find out the effect of smoking on the lungs.
- Possible answers in discussions:

a) If your sitting score is under 20, your lungs are not working to the best of their ability. A pupil might have a respiratory problem or disease.

b) If a person was a smoker, their scores would be very low because smoking and alcohol damage the lungs.

c) If a person eats food rich in vitamins, their score would be high, because they would have a healthy respiratory system.

Possible written response:

Poor behaviors can affect the lungs by damaging them or causing respiratory problems. A damaged or problem lung would cause the lung capacity to go down, as shown during the application activity.

Guidance to the application activity 11.3

i. This application activity 11.3 targets the critical thinking of pupils on the learnt content.

ii. Allow all pupils to participate and note their progress. Make sure you are being gender sensitive and inclusive for pupils with special needs.

iii. Make sure that everyone in the class is participating to ensure communication skills.

iv. During the activity, keep facilitating and provide the reason to the correct answer for those 'True' questions. You can do the same and explain the reason for the 'false' questions.

v. At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarise the content.

Answers to the application activity 11.3

1. False
2. True
3. False
4. True
5. True

11.4. Diseases of respiratory system

a) Learning objectives

To identify main diseases of the respiratory system and state their causes, signs & symptoms and how to prevent them.

b) Teaching and learning resources

- Reference materials on diseases of the respiratory system such as textbooks.
- Teacher may improvise learning resources such as charts or pictures of respiratory diseases based on available materials.

c) Guidance and answers to the activity 11.4

Guidance to the activity 11.4

i. Ask pupils to look at the pictures on the activity 11.4 and let them discuss what they see.

ii. In case you have a pupil, who has visual impairment or disability, let the other pupils support in the description of the image by telling them what they can see. For example: In picture A, a child is coughing and has a cloth to minimize the fever, in picture B, another child is having an asthma pump (asthma inhaler).

iii. Ask the pupils to answer the questions that follow. N.B: Remember that there is no right or wrong answer here, the purpose of the learning activity is to introduce pupils' prior knowledge on the content to be learnt.

iv. Build on the pupils' prior knowledge and introduce them to the possible dangers and safety in using simple machines.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 11.4

1. The child in A is coughing and has a cloth to reduce the fever.
2. The child in B, is having the asthma inhaler (The asthma pump).
3. No, they are sick of some respiratory diseases.

Guidance to the application activity 11.4

i. This application activity involves matching the diseases with their correct symptoms.

ii. Guide the pupils in pairs to conduct the application activity 11.4.

iii. While pairing your pupils, make sure that they are paired according to their needs and abilities so that they could help one another. N.B: Be gender responsive and inclusive in your pairing.

iv. Provide the answers to the table on how the pupils should have matched diseases versus their symptoms.

v. During the activity, keep on facilitating how pupils to do and answer the questions, and note their progress. At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarise the content.

Answer to the application activity 11.4

The answers are:

1. c
2. a
3. d
4. b

11.5. Suffocation

a) Learning objectives

To define suffocation, state its causes and do first aid to a suffocation victim.

b) Teaching and learning resources

- Reference materials on diseases of the respiratory system such as pupil's textbook.
- Teacher may improvise teaching resources (e.g., pictures or demonstrations) based on available materials.

c) Guidance and answers to the activity 11.5

Guidance to the activity 11.5

i. Introduce the lesson using probing questions such as 'Have you or your friend ever been shocked by the water? What did you do in such situation?

ii. Let pupils look at the picture and say what is going on in the picture of the activity 11.5 in the pupil's book. For the pupils with visual impairment assist them or pair them with the advanced leaners to help them in the description of the picture.

iii. This is an individual task that requires pupils to reflect on the picture. Let each pupil share what their saw with the whole class.

iv. Guide the pupils to reach on the intended answers. But do not allow chorus answers, let them answer one-by-one.

v. Build on the pupils' prior knowledge and introduce them to 'Suffocation'.

vi. Summarise the lesson by highlighting the main points as pupils write summary notes.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 11.5

1. The one in a green t-shirt looks weaker.
2. Holding his chest, and he is about to hit him in the back/shoulders.
3. This is an open question for each pupil to be engaged in a lesson (Take an advantage of this to introduce the concepts on suffocation). Answers are in the content of subheading 11.5 point b.

Guidance to the application activity 11.5

i. This application activity 11.5 aims to assess pupils' comprehension on the learnt content.

ii. Provide time to do the exercise individually and note down their progress. It might also be provided as a homework and answers will be given in the next lesson.

iii. Make sure that the instructions on how to complete the activity are clear for every pupil.

iv. During the activity, keep facilitating pupils on how to do and answer the questions, and note their progress.

v. At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarise the content.

Answer to the application activity 11.5

1. It is when someone cannot breathe properly because of no air or something blocking the air.
2. Food or something blocking the throat/trachea.
3. Gently hit between their shoulders.
4. They can faint, get brain damage, or die.
5. Turn them on their side with head down and give mouth-to-mouth if not breathing.

11.6. End of unit assessment

I. Guidance on End unit assessment

4. This part provides the guidance and answers for end of unit assessment with cross reference to the textbook.
5. Tell the pupils that the last part of the unit is the end of unit assessment comprising the project where they apply what they learnt and questions that check pupils' knowledge, understanding, skills, attitudes and values related to the whole content of the unit.
6. **For the project:**
   - Request every pupil to do the project either at school or at home. As a teacher, you will decide if it is possible to do the project at school or home and give them guidance. They will use materials available in their community.
   - Request pupils to submit their work, organize a class display table or mini exhibition and allow pupils explain how they made their objects and how it is useful.
   - Reward the objects made or work of pupils by providing feedback and marks.
   - Select some of good objects made by pupils and display them in class.
7. **For the quiz,** choose some questions for the quiz depending on available time for the assessment. The quiz must be done individually whereby pupils will submit their responses on papers to be marked.
8. As all questions may not be done during the period(s) allocated to the end of unit assessment, request them to answer the questions during their self-study time or as homework.
9. Mark their answers and keep records of every student's achievement.
10. Provide the remedial, consolidation and extended activities when necessary.

II. Answers to the end of unit assessment

A. Knowledge and understanding

1. a) To increase the supply of oxygen to muscle tissues to meet increased demand.
   b) Because our body tissues are starved of oxygen when we hold our breath.
2. a) Chest – ribs move upwards and outwards thereby increasing the volume of the chest cavity. Lungs – expand allowing air in.
   b) Chest – Ribs move downwards and inwards thereby reducing the volume of the chest cavity. Lungs - contract thereby expelling the air.
3. Diaphragm
4. Refer to the activity in the pupil's book.
5. Carbon dioxide, oxygen.
6. Inhalation, exhalation
7. c. Oesophagus
8. b) Trachea
9. c) To bring in oxygen and remove carbon dioxide
10. a) Nose
11. Lungs
12. Diaphragm
13. Oxygen, carbon dioxide

B. Skills

14. a) Balloons
15. b) Eat food rich in vitamins
16. exercise
17. gas mask
18. trachea

C. Attitudes and Values

19. It keeps your lungs healthy
20. Responsibility
21. Check-ups

11.7. Additional activities

A. Remedial activities

1. Name two gases involved in respiration _____ and _____
2. The trachea is also called _____
3. Gaseous exchange in the lungs takes place inside the _____
4. (a) What is most likely to happen if the string is pulled down in the diagram below?_____ (b) What will happen if the rubber sheet is pushed upwards_____?

Answers to the remedial activities

1. Oxygen and carbon dioxide.
2. Wind pipe.
3. Air sacs or alveoli.
4. (a) Balloons will expand.

   (b) Balloons will reduce in size.

B. Consolidation Activities

1. What is the difference between respiratory system and circulatory system?
2. What makes up the respiratory system
3. Suppose we lacked a respiratory system, what will happen?
4. Which organ allows air to get into the breathing system?
5. When the diaphragm flattens during breathing in chest cavity _____.
6. What is the equivalent of these things in a breathing model? (a) Y-shaped tube _____ (b) Bell jar _____
7. What happens in the alveoli during breathing?
8. What is hygiene of the respiratory system?
9. List three practices that will enhance a healthy respiratory system.
10. How will washing hands before every meal help in keeping respiratory system healthy?
11. Give four diseases that affect the respiratory system.
12. Which signs would you look out for in order to know when someone is suffering from: (a) Tuberculosis? (b) Asthma? (c) Bronchitis?)
13. How would you prevent pneumonia in your locality?
14. To suffocate is to _____?
15. What causes suffocation?

Answers to the consolidation activities

1. Respiratory system helps in bringing fresh air into the lungs and removing stale air out of the body. Circulatory system on the other hand helps in transportation of substances in the body.
2. Lungs, trachea, bronchi, bronchioles, diaphragm, alveoli and ribs.
3. We will not survive because our body tissues will lack oxygen, and also, carbon dioxide will not be removed from the body leading to poisoning of body cells.
4. When the diaphragm flattens during breathing in chest cavity _____. (Ans: increases) in volume. This reduces the _____ (Ans: pressure) inside the chest

cavity thereby leading to air rushing into the lungs.

5.(a)Y-shapedtube_____(Ans:Trachea)(b)Belljar_____
(Ans: chest cavity)

6. Gaseous exchange occurs i.e. oxygen is introduced into the bloodstream while carbon dioxide is released into the alveolus airspace

7. Practices that keep the respiratory system healthy

8. Balanced diet, exercising, not smoking or taking alcohol, among others

9. Prevents infection of the lungs

10.Heart attack , stroke, antherosclerosis, high blood pressure among others)

11. Refer to Table 11.4 of Pupil's book

12.: Refer to table 11.4 of Pupil's book

13. To cause to die by to depriving air or oxygen)

14. Chocking, drowning, asthma, inhaling poisonous substances

15. Refer to pupil's book under the summary 11.5

C. Extended Activities

1. Relate the structure of the respiratory system to its functions.

2. Account for darkened lungs in smokers?

3. Using diagrams show the difference between breathing in and out.

4. Would you rather eat apples and grapes or chew sugarcane? Why?

Answers to the extended activities

1. (a) Trachea, bronchi and bronchioles are lined with tough substance made of cartilage, which prevents them from collapsing.

   (b) The nose and nostril have hairs and mucous which filter dust particles and germs.

   (c) There are numerous alveoli, which increase surface area for gaseous exchange.

   (d) The walls of alveoli are very thin to reduce distance travelled by gases during gaseous exchange. (e) Alveoli are surrounded by a network of blood capillaries which supply them with blood.

2. The smoke gets into the lungs, interfere with or stain lung tissues which makes them appear dark. This reduces efficiency of gaseous exchange.

3. For a diagram, refer to the pupil' book.

4. Eat apples and grapes. They are good for healthy lungs as opposed to sugar from sugar cane which affects the lungs.

Unit 12

HUMAN REPRODUCTIVE SYSTEM



i. Key unit competence:

Pupils will be able to identify the parts and functions of the human reproductive system and demonstrate appropriate hygiene practices, prevention and responsible behaviors toward sexually transmitted infections and diseases.

ii. Prerequisites

Pupils have learned the parts of human body and their functions in lower primary. They also know that adult people give birth to children. They observed in their life that children, youth, adults and old people demonstrate different behaviors and characteristics. As teacher, you may ask some questions to check what pupils know on reproduction before starting the new unit.

Introductory activity

Guidance for introductory activity.

i) Ask pupils to observe introductory pictures in Pupil's Book on unit 12. Let the pupils describe each picture.

ii) Help pupils to predict what they are going to learn in the unit.

iii) Read the content in the pupils' books and other relevant texts before the lesson.

iv) Let pupils present their findings from their research and guide them in the discussions.

v) Obtain the relevant charts, models and pictures. Let the pupils to name some parts of their bodies. Encourage them to name reproductive parts well.

vi) Ask pupils to give the functions of some body parts.

vii) Lead the pupils in the discussion of the male and female reproductive systems.

viii) Display charts and pictures of male and female reproductive organs (see pupil book).

ix) Let pupils observe and identify the external parts of the reproductive systems.

x) Display charts of male and female reproductive organs. Let pupils observe and identify their external parts.

Assess pupils' ability to:

- Draw and label external parts of the male and female reproductive system.
- Explain the functions of the male and female genital organs.

Answers for introductory activity,

i) **Picture a**: Female reproductive system

Picture b: Male reproductive system

Picture c: A wife breastfeeding her child.

Picture d: Parents and childdren discussing on different needs.

ii) 8 persons

iii) No, the memberws of the family are too many to find everything.

iv) The family members should work hard

iii. Generic competences:

- Critical thinking: Developed through identifying and explaining the functions of male and female reproductive organs, analyzing changes during puberty, evaluating the causes and consequences of early pregnancy, and discussing ways of preventing STIs and HIV.
- Research and problem solving: Developed through conducting research on reproductive health, investigating causes and prevention of STIs and HIV, analyzing case studies, and proposing responsible behaviours for maintaining reproductive health.
- Communication (Literacy and Numeracy): Developed through group discussions, role plays, presentations, and sharing ideas about puberty changes, hygiene practices, prevention of unintended pregnancy, and respectful ways of supporting people living with HIV.

iv. Basic values:

- Dignity and integrity: Developed through respecting personal dignity, practicing proper hygiene, making responsible decisions related to reproductive health, and showing respect towards people living with STIs or HIV.
- Respect for others and human rights: Developed through promoting non-discrimination, empathy, and support for people living with HIV and AIDS, while respecting individual differences and rights.
- Self-reliance: Developed through encouraging learners to take responsibility for their personal hygiene, make informed choices, protect themselves from infections, and seek appropriate health information and support.

v. Cross-cutting issues to be addressed

1. Gender Education:

Pupils should appreciate and respect the opposite gender.

Teachers should guide them to positively accept and cope with body changes.

2. Comprehensive Sexuality Education:

Adolescents must be taught to avoid risky sexual behaviour to prevent HIV/AIDS and teenage pregnancies.

Emphasize abstinence and responsible behaviour.

3. Inclusive Education:

Both able-bodied and disabled pupils experience physical changes.

All pupils should be included in open, supportive discussions about these changes.

4. Peace and Values Education:

Understanding shared changes promotes mutual respect.

Encourages harmony, friendly relationships, and prevents teasing or discrimination.

vi. List of lessons/sub-heading

#	Lesson title	Learning objectives	Number of periods: 16
1	Functions of human reproductive system	• To give functions of human reproductive system	2
2	Major internal and external parts of the male reproductive organ and their functions	• Identify the male and female reproductive organs. • State the function of genital organs.	1
3	Major internal and external parts of the female reproductive system	• To recognise the female reproductive systems. • To give the functions of the female reproductive system's parts.	2
4	Hygiene of the male and female genital organs.	• To explain how to maintain the hygiene of male \and female reproductive organs. • To show concern to care for the genital male and female organs. • Demonstrate proper hygiene practices for external genital organs • Show care for genital health to prevent diseases.	1
5	Sexual characteristics at puberty	• To identify sexual characteristics of boys and girls at puberty. • To identify principles of responsible sexual behaviour of boys. • To appreciate the importance of talking about changes at puberty. • To show concern to care for genital organs	2

6	Safe and responsible sexual behavior and choices	• To explain responsible sexual behaviour. • To practice the principles of responsible sexual behavior. • To be aware of responsible sexual behaviour	2
7	Prevention of early pregnancy	• Describe unplanned pregnancy to make safe and responsible choices to prevent it. • Describe how to prevent unintended pregnancy.	1
8	Sexually transmitted infections (STIs)	• Describe what STIs are. • Name common sexually transmitted infections (STIs) • Be aware of ways to reduce the risk of infection.	1
9	Means of transmission of common STIs/HIV	• Identify the ways common STIs and HIV are transmitted • Show personal responsibility to protect themselves from STIs and HIV.	1
10	Prevention and treatment of Sexually Transmitted Infections (STIs)	• To explain available treatment for HIV and AIDS and how to prevent it. • Demonstrate how to stay safe and healthy	1
11	Living positively with HIV and AIDS	• Explain how to live positively with HIV and AIDS. • To state how to live positively with HIV and AIDS patients. • Demonstrate ways to support and care for someone living with HIV/AIDS. • Respect people with STIs or HIV without judging them	1
12	End of unit assessment	• Assess the acquired competences • Demonstrate the achievement of the key unit competence	1

12.1: Functions of human reproductive system

a) Learning objectives

To recognize the male and female reproductive systems.

To give the functions of the male female reproductive systems.

b) Teaching and learning resources

- Charts of reproductive system, Picture in Pupil's books, Internet, Relevant textbooks, Models
- The teacher may improvise or use locally available teaching and learning resources based on the context wherever necessary.

c) Guidance and answers to the activity 12.1.

Guidance for human reproductive system.

i) Help identify the main organs of the male and female reproductive systems.

ii) Help pupils to explain the functions of reproductive organs.

iii) Brainstorm pupils to understand the role of hormones in reproduction.

iv) Discuss the importance of reproductive health.

v) Ensure a comfortable, respectful learning environment.

vi) Address maturity levels and encourage open but sensitive discussions.

vii) Set ground rules for respectful communication about the topic.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answer to learning activity 12.1

1. The major function of the reproductive system is **reproduction.**
2. Encourage pupils to speak and brainstorm.

Guidance on application activity 12.1

- Ask pupils to do this activity in pairs.
- Monitor the activity.
- Encourage every pupil to write the answers in their notebooks.
- Mark the pupils and provide feedback.

Answers to the application activity 12.1

The reproductive system is important because it helps people to have babies and continue human life. It also helps boys and girls grow and develop into adults.

12.2. Major internal and external parts of the male reproductive organ and their functions

a) Learning objectives.

- Identify and name the main external reproductive parts of males.
- Describe the basic functions of each external and internal reproductive part.

b) Teaching and learning resources.

- Illustrations of puberty changes.
- Labelled diagrams.
- Charts and flash cards.
- Teachers books, pupils' book and textbook.

Guidance and answers to the activity

Guidance for the activity 12.2.

i) Create a supportive classroom atmosphere where pupils feel comfortable for asking questions.

ii) Make sure to answer any questions clearly and kindly using real facts from science.

iii) Simplify terminology so pupils can understand.

iv) Ensure that pupils know learning about human body changes is normal and important.

v) Use clear diagrams and illustrations of male and female external reproductive parts.

vi) Talking in groups or listening to stories can make learning easier and more fun.

vii) Encourage pupils that they should learn from books, teachers, and other trusted sources to get the right information, even when they are not in school.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 12.2

1. Penis
2. Penis is a male reproductive organ

Guidance on application activity 12.2

- Ask pupils to do this activity in pairs.
- Monitor the activity.
- Encourage every pupil to write the answers in their notebooks.
- Mark the pupils and provide feedback.

Answers to the application activity 12.2

1. Find the answer in pupil's book fig 12.1
2. a) Male external reproductive organs
   b) R: Penis, Q: Testis, P: Pubic hairs
   c) - Penis: Used for sexual intercourse to release the sperm cells to the female vagina
   - Testis: Production of sperm
   - Pubic hairs: keeping the area clean and dry
3. For diagram of a male reproductive system, refer to the Fig. 12.1 of the Pupil's book

12.3. Female external and internal reproductive parts

a) Learning objectives.

- Identify and name the main external and internal reproductive parts of females.
- Describe the basic functions of each external and internal reproductive part.

b) Teaching and learning resources.

- Illustrations of puberty changes.
- Labelled diagrams.
- Charts and flash cards.
- Teachers' books, pupils' book and textbook.

c) Guidance and answers to the activity 12.3.

Guidance for the activity 12.3.

i) Create a supportive classroom atmosphere where pupils feel comfortable asking questions.

ii) Make sure to answer any questions clearly and kindly using real facts from science.

iii) Simplify terminology so pupils can understand.

iv) Ensure that pupils know learning about human body changes is normal and important.

v) Use clear diagrams and illustrations of male and female external reproductive parts.

vi) Talking in groups or listening to stories can make learning easier and more fun.

vii) Encourage pupils that they should learn from books, teachers, and other trusted sources to get the right information, even when they are not in school.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 12.3.

1. Vulva
2. Vulva is a female external reproductive organ

Guidance on application activity 12.3

- Ask pupils to do this activity individually.
- Monitor the activity.
- Encourage every pupil to write the answers in their notebooks.
- Mark the pupils and provide feedback.

Answers to the application activity 12.3

1. Internal parts: ovary, uterus, fallopian tubes, cervix, vagina.

   External parts: labia majora, labia minora, clitoris, urethra, hymen.

2. Refer to the Pupil's book, under the subheading 12.3 (Major internal and external parts of the female reproductive system)

12.4. Hygiene of the male and female genital organs

a) Learning objectives

To explain how to maintain the hygiene of male and female reproductive organs.

b) Teaching and learning resources

Charts, Picture in Pupil's books, Internet, Relevant textbooks, Models, Soap, water

c) Guidance and answers to the activity 12.4.

Guidance for to the activity 12.4.

i) Read the content in the pupils' book and other relevant texts before the lesson.

ii) Obtain the relevant charts, models and pictures before the lesson.

iii) Group the pupils according to their gender.

iv) Let pupils of male gender discuss how they practice hygiene of their genital organs.

v) Lead the pupils in the discussion

vi) Let pupils of female gender discuss how they practice hygiene of their genital organs.

vii) Lead the pupils in the discussion

viii) Demonstrate how underwear should be washed and stored as shown in the activity 12.4 of the pupil's book

ix) Let all pupils practice washing and keeping the underwear properly.

x) Let them write short notes in their notebooks.

xi) Assess pupils' ability to clean their underwear using soap and water.

xii) Assess pupils' ability to explain how to practice hygiene of genital organs.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to Activity 12.4

i) One image includes a sanitary pad, highlighting the importance of menstrual hygiene. Changing pads regularly ensures cleanliness and prevents discomfort or infections.

- Proper storage of underwear.
- Regular hygiene practices.

ii) Refer to lesson 12.4 under subheading 12.4 in pupil book.

Guidance on application activity 12.4

i) This activity will help pupils apply their knowledge about male and female external reproductive parts and practice hygiene habits.

ii) Begin by reviewing key concepts about external reproductive parts and hygiene.

iii) Explain the importance of proper hygiene in maintaining health.

iv) Ensure the class feels safe and comfortable discussing the topic.

v) Have each group share key takeaways from their discussion?

Answers to application activity 12.4

1. See lesson 12.4 hygiene for male and female genital organs.

   Pull back the foreskin and wash underneath with warm water daily to prevent bacterial build-up.

2. Change regularly the underwear after washing the body

12.5. Sexual characteristics at puberty

a) Learning objectives

To identify and name Primary Sexual characteristics in human

b) Teaching and learning resources

- Charts, pictures, reference books, XO laptops
- The teacher may improvise or use locally available teaching and learning resources based on the context wherever necessary.

c) Guidance and answers to the activity 12.5.

Guidance to the activity 12.5 a & b

Guidance sexual characteristics at puberty

i) Read the content before the lesson.

ii) Obtain the necessary pictures and charts for the lesson.

iii) Put pupils in groups of 5 and let them discuss their findings from research

iv) Guide them in their discussions to identify the primary sexual characteristics in boys and girls.

v) Provide additional information that pupils have not discussed.

vi) Assess pupils' ability to identify and name primary sexual characteristics in human.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to activity 12.5 a & b

See the content under the sub-headings 12.5. A (Secondary sexual changes in girls) and 12.5. B (Secondary sexual changes in boys)

Guidance on application activity 12.5.a

- Read the story carefully.
- Identify what is changing in Halima's body.
- Compare her with Liana.
- Compare Halima and Liana.
- Use age and physical changes to reason.
- Think about where Halima learned about puberty.
- Mention the role of adults and teachers
- Let pupils draw a simple picture of a girl before puberty
- Pupils now draw the girl after puberty
- Let them exchange and discuss what they see has changed.

Answers to application activity 12.5.a

i) 1. Breast development – Halima's breasts are growing and sometimes itch.

2. Widening of hips – Halima mentions that her hips are growing bigger.

3. Increased sweating – She says she sweats a lot and needs to wash twice a day.

4. Menstruation (periods) – Her teacher gave them sanitary towels and talked about periods

5. Body shape change – Halima is no longer slim like Liana; her body is becoming curvier.

ii) No, all girls do not enter puberty at the same age.

Evidence: Halima and Liana are the same age (11), but Halima is already going through changes like breast growth and sweating, while Liana is still slim and has not shown these changes.

iii) Girls can get information from teachers and school lessons.

Evidence: Halima's Science and Elementary Technology teacher explained puberty to the class, gave them sanitary towels, and taught them how to use them.

iv) Accept various answers

v) Accept various answers

Guidance on application activity 12.5.b

- Read the story carefully.
- Identify changes that are happening to Franc and Edgar.
- Use a pencil and paper.
- Draw a simple figure of a boy before puberty.
- Show childlike features – small body, thin arms and legs, smooth face, short.
- Draw the same boy, but now after puberty – as a young man.

Answers to application activity 12.5.b

i)

- In height and strength: Franc is taller and stronger than Edgar.
- Broadening of chest and shoulders: Franc has broader shoulders and chest.
- Appearance of body hair: Franc has hair under his armpits and chest.
- Voice deepens: Franc has a deep voice; Edgar's voice is starting to break.
- Oily skin and pimples: Franc has an oily face and some pimples.

ii) It is not always good for boys to get puberty information from peers.

iii) Accept various answers

iv) Accept various answers

12.6. Safe responsible sexual behaviour

a) Learning objectives

- To explain responsible sexual behaviour.
- To practice the principles of responsible sexual behavior.
- To be aware of responsible sexual behaviour

b) Teaching and learning resources

- Charts, Pictures, Illustrations, Models
- The teacher may improvise or use locally available teaching and learning resources based on the context wherever necessary

c) Guidance and answers to the activity

Guidance to the activity 12.6

- Ask pupils to do this activity in pairs.
- Monitor the activity.
- Encourage every pupil to write the answers in their notebooks.
- Mark the pupils and provide feedback.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 12.6

1. A girl refusing gift from sugar daddy.
2. He has a gift
3. Yes, because she is thinking the consequences of receiving the gift.

Guidance on application activity 12.5.b

- Ask pupils to do this activity in pairs.
- Monitor the activity.
- Encourage every pupil to write the answers in their notebooks.
- Mark the pupils and provide feedback.

Answers to the application activity 12.6

1. Friends in picture B are doing correct things because they are playing friendly. Contrarily, the boy who is defending in picture A is being brutal.
2. They should avoid brutality while playing, pushing or kicking others, using bad or rude words, fighting, not sharing the ball with teammates, and refusing to listen to the referee or teacher.
3. Playing safely and follows the rules, respecting others and plays fairly, saying "no" to fighting or dangerous games, choosing kind words and actions, telling a teacher or adult if there is a problem.

12.7. Prevention of unplanned pregnancy

a) Learning objectives

To be aware of the methods to use in order to prevent unplanned (unwanted) pregnancy.

b) Teaching and learning resources

- Contraceptives such as condoms, pills, diaphragms and coils, pipe, string, some water in a tin, charts on vasectomy and tube ligation, textbooks or charts showing the contraceptives.
- The teacher may improvise or use locally available teaching and learning resources based on the context wherever necessary

c) Guidance and answers to the activity 12.7

Guidance to the activity 12.7

i) Begin the lesson by probing pupils with some questions like 'Do you know any friends of yours or neighbors who have kids who got pregnant before marriage? What do you know about early pregnancy?'.

ii) Ask pupils to look the pictures in the activity 12.7 and let them brainstorm for a few minutes.

iii) Select one-by-one technique to guide pupils to make a story from the pictures.

iv) Guide their story to reach the intended result, as they will use the story to answer the questions that follow.

v) Emphasize the fact that this topic is important because many young girls and boys are the victim of unplanned pregnancy. Add that this is another aspect of the cross-cutting issue of Comprehensive Sexuality Education.

vi) At this point, you can introduce the topic on prevention of unplanned pregnancy.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 12.7

1. The stories from pupils will vary, but the main idea is a friendship of young teenagers that resulted into pregnancy and rejection and suffering.
2. Yes, it will.
3. The girl is not ready to be a mother, and the boy is not ready to be a father. This may result to the drop out of school or even death to the girl due to pregnancy complications at a younger age, etc.

Guidance to the application activity 12.7

This application activity 12.7 aims to assess pupils' comprehension on the learnt content.

i) Provide clear instructions on how to complete the activity because this is an individual task, and it may serve as a homework.

ii) During the activity, keep facilitating pupils on how to do and answer the questions, and note their progress.

iii) At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarise the content.

Answers to the application activity 12.7

1. When the girl got pregnant under 18 years old and before marriage.
2. To abstain from early sexual intercourse and wait for marriage.
3. Refer to the content summary under sub-heading 12.4.

12.8. Sexually transmitted infections (STIs)

a) Learning objectives

- To describe common STIs.
- To explain the transmission, prevention and treatment of common STIs.

b) Teaching and learning resources

Reference materials such as textbooks on sexually transmitted infections, their causes, signs and symptoms and prevention/control.

c) Guidance and answers to the activity 12.8

Guidance to the activity 12.8

i) Introduce the lesson by reminding pupils about the previous lesson. Ask them whether they recall any of the sexually transmitted infections (STIs).

ii) This is task to be done with the whole class that requires students to reflect on the pictures.

iii) Guide the pupils to reach on the intended answers. But do not allow chorus answers, let them answer one-by-one.

iv) Build on the pupils' prior knowledge and introduce them to STIs.

v) At this point you can explain the concept of sexually transmitted infections (STIs), and let the pupils take notes.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 12.8

1. At their reproductive genitalia
2. No, there are not.
3. They are suffering from diseases that attack the genital parts.

Guidance to the application activity 12.8

i) This application activity 12.8 involves multiple choice questions to choose the correct answer on the transmission of common STIs/HIV; it focuses on communication skills.

ii) Guide the pupils in pairs to do the activity, and this will improve their communication, collaboration and cooperation skills.

iii) While pairing your pupils, make sure that they are paired according to their needs and their abilities so that they could help one another.

iv) Provide the answers to the provided application activity.

v) During the activity, keep facilitating pupils on how to do and answer the questions, and note their progress. At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarise the content.

Answers to the application activity 12.8

1. c
2. b
3. c
4. c
5. a

12.9. Means of transmission of common STIs/HIV

a) Learning objectives

- To state the meaning of HIV and AIDS.
- To explain the cause of HIV and AIDS.
- To describe how HIV and AIDS is transmitted.
- To describe the symptoms of HIV and AIDS.

b) Teaching and learning resources

Reference materials on HIV and AIDS – may include textbooks, magazines, and any other improvised teaching and learning resources that can demonstrate the means of sexually transmitted infections.

c) Guidance and answers to the activity 12.9

Guidance to the activity 12.9

i) Introduce the lesson by reminding pupils about the previous lesson. Ask them whether they recall any of the sexually transmitted infections (STIs).

ii) This is task to be done with the whole class that requires pupils to reflect on the picture.

iii) Guide the pupils to reach on the intended answers. But do not allow chorus answers, let them answer one-by-one.

iv) Build on the pupils' prior knowledge and introduce them to the means of STIs.

v) Emphasize the need to be very cautious when sharing some shaving materials and razor blades as they can transmit STIs.

vi) At this point you can explain the means of sexually transmitted infections (STIs), and let the pupils take notes.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 12.9

1. A razor blade, a scissor, and a syringe.
2. Yes, they can.
3. Through transfer of blood, from a sick person to a normal person.

Guidance to the application activity 12.9

This application activity 12.9 involves true or false questions on the means of transmission of common STIs/HIV; it focuses on communication skills.

i) Guide the pupils in groups to do the activity, and this will improve their communication, collaboration and cooperation skills.

ii) While grouping your pupils, make sure that they are grouped according to their needs and their abilities so that they could help one another.

iii) Provide the answers to the provided application activity.

iv) During the activity, keep facilitating pupils on how to do and answer the questions, and note their progress. At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarise the content.

Answers to the application activity 12.9

a. True

b. False

c. False

d. True

12.10. Prevention and treatment of Sexually Transmitted Infections (STIs)

a) Learning objectives

To explain available treatment for HIV and AIDS and how to prevent it.

b) Teaching and learning resources

Reference materials on HIV and AIDS – may include textbooks, magazines on the prevention and treatment of common STIs and HIV.

c) Guidance and answers to the activity 12.10.

Guidance to the activity 12.10

i) Introduce the lesson by reminding pupils about the previous lesson. Ask them whether they recall the means of STIs.

ii) This is task to be done with the whole class that requires students to reflect on the pictures.

iii) Guide the pupils to reach on the intended answers. But do not allow chorus answers, let them answer one-by-one.

iv) Build on the pupils' prior knowledge and introduce them to the prevention and treatment on STIs/HIV.

v) Emphasize the need to learn about the prevention and treatment, that it is necessary to safeguard their future and prevent themselves from STIs and HIV. Remind them that the government of Rwanda is mobilizing the youth to avoid anything that can lead to transmission of HIV to young people as they are the future of the country.

vi) Highlight the key points and let the pupils take notes.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 12.10

1. Blue pills
2. Condom
3. Yes, they can.

Guidance to the application activity 12.10

i) This application activity 12.10 targets the critical thinking of pupils on the content learnt.

ii) Allow all pupils to participate, and note their progress. Make sure you are being gender sensitive and inclusive for pupils with special needs.

iii) Make sure that everyone in the class is participating to ensure communication skills.

iv) During the activity, keep facilitating pupils and provide assistance as it involves some short case studies.

v) At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarise the content.

Answers to the application activity 12.10

1.Abstaining from sexual intercourse is the best and the first advice for young people or use condom if they cannot abstain.

2. She should seek medical assistance to ensure the baby is not infected and stays healthy.

3. Suspicious of infection means starting to think you are infected. You seek medical help immediately.

12.11 Living positively with HIV and AIDS

a) Learning objectives

- To explain how to live positively with people living with HIV and AIDS.
- To show how to be accepting of all people.

b) Teaching and learning resources

- Reference materials on HIV and AIDS (textbooks, magazines, etc.)
- Poster paper
- Optional: Crayons, marker pens, paint

c) Guidance and answers to the activity 12.11

Guidance on the activity 12.11

i) Tell pupils that they will be answering the following science question: How can you love and accept all people, including those with HIV and AIDS?

ii) Allow pupils to observe the pictures in activity 12.11. For pupils with visual impairment provide assistance or pair them with the advanced pupils to help them in describing what they see in the pictures.

iii) Guide pupils to answer the questions and by guiding them to the intended ones about living and positively with HIV and AIDS.

iv) Highlight the key points as pupils write down short notes.

Then, proceed as follows for hands-on investigation part:

This part involves creating advocacy posters to encourage the love and acceptance of all people

v) Guide pupils to work in small groups of 3-4 pupils, as this will improve their communication, collaboration, and cooperation skills. When placing pupils into groups, make sure they are placed according to their needs and abilities so that they could help one another.

vi) Instruct pupils to think of messaging or slogans they can use for their poster, so that all people are accepted by family and the community.

vii) Instruct pupils to think of images that would visually show the acceptance of all peoples by their family and the community. If necessary, remind pupils of the second picture in activity 12.11 and explain how this is not an example of this.

viii) Direct pupils to create their poster, using their slogan and visual images, to display the meaning of accepting all people in families and the community.

ix) If allowed, post the posters around the school campus.

Answers to the activity 12.11

1. The couple should receive counselling services to help them accept their situation and avoid denial.
2. In picture (b), the young lady is being isolated and not loved.

Note: On creating posters, be aware that posters will vary. Accept all innovations. Posters should include ideas like love, balanced diet, exercise, sleep, and counselling.

Guidance to the application activity 12.11

This application activity 12.11 involves filling the blank spaces with the correct word that fit the with how to live positively with HIV and AIDS; it focuses on communication skills.

i) Guide the pupils in pairs to do the activity, and this will improve their communication, collaboration and cooperation skills.

ii) While pairing your pupils, make sure that they are paired according to their needs and their abilities so that they could help one another. N.B: Be gender responsive and inclusive in your pairing.

iii) Provide the answers to the provided application activity.

iv) During the activity, keep facilitating pupils on how to do and answer the questions, and note their progress. At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarise the content.

Answers to the application activity 12.11

1. Love
2. Balanced diet
3. Counselling
4. Exercise
5. Sleep
6. Alcohol

12.12. End of unit assessment

I. Guidance on End of unit assessment

1. This part provides the guidance and answers for end of unit assessment with cross reference to the textbook.
2. Tell the pupils that the last part of the unit is the end of unit assessment comprising the project where they apply what they learnt and questions that check pupils' knowledge, understanding, skills, attitudes and values related to the whole content of the unit.
3. **For the project:**
   - Request every pupil to do the project either at school or at home. As a teacher, you will decide if it is possible to do the project at school or home and give them guidance. They will use materials available in their community.
   - Request pupils to submit their work, organize a class display table or mini exhibition and allow pupils explain how they made their objects and how it is useful.
   - Reward the objects made or work of pupils by providing feedback and marks.
   - Select some of good objects made by pupils and display them in class.
4. **For the quiz,** choose some questions for the quiz depending on available time for the assessment. The quiz must be done individually whereby pupils will submit their responses on answer sheets.
5. As all questions may not be done during the period(s) allocated to the end of unit assessment, request them to answer the remained questions during their self-study time or as homework.
6. Mark their answers and keep records of every student's achievement.
7. Provide the remedial, consolidation and extended activities when necessary.

II. Answers to the end of unit assessment

A. Knowledge and understanding

1. b) To help living things have babies
2. b) Penis
3. c) Fallopian tubes
4. Ovary
5. Uterus
6. Penis
7. Four possible effects of irresponsible sexual behavior:
   - Unwanted teenage pregnancy: This can disrupt a learner's education and future plans.
   - Sexually Transmitted Infections (STIs), including HIV/AIDS
   - Emotional stress: Guilt, fear, or depression may result from early or unprotected sexual activity.
   - Stigma and social rejection: discrimination from peers, family, or the community.
8. Appearance of public hair, genitals develop, sometimes pimples appear.

B. Skills

9. a) Bath regularly with clean water and mild soap
10. b) Two or more times a day
11. Report
12. Hygiene
13. True
14. False

C. Attitudes and Values

15. b) Because they need love and support like everyone else
16. Compassion
17. Confidentiality
18. True
19. False

12.13. Additional activities

Remedial Activities:

- Suggestion of Questions and Answers for remedial activities for slow pupils.
- Using charts with unlabelled pictures, ask pupils to identify by saying aloud the parts of the male and female reproductive system.
- Provide pictures that are partially labelled for pupils to complete.

Example of Questions

1. External parts of males: P _ n _ _

2. External parts of females: Mo_ _ P_ b _ _

- Give 'Yes' and 'No' questions to help pupils understand hygiene of male and female genital organs.
- Provide pictures/pictures of male and female for pupils to observe and identify changes during puberty.
- Allow pupils to read customised short stories on responsible behaviour.
- Give pupils leading questions for them to answer orally e.g.

3. The urethra allows _____ and _____ to pass through it.

Answers to Remedial Activities

1. Penis
2. Mons pubis
3. Sperms and urine

Consolidation activities:

- Suggestion of questions and answers for deep development of competences.
- Give pictures for male and female reproductive systems for pupils to label the major parts.
- Give tables to be filled in by the individual pupils after discussion in groups.

Example of Questions

1.

Hygiene of female genitals	Hygiene of male genitals
1	
2	
3	
4	
5	

2• Provide charts for pupils to identify parts of the male and female anatomy that are similar.

Female organ	Male organ
Clitoris	Penis
Ovary	Testis
_____	_____

Answers to Consolidation Activities

1.

Hygiene of female genitals	Hygiene of male genitals
1. Bath regularly.	• Bath regularly.
2. Change into clean underwear.	• Wear clean underwear
3. Wipe genitals.	• Shake penis gently after urinating.
4. Do not insert objects into vagina.	• Uncircumcised boys clean the area under foreskin.
5. Visit the doctor in case of abnormal discharge.	• Seek medial attention in case of abnormal discharge.

2.

Female organ	Male organ
Vaginal opening	Urethra opening

(c) Extension Activity

Discuss and identify other safe responsible behaviour with your parents at home and list to them down.

Answers to Extension Activities

1. Clear water, air, glass window.
2. Transparent, translucent, opaque.
3. Because of the refraction of light at the surface of water

Unit 13

ENERGY MANAGEMENT



i. Key unit competence:

Pupils will be able to identify forms and sources of energy, explain energy transformations, and demonstrate responsible use of clean energy to protect the environment.

ii. Prerequisites

Under energy management pupils will be taught about what energy is, the various forms of energy, its importance and the various energy transformations in our daily lives. Remember pupils have already tackled some aspects of energy in primary 5. For example, they learnt about light, its properties and electricity as a form of energy. Baes on the pupils' experiences in these areas to introduce the concepts in this topic. Ask some questions on the Prerequisites so that you may assess their level of understanding to help you how to proceed with the new content.

Also during the lessons, strive to bring to the awareness of pupils the fact that this topic is related to forms of energy in the subjects they will learn in their secondary studies such as Physics. They may only need the basic information otherwise; details of energy generation and usage are taught in Physics in more.

Introductory activity and guidance

A. Guidance on the introductory activity

i) In this unit, you will teach about the various forms of energy, their sources, energy transformations and renewable and non-renewable energy sources.

ii) Ask pupils to look at the picture of the introductory activity and let them discuss what they see.

iii) Let them brainstorm in five minutes to discover what is taking place in the picture of the introductory activity.

iv) Ask them questions under the picture and try to orient their answers to the right ones.

v) What topics do they think this unit will include based on the picture?

vi) Give time for some brainstorming and after share the main sub-units.

vii) Let them brainstorm and give alternative energy as a solution to the problem situation in the picture.

B. Answers to the introductory activity

1. They are studying
2. They are using a tin lamp.
3. The light is not sufficient. It produces a lot of smoke.
4. The parent should aware that there are other sources of light like solar and biogas that give clean light.

iii. Generic competences:

- Critical thinking: Developed through identifying different forms and sources of energy, analysing energy transformations, comparing renewable and non-renewable energy sources, and evaluating the effects of energy use on the environment.
- Research and problem solving: Developed through researching different energy sources, investigating renewable energy technologies, analysing energy challenges, and suggesting ways of promoting efficient and sustainable energy use.
- Creativity and innovation: Developed through designing models, demonstrating energy transformations, and developing creative ideas for conserving energy and promoting the use of renewable energy.

iv. Basic values:

- Responsibility: Developed through promoting careful use of energy, reducing energy wastage, and understanding individual responsibility in protecting energy resources and the environment.
- Self-reliance: Developed through encouraging learners to make informed choices about energy use, identify suitable energy sources for different needs, and apply energy-saving practices in daily life.

v. Cross-cutting issues to be addressed

1. Gender education

Both boys and girls can assemble biogas digester and a solar facility. It is not a preserve of the boy gender. Also, emphasize to pupils that anybody irrespective of their gender can pursue a carrier in engeering and energy conservation specialist. Give examples of role models who are successful engineers in the area where the pupils come from.

2. Standardization culture

Emphasize the need to use certified energy equipment. Caution pupils against using counterfeits which may lead to accidents or injury of users.

3. Inclusive education

All pupils should be encouraged to participate during lessons and practical. Special arrangement should be made to take care of pupils with special needs. For example,

provide brail for blind pupils, large print text for those with sight problems and allocate physically challenged pupils to others to assist them during field trips and practical activities. Further, this category should be given tasks that they can manage during the practical sessions.

4. Financial education

Pupils should be made aware of the fact that they need to save energy. This they can do by switching of energy utilizing equipment when not in use. Also, buying quality equipment will lead to saving money in the long run as they stay for long and costs on repair are reduced.

5. Environment and sustainability education

Bring to the attention of pupils the fact that some of the devices in this unit such as broken bulbs, used dry cells, spoilt car batteries, solar panels and generators are environmental hazards. They should appropriately be disposed of including some being taken for recycling.

vi. List of lessons

#	Lesson title	Learning objectives	Number of periods:7
1	Definition and forms of energy	• Define the concept of energy • List the forms of energy • Make relevant choice of the best form of energy to use	1
2	Energy transformation/ energy conversion	• Explain ways of energy transformation/ conversion • To representing diagrams on the transformation of energy in different forms • Draw diagrams that show energy transformation.	1
3	Importance of energy	• Explain the importance of energy • Explain why energy is important. • Be aware of using energy wisely and carefully. • Show concern about how energy use affects the environment.	1
4	Sources of energy	• Identify the main sources of energy	1

5	Renewable and non-renewable energy	• Identify and explain renewable energies andstate some examples • To do advocacy ofthe use of renewable energy	1
6	Advantage of using renewable energy	• Explain advantages of using renewable energy. • Encourage the use of clean, renewable energy.	1
7	End of unit assessment	• Assess the acquired competences • Demonstrate the achievement of the key unit competence	1

Guidance on different lessons

13.1. Definition and forms of energy

a) Learning objectives

- Define the concept of energy
- List the forms of energy
- Determine the difference between kinetic and potential energy

b) Teaching and learning resources

- Flat surface to use as a ramp (e.g., stiff cardboard, hard-back book, wooden board)
- Items that will roll down a ramp (e.g., balls, rolled up paper, pencils, cylindrical pens, marker pens, water bottles)

c) Activities' guidance and answers

Guidance on the activity 13.1

i) Introduce the lesson by telling pupils the science question they will be investigating: How does the height of an object affect the energy of the object as it travels down a ramp?

ii) You may begin the lesson by asking pupils probing questions such as: Have you ever heard of the word "energy"? What do you think the word "energy" means?

iii) Show pupils a variety of pictures (a magnet, a stove, a person running, a rock on a cliff). Ask pupils to predict if these objects have energy and write their thoughts in their exercise books.

iv) Divide the class into groups of 3-4 pupils to complete this investigation.

v) Instruct pupils to set up a ramp on their desks using their pupil books and the flat surface.

vi)Instruct pupils to create a data table (from pupil's book) in their exercise book to collect data.

vii) Instruct pupils to release their object at the top of the ramp to let it roll down. Do not give it a push. Count how many seconds it takes to roll down the ramp.

viii) Instruct the pupils to try different heights. Record the heights and the seconds each time.

Then, proceed as follows:

ix) Discuss with pupils:

a) Which height had the fastest time?

b) Which height had the slowest time?

c) What does this tell you about the energy of the object as it was rolling down the ramp?

x) Highlight for pupils that there are many types of energy.

a) **Kinetic energy** is when an object is moving. Ask pupils which height had the most kinetic energy?

b) **Potential energy** is when an object has stored energy, for example when the object was at the top of the ramp. Ask pupils which height had the most potential energy?

c) When the object starts moving, it converts its **potential energy to kinetic energy.**

d) Instruct pupils to demonstrate with the learning resources at their desk when an object would have high **potential energy**, and when an object would have high **kinetic energy.**

xi) Highlight the other forms of energy: chemical, thermal (heat), electrical, and magnetic.

xii) Restate the science question that was introduced at the beginning of the lesson: How does the height of an object affect the energy of the object as it travels down a ramp? Have pupils answer the science question in their own words. Encourage them to use information from the lesson.

Answers to the activity 13.1

- Energy can mean the effort needed to do work.
- Pupil predictions of the pictures will vary.
- Possible answers in discussions:

a) The tallest height should have the fastest time.

b) The shortest height should have the slowest time.

c) The higher the ramp, the more energy the object will have while rolling down.

- Possible written response to science question:

The higher the ramp, the more potential energy the object stores at the top of the ramp. When the object is released, the potential energy is converted to kinetic energy as it travels down the ramp. The higher ramps have more kinetic energy in the objects as they travel down the ramp.

Guidance on the application activity 13.1

i) This application activity boosts the critical thinking of pupils and this activity might be given as homework.

ii) Let pupils reflect on the learnt content and answer the questions.

iii) Help pupils with visual impairment and other disabilities on the picture that is under this activity.

iv) During the activity, keep facilitating pupils on how to do and answer the questions, and note their progress.

v) At the end of the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarise the content.

Answers to the application activity 13.1

1. Energy is the ability to do work.
2. a. The matchstick will lighten up;

   b. The matchstick contains chemicals that store energy. When you strike the match, this chemical energy changes into heat and light.
3. Chemical, thermal (heat), electrical, mechanical, elastic, electromagnetic, etc
4. TV, DVD, refrigerator, electric cooker, electric iron, radio.

13.2. Energy transformation (energy conversions)

a) Learning objectives

Explain way of energy transformation/conversion.

b) Teaching and learning resources

Torch, dry cells, matches and match boxes, pen or comb, pieces of paper, file, metal bar, rubber bands, bulb, switch, kettle, turbines, or charts showing these items, computers connected to the internet.

c) Activities' guidance and answers

Guidance on the activity 13.2

i) You may remind pupils about what they learnt in the previous lesson on forms of energy. Ask Questions such as: (i) What forms of energy do you know?

ii) Let pupils observe the picture in the activity 13.2, and make sure to support pupil with special needs.

iii) Ask pupils whether they think the various forms of energy are inter- convertible and give examples.

iv) Guide pupils to give an explanation of what happened in every activity above. Emphasize

the fact that energy can NEVER be created but can only be transformed from one form to another.

v) At this point, you can introduce the concept of energy transformation as the conversion from one form of energy to another.

vi) Wrap up the section as you provide a summary of notes to take.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 13.2

1. Pupils will name the types of energy as they appear on the picture. They are for example: - Chemical energy
   - Thermal energy
   - Electrical energy
2. Electrical to sound and light in the TV; Chemical energy from food to mechanical in the cyclist; Chemical to light in the torch, Mechanical from the generator to light in the bulb; and chemical from fuel to mechanical in the car.

Guidance to the application activity 13.2

i. Let pupils observe the things happening in this activity as shown. They should then analyze them case-by-case.

ii. This might serve as a homework or an individual work in class.

iii. Guide pupils to understand that the energy transformation that takes place in each case. An example is ironing a piece of cloth using an electric iron.

iv. During the activities, keep on observing and understanding what pupils do and answer, and note their progress. At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarise the content.

Answers to the application activity 13.2

1. Electrical energy, mechanical energy and chemical energy
2. Electrical to thermal energy.
3. Solar energy to electrical energy; and from solar energy to thermal energy.

13.3. Importance of energy

a) Learning objectives

Explain the importance of energy

b) Teaching and learning resources

- Charts of importance of energy, computers connected to the internet for videos.
- The teacher may improvise or use locally available teaching and learning resources based on the context wherever necessary

c) Activities' guidance and answers

Guidance on the activity 13.3

i) You may begin this lesson by asking pupils whether they recall the forms of energy they learnt. Build on this and introduce Activity 13.3

ii) Let them observe the pictures in the activity 13.3. For the pupils with visual impairment, provide assistance or pair them with the advanced pupils to help them in describing what they can see in the pictures.

iii) Guide pupils to answer the questions and by guiding them to the intended one about the importance of energy.

iv) Wind up the lesson by summarizing the lesson and highlight the key points as pupils write down short notes.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 13.3

1. Pupils (in picture a) are studying. They are using electrical energy.
2. The plant (in picture b) is consuming the sunlight energy. The energy being used is solar energy.
3. The radio (in picture c) is on charger. The energy being used is electrical and electromagnetic energy.
4. People (in picture d) are in a picnic and hiking (Climbing mountains). Mechanical energy is being used.
5. The man (in picture e) is cooking. The energy being used is thermal/heat energy.

Guidance to the application activity 13.3

i) This application activity 13.3 involves filling the blank spaces with the correct importance of energy and it focuses on communication skills.

ii) Guide the pupils in pairs to do the activity, and this will improve their communication, collaboration and cooperation skills.

iii) While pairing your pupils, make sure that they are paired according to their needs and their abilities so that they could help one another. N.B: Be gender responsive and inclusive in your pairing.

iv) Provide the answers to the provided application activity.

v) During the activity, keep facilitating pupils on how to do and answer the questions, and note their progress. At the end the lesson, ask again some questions on the performed

activities or content learnt to assess the learning and summarise the content.

Answers to the application activity 13.3

1. Move
2. Sun
3. Devices
4. Charcoals
5. Agriculture

13.4. Sources of energy

a) Learning objectives

Identify the main sources of energy.

b) Teaching and learning resources

Charts on sources of energy such as biogas installations, solar panels, generators, windmills, textbooks, or computers connected to the internet to shows some videos on the sources of energy. N.B: On the application activity 13.4 you will need specific materials refer to the pupil's book under the sub-topic 13.4.

c) Activities' guidance and answers

Guidance on the activity 13.4

i) Put the pupils into groups depending on the class size and their abilities.

ii) Let them observe the pictures in the activity 13.4. For the pupils with visual impairment, provide assistance or pair them with the advanced pupils to help them in describing what they can see in the pictures.

iii) Guide pupils to answer the questions and by guiding them to the intended one about the main sources of energy.

iv) Inform them that in Rwanda, electricity mainly comes from hydro-power generation plants such as WASAC. You might even plan for a visit to this facility or any nearby power generation facility.

v) Wrap up the lesson by highlighting the main points and allowing pupils to write short notes.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 13.4

1. Pupils might see: Hills, the Sun, a jerrycan with a tag of 'Kerosene', a sack of charcoal, a stove of biogas, windmills, and a house with a solar panel and the Sun shining on it.
2. Thermal energy (Sun, charcoal, biogas, solar panel), chemical energy (Charcoal, kerosene), electrical energy (solar panel, windmills).
3. All the pictures show the sources of energy.

Guidance on the application activity 13.4

i) Let pupils assemble the materials as shown in this application activity 13.4. They should then connect them to the bulb and observe what happens.

ii) Guide pupils to understand that the energy being generated here is from the chemical sources and is being transformed to electrical which causes the bulb to light.

iii) Ask them to give an explanation on this.

iv) Allow pupils to brainstorm of other cases where this kind of sources of energy. They should come up with a list and share with other class members.

v) During the activities, keep on observing and understanding what pupils do and answer, and note their progress. At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarise the content.

Answers to the application activity 13.4

- Step 1-3, assembling the materials.
- Q4, the bulb lights on.
- a. Electrical energy
- b. A lemon was the source of energy.

13.5. Renewable and non-renewable energy.

a) Learning objectives

- Identify and explain renewable and non-renewable energies and state some examples.
- To do advocacy of the use of renewable energy.

b) Teaching and learning resources

Solar panels, solar battery, bulbs, chart on an installed solar power plant, computers connected to the internet for videos.

c) Activities' guidance and answers

Guidance on the activity 13.5

i) Let pupils study the two pictures, for the pupils with visual impairment, provide assistance or pair them with advanced pupil to support in describing the pictures.

ii) Initiate a discussion on what happens in the second picture (B) when the charcoal gets finished before food is not yet cooked. The answer will be: 'more charcoal will be

added'. As for the second picture, the cooking process goes on un- interrupted.

iii) Guide pupils to engage in more discussions about the advantages and disadvantages of each. For example, A is clean energy, B - can cause death if used in a room without enough ventilation due to carbon monoxide poisoning, etc. Based on these pupils should feel free to choose which cooking method they prefer.

iv) Narrow down to solar energy as an example of renewable source of energy.

v) Let pupils write summary notes as you do the explanation.

Answers to the activity 13.5

1. In picture A, cooking is taking place using the solar energy.
2. In picture B, cooking is taking place using the charcoal.
3. The charcoal will get finished.
4. The food will not be fully cooked when the source of energy (charcoal) gets finished, and it will require to add more charcoals.

Guidance on the application activity 13.5

i) Introduce the activity by asking pupils to reflect on the classification of renewable and non-renewable energies. Allow them to give reasons for their choice of answer.

ii) Guide pupils to form groups depending on class size and their abilities. Let them carry out the application activity 13.5 in their groups.

iii) Let them answer the question and fill the table in the application activity 13.5

iv) During the activities, keep on observing and understanding what pupils do and answer, and note their progress.

v) At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarise the content.

Answers to the application activity 13.5

Renewable	**Non-renewable**
Wind	Batteries
Solar	Charcoal
Biofuels	Petrol
Ocean	Natural gas
Geothermal	-

13.6. Advantage of using renewable energy

a) Learning objectives

Explain advantages of using renewable energy.

b) Teaching and learning resources

Solar panels, solar battery, bulbs, chart on an installed solar power plant, computers connected to the internet for videos.

c) Activities' guidance and answers

Guidance to the activity 13.6

i) You may introduce the lesson using probing questions such as 'From what just finished learning on the renewable and non-renewable energies, which of the two do you think is appropriate to use?

ii) Let pupils look at the picture and say how the cooking is taking place in the pictures of the activity 13.6 pupil's book. For the pupils with visual impairment assist them or pair them with the advanced pupils to help them in the description of the pictures.

iii) Guide the pupils to reach on the intended answers.

iv) Guide them to discover the importance of using renewable energies as listed in their pupil's books.

v) Summarise the lesson by highlighting main points as pupils write summary notes.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 13.6

1. In picture a, cooking is taking place on firewood.
2. In picture b, cooking is taking place using biogas.
3. The most appropriate to use is biogas, because it is not renewable.
4. The cooking in (b) is environment friendly.

Guidance to the application activity 13.6

i) This application activity 13.6 targets the critical thinking of pupils on the learnt content.

ii) Allow all pupils to participate, and make sure you are **being gender sensitive and inclusive** for pupils with special needs.

iii) Make sure that everyone in the class is participating to ensure communication skills.

iv) During the activity, keep facilitating and provide the reason to the correct answer for those 'True' questions. You can do the same and explain the reason for the 'false' questions.

v) At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarise the content, note their progress.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the application activity 13.6

1. True
2. False
3. True
4. False
5. False

13.7. End of unit assessment

I. Guidance to the end unit assessment

1. This part provides the guidance and answers for end of unit assessment with cross reference to the textbook.
2. Tell the pupils that the last part of the unit is the end of unit assessment comprising the project where they apply what they learnt and questions that check pupils' knowledge, understanding, skills, attitudes and values related to the whole content of the unit.
3. **For the project:**
   - Request every pupil to do the project either at school or at home. As a teacher, you will decide if it is possible to do the project at school or home and give them guidance. They will use materials available in their community.
   - Request pupils to submit their work, organize a class display table or mini exhibition and allow pupils explain how they made their objects and how it is useful.
   - Reward the objects made or work of pupils by providing feedback and marks.
   - Select some of good objects made by pupils and display them in class.
4. **For the quiz,** choose some questions for the quiz depending on available time for the assessment. The quiz must be done individually whereby pupils will submit their responses on papers to be marked.
5. As all questions may not be done during the period(s) allocated to the end of unit assessment, request them to answer the questions during their self-study time or as homework.
6. Mark their answers and keep records of every student's achievement.
7. Provide the remedial, consolidation and extended activities when necessary.

II. Answers to the end of unit assessment

A. Knowledge and understanding

1. B) The ability to do work or make things happen.
2. A) Chemical energy
3. B) Electrical
4. i.
5. Chemical
6. Kinetic
7. B. Heat and motion
8. When there is no fuel there is no chemical energy to move the vehicle. Chemical energy stored in fuel is released during burning and converted to mechanical energy which is used to drive the vehicle.

B. Skills

Choose the correct answer

9. B) Heat energy
10. A) Electrical to kinetic
11. Diagram
12. Experiment
13.

B	**C**	**E**	**L**	**E**	**C**	**T**	**R**	**I**	**C**	**A**	**L**
R	**H**	V	M	Q	V	U	Y	Z	P	J	E
H	**E**	**A**	**T**	L	**S**	**O**	**U**	**N**	**D**	K	L
W	**M**	**E**	**C**	**H**	**A**	**N**	**I**	**C**	**A**	**L**	A
X	**I**	Z	L	G	E	G	F	J	I	Q	S
Y	**C**	Q	D	X	O	R	V	W	L	Z	T
O	**A**	X	O	V	M	N	M	S	F	T	I
W	**L**	**I**	**G**	**H**	**T**	S	U	A	L	D	C
D	M	N	K	Y	R	X	Q	W	L	P	H

C. Attitudes and Values

14. B) It helps protect the environment
15. Positive
16. Wise
17. True

13.8. Additional activities

A. Remedial Activities

1. Name the common sources of fuel in your community.
2. Identify some domestic uses of heat energy.
3. Draw a simple chart to illustrate the energy transformation of fuels during burning.
4. Solar water heater uses energy from the……
5. Compare and contrast solar energy and biogas.

❖ **Answers to the remedial activities**

1. Firewood, paraffin, charcoal.
2. Cooking, warming, heating.
3. It is a stored energy
4. Sun
5. Solar energy is from the Sun, biogas is from cow dung or kitchen wastes; solar energy is safe, biogas may be dangerous in case of leakages; both are clean sources of energy, both are cheap, both are convenient to use.

B. Consolidation activities

1. What is energy transformation?
2. Give two examples of:
   a) Renewable sources of energy
   b) Non-renewable sources of energy
3. Where does energy come from?

❖ **Answers to the consolidation activities**

1. Conversion of energy from one form to another
2. a) Solar, water, wind.
   b) Coal, crude oil, nuclear, natural gas
3. From various sources such as Sun, wind, water, biomass, etc

C. Extended activities

1. Food is often referred to as fuel of the body. Explain the statement.
2. Name two things that are produced when fuels burn.
3. What is potential energy?
4. Mention ways you can use to conserve energy in your community.

❖ **Answers to the extended activities**

1. It is because food has stored energy which is produced when food is digested to make the body move and feel warmth.
2. Heat and light

3. Chemical heat + light

4. – Using energy sparingly.
   - Using energy efficient devices
   - Prioritizing using renewable sources of energy.
   - Emphasizing on the 3Rs of conservation that is, Reduce, Recycle and Re- use.

Unit 14

MAGNETISM



i. Key unit competence:

Pupils will be able to explain how magnets work, identify their types and uses, make a temporary magnet, and demonstrate interest in exploring magnets in daily life.

ii. Prerequisites

This is a completely new concept to the pupils except for having been mentioned in the previous unit as a form of energy. Take advantage of the last unit on energy as you introduce the topic on magnetism, and ask pupils some questions on the what they remember on the form of energy called 'Electromagnetic energy', so that you can build on their prior knowledge while teaching this new topic. Here, you will guide the learning process on the definition of magnetism, types of magnets, definition of magnetic field and its effects, the magnetic compass and finally the uses of magnets.

During the lessons, strive to bring to the awareness of pupils the fact that this topic is related to magnetism that will be furthered in their secondary studies in the subject called Physics. Let pupils understand that at this level, they may only need the basic information otherwise; bring to their attention as well the fact that magnetism is heavily applied in engineering therefore taking this topic is seriously of importance.

Introductory activity and guidance

A. Guidance on the introductory activity

i) In this topic, you will teach about what magnetism is, the various types of magnets, their uses and magnetic field.

ii) As a way of introducing these concepts, refer pupils to the diagram on the introductory activity in their pupil's book.

iii) Let them brainstorm in five minutes to discover what is taking place in the picture of the introductory activity.

iv) Ask them questions under the picture and try to orient their answers to the right ones.

v) What topics do they think this unit will include based on the picture?

vi) Give time for some brainstorming and after share the main sub-units.

vii) Ask pupils to give a quick solution to the mother of the child.

B. Answers to the introductory activity

1. In picture A is Kalisa has accidentally spilled small metallic nails into the flour.
2. In picture B, the mother is having a problem of how she will remove the small metallic nails from the flour and it is proving to be difficult. She is stressed.
3. The mother should be advised to use a magnet to attract these small nails.

iii.Generic competences:

- Critical thinking: Developed through observing and analysing the properties of magnets, comparing magnetic and non-magnetic materials, investigating magnetic forces, and interpreting results from practical experiments.
- Research and problem solving: Developed through researching types and uses of magnets, carrying out investigations to identify magnetic materials, testing magnetic properties, and applying knowledge to explain the use of magnets in daily life.
- Creativity and innovation: Developed through making temporary magnets, designing or imagining useful devices that use magnets, and exploring creative applications of magnets in everyday situations.

iv. Basic values:

- Self-reliance: Developed through encouraging learners to explore magnetic properties independently, conduct simple investigations, make temporary magnets, and apply knowledge of magnets to solve practical problems.
- Hard work: Developed through active participation in experiments, careful observation, recording results, and persistence in investigating how magnets behave under different conditions.

v. Cross-cutting issues to be addressed

1. Gender education

Both boys and girls should participate equally in the activities. Also, emphasize to pupils that anybody irrespective of their gender can pursue a carrier in engineering. Give examples of role models who are successful engineers in the area where the pupils come from.

2. Standardization culture

Emphasize the need to use certified magnetic equipment. Caution pupils against using counterfeits which may lead to accidents or injury of users.

3. Inclusive education

All pupils should be encouraged to participate during lessons and practical. Special arrangement should be made to take care of pupils with special needs. For example, provide brail for blind pupils, large print text for those with sight problems and allocate physically challenged pupils to others to assist them during field trips and practical

activities. Further, this category should be given tasks that they can manage during the practical sessions.

4. Financial education

Emphasize the need to buy quality equipment and devices made of magnets. By buying quality equipment, people save money in the long run as the equipment stay for long and costs on repair are reduced.

5. Environment and sustainability education

Bring to the attention of pupils the fact that some of the devices made of magnets can be environmental hazards. They should appropriately be disposed of or should be recycled.

vi. List of lessons

#	Lesson title	Learning objectives	Number of periods: 7
1	Definition and types of magnets	• Define a magnet • Explain what a magnet is made of • Explain different types of magnets and magnetic fields. • Differentiate types of magnets	1
2	Making a temporary magnet	• Make a temporary magnet	1
3	Characteristics of magnets	• Identify the characteristics of magnets	1
4	Magnetic forces and magnetic materials	• Classify material according to magnetic force • Compare and classify the types of magnets, non-magnets and magnetic materials.	1
5	Magnetic field	• Explain magnetic field	1
6	Magnetic compass, and uses of magnets	• List some uses of magnets in everyday objects. • Show curiosity about how magnets work and how they are used every day.	1

7	End of unit assessment	• Assess the acquired competences • Demonstrate the achievement of the key unit competence	1

Guidance on different lessons

14.1. Definition and types of magnets

a) Learning objectives

- Define a magnet
- Differentiate types of magnets

b) Teaching and learning resources

Bar magnets, pins, paper clips rubber, pencils or pieces of wood, coin, string, charts of other materials on magnetism, computers connected to the internet for videos regarding magnetism.

c) Activities' guidance and answers

Guidance to the activity 14.1

i) Put the pupils in pairs and let them carry out the activity 14.1 in Pupil's book.

ii) Let them observe the pictures and make discussion about the questions provided. For the pupils with the visual impairment, provide assistance or pair them with the advanced pupils to help in the description of the pictures.

iii) Let pupils answer the questions.

iv) Take advantage of the pupils' answers and give the definition and types of magnets.

v) Emphasize the fact that a magnet is a special metal with the ability to pull or push other objects. Let them know that this is what magnetism is about.

vi) Wrap up this lesson by highlighting the various types of magnets as pupils write summary notes.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 14.1

1. Objects in the pictures include nails, a stone (a natural magnet), and a red-and-black material (an artificial magnet).
2. Nails are being attracted by a natural magnet, and in the other picture, the nails are being attracted by an artificial magnet.
3. Similarity is the attracting of nails.

Guidance to the application activity 14.1

i) This application activity 13.1 involves reading and understanding the questions, and it enhances the reading skills of pupils as well as critical think skills.

ii) Allow pupils to read the questions in groups and make sure that by forming groups you are being gender sensitive and inclusive for pupils with special needs.

iii) Make sure that everyone in the group is participating to ensure collaboration and cooperation.

iv) During the activity, keep facilitating pupils on how to do and answer the questions, and note their progress.

v) At the end of the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarise the content.

Answers to the application activity 14.1

i) c. An object that attracts materials made of iron or steel.

ii) a. Magnetite

iii) b. People

iv) c. In a radio or television

v) c. Natural and artificial magnets

14.2. Making a temporary magnet

a) Learning objectives

Make a temporary magnet.

b) Teaching and learning resources

Nail, piece of copper wire, a dry cell, pins or paper clips, masking tape, switch, permanent magnet, and needle, charts, textbooks, handouts and other materials on magnetism, computers connected to the internet.

c) Activities' guidance and answers

Guidance to the activity 14.2a

i) This activity involves critical thinking skills, let pupils observes the pictures and answer the question that follow. **N.B:** Make sure that pupils with **special needs are assisted** and discourage **chorus answers**.

ii) Ask pupils to observe the pictures in the activity 13.2a in pupil's book and answer the questions that follow.

iii) Guide pupils to the intended answers, and take advantage of this to tell them how they can make temporary magnets from the pictures.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 14.2a

1. A person having a magnet in the hand, and iron bar.
2. The iron bar is being rubbed with a magnet.
3. The person in the picture is rubbing the magnet in one same direction.

Guidance to the activity 14.2b

i) Put the pupils in groups of five or any other number depending on class size and let them follow up the procedures as shown in the figure in the activity 14.2b in Pupil's book.

ii) Let them try to pick up pins or paper clips when the nail was rubbed with a magnet for more than 50 times. Let them share their observations.

iii) Guide the pupils to give explanations to their observations above.

iv) Wrap up this topic by highlighting the various types of magnets i.e. natural verses artificial and permanent verses temporary as pupils write summary notes.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 14.2b

a) Rubbing a magnet a few times over an unmagnetized piece such as an iron nail
b) the nail was magnetized.

Guidance to the activity 14.2c

i) Put the pupils in groups of five or any other number depending on class size and let them connect the items as shown in the figure in the activity 14.2c in Pupil's book.

ii) Let them try to pick up pins or paper clips when the wires are connected to the battery and when they are disconnected. Let them share their observations.

iii) Guide the pupils to give explanations to their observations above.

iv) Wrap up this topic by highlighting the various types of magnets i.e. natural verses artificial and permanent verses temporary as pupils write summary notes.

Answers to the activity 14.2c

When the wires are connected to the battery, the nail become a magnet and attracts paper clips or pins, but when they are disconnected, the nail lost the magnetism and does not attract paper clips or pins. i.e the nail becomes a temporary magnet.

Guidance to the application activity 14.2

i) This application activity 14.2 aims to assess pupils' comprehension on the learnt content.

ii) Provide time to do the exercise individually, and note down their progress. It might also be provided as a homework and answers will be given at the beginning of the next lesson.

iii) Make sure that the instructions on how to complete the activity are clear for every pupil.

iv) During the activity, keep facilitating pupils on how to do and answer the questions, and note their progress.

v) At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarize the content.

Answers to the application activity 14.2

1. Temporary magnet loses its magnetic properties, whereas a permanent magnet keeps its magnetic properties.
2. Both temporary and permanent magnet attract magnetic small metals (Keep in mind that the word 'magnetic materials' is not yet learnt).
3. Magnet and iron nail

14.3. Characteristics of magnets

a) Learning objectives

Identify the characteristics of magnets

b) Teaching and learning resources

Bar magnets, pins, paper clips rubber, pencils or pieces of wood, coin, string, charts, textbooks, handouts and other materials on magnetism, computers connected to the internet for videos on earth's magnetism.

c) Activities' guidance and answers

Guidance to the activity 14.3

i) Introduce the lesson by reminding pupils about the previous lesson. Ask them whether they recall the parts of a magnet and what it does to metallic materials.

ii) This is task to be done with the whole class that requires pupils to reflect on the pictures.

iii) Guide the pupils to reach on the intended answers. But **do not allow chorus answers**, let them answer one-by-one.

iv) Build on the pupils' prior knowledge and introduce them to the characteristics of a magnet.

v) Emphasize the need to identify the north pole of the earth's magnet, and make sure that the leaners understand the difference between normal magnet and the earth's magnet.

vi) At this point you can explain the various features of a magnet and its characteristics, and let the pupils take notes.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 14.3

1. Like poles are repelling and unlike poles are attracting.
2. A suspended magnet, the earth and its magnet.
3. A suspended magnet's north pole is being attracted by the south pole of the earth's magnet.

Guidance to the application activity 14.3

i) This application activity 14.3 targets the critical thinking of pupils on the learnt content.

ii) Allow all pupils to participate, and note their progress. Make sure you are **being gender sensitive and inclusive** for pupils with special needs.

iii) Make sure that everyone in the class is participating to ensure communication skills.

iv) During the activity, keep facilitating and provide the reason to the correct answer for those 'True' questions. You can do the same and explain the reason for the 'false' questions.

v) At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarize the content.

Answers to the application activity 14.3

1. False
2. True
3. False
4. False
5. False

14.4. Magnetic forces and magnetic materials

a) Learning objectives

- Compare and classify materials according to magnetic force (magnetic vs. non-magnetic)

b) Teaching and learning resources

- Bar magnets
- Items that are magnetic (e.g., safety pins, sewing needles, nails, iron fillings, coins)
- Items that are non-magnetic (e.g., paper, rubber, string, biro pen casing, glass, cork)

c) Activities' guidance and answers

Guidance to the activity 14.4

i) Introduce the lesson by telling pupils the science question: What types of materials attract magnets?

ii) You may begin the lesson by asking pupils probing questions such as: Do you think all materials found on Earth can be attracted to a magnet?

iii) Let pupils observe the pictures on page 200 of the pupil's book. Ask pupils to identify which objects they are and what materials they are made out of.

iv) Divide the class into groups of 3-4 pupils for this lesson.

v) Ask pupils to make a data table in their exercise book to collect data from the investigation. The data table might look something like this:

Object	Prediction (Magnetic or Non-Magnetic)	Result (Magnetic or Non-Magnetic)

vi) Pass out the magnetic and non-magnetic learning resources to each group. Do not pass out the bar magnets yet. Instruct pupils to name each object, make a prediction if the object will attract a magnet (be magnetic or non-magnetic).

vii) After all pupils have made their predictions, pass out the bar magnets. Allow the groups to test their objects to see which are magnetic and which are not. Pupils should record their results in their data tables.

Then , proceed as follows:

viii) Discuss with pupils

a) Which objects attracted the magnets? What patterns do you notice about the material(s) the objects are made of?

b) Which objects did not attract the magnets? What patterns do you notice about the material(s) the objects are made of?

ix) Highlight for pupils what a **magnetic force** is and what types of materials are attracted and not attracted by magnets.

x) Restate the science question that was introduced at the beginning of the lesson: What types of materials attract magnets? Have pupils answer the science question in their own words. Encourage them to use information from the lesson.

Answers to the activity 14.4

- Pictures: metal nails, metal paper clips/wire, metal screws, metal pins, plastic straws, plastic spoons, masking tape, plastic hanger, and wooden pencils.
- Discussion question answers will vary based on the objects given to the pupils.
- Possible written response to science question:

  Magnets are attracted to items made of iron, steel, nickel, or cobalt. These materials have a strong magnetic force.

Guidance to the application activity 14.4

i) This application activity 14.4 involves filling the materials in their corresponding category in the table and it focuses on communication skills.

ii) Guide the pupils in pairs to do the activity, and this will improve their communication, collaboration and cooperation skills.

iii) While pairing your pupils, make sure that they are paired according to their needs and their abilities so that they could help one another. N.B: Be gender responsive and inclusive in your pairing.

iv) Provide the answers to the table provided.

v) During the activity, keep facilitating pupils on how to do and answer the questions, and note their progress. At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarize the content.

Answers to the application activity 14.4

1. Magnetic force is the pulling or pushing effort of a magnet to attract certain materials.
2.

Material	Magnetic force
Iron	Attracted
A sewing needle	
Iron fillings	
Paper clips	
Staples	
A piece of paper	Not attracted
Pieces of glass	
Sugar	
Copper	

3. Use a magnet to attract and remove the iron filings from the sand.

14.5. Magnetic field

a) Learning objectives

- Explain magnetic field.
- Draw the magnetic fields of various types of magnets.

b) Teaching and learning resources

Bar and horseshoe magnets, iron fillings, a piece of white paper, charts showing magnetic fields of different magnets.

c) Activities' guidance and answers

Guidance to the activity 14.5

i) Put the pupils in pairs or groups depending on the size of the class. Make sure you are gender sensitive and inclusive as you group your leaners.

ii) Provide each group with the materials listed under the activity 14.5. Let them carry out the activities as described in the procedures.

iii) Let them make their observations and record them in their notebooks. They should draw the pattern obtained when the magnet was used.

iv) Guide the pupils to discover the magnetic field lines of bar magnet. If you want, you can give different types of magnets to different groups (A bar magnet to some groups and a horse-shoe magnet to other groups).

v) Did they get it right? If not, take advantage of this and introduce the magnetic field to the pupils.

vi) Wrap up this lesson by highlighting the meaning of magnetic field and clarifying the differences between the various magnetic fields formed by different types of magnets and when like and unlike poles are brought close together.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 14.5

a) The iron fillings are attracted by tthe magnet and form the pattern

b) The pattern should appear as shown in the content under this activity 14.5 in the pupil's book.

Guidance to the application activity 14.5

i) Put the pupils in pairs or groups depending on the size of the class.

ii) Provide each group with the materials listed. Let them carry out the activities described under the application activity 14.5 in the procedures.

iii) Give some groups to do the activity under case 1 and other groups should do the activity under case 2.

iv) One group should proceed by bringing like poles of same magnet and the other group should proceed by bringing the unlike poles together. They should observe the pattern formed by the iron fillings and come up with a drawing.

v) Let them compare their drawings to those in case 1 and in case 2 for bar magnet. Guide them to correct their drawings accordingly.

vi) During the activities, keep on observing and understanding what pupils do and answer, and note their progress. At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarize the content.

Answers to the application activity 14.5

The patterns for both case 1 and case 2 should appear as shown in the content under the activity 14.5 in the pupil's book. And the answers for the reflection questions are also under the same content.

14.6. Magnetic compass and uses of magnets

a) Learning objectives

- Explain magnetic compass and its uses.
- List some uses of magnets in every day tools.

b) Teaching and learning resources

Bar or horseshoe magnet, clear plastic jar with water, three needles, thread, pencil, piece of manila paper, clay or plasticine, tape, magnetic compass, a piece of needle or pin, sand, grass, charts or textbooks on uses of magnets.

c) Activities' guidance and answers

Guidance to the activity 14.6

i) This activity involves critical thinking skills, let pupils observes the pictures and answer the question that follow. N.B: Make sure that pupils with special needs are assisted and discourage chorus answers.

ii) Ask pupils to observe the pictures in the activity 13.6 in pupil's book and answer the questions that follow.

iii) Guide pupils to the intended answers, and take advantage of this to discuss how magnets are used in items from the pictures.

iv) Facilitate the pupils to summarize the lesson by providing the main points from their findings based on the uses of the magnets in everyday life.

v) Let them take notes of the summary.

Note: Ensure to develop the values and generic competencies related to this lesson.

Answers to the activity 14.6

1. A compass
2. It indicates direction (Keep this answer short as the pupils might have limited knowledge on this before learning the lesson).
3. Microphone, speakers, TV, casette player, sound mixer.
4. Yes, they are useful in our lives.
5. The all use electromagnetic energy to function (Remind pupil of Unit 12 as a Prerequisites for you new lesson)

Guidance to the application activity 14.6

i) Put the pupils in groups depending on the size of the class. Provide each group with the materials listed.

ii) Guide the pupils to carry out the activities described under this application activity 14.6. At the end of the application activity, they would have made a home-made magnetic compass and tested how it works. Let them make their observations and record them in their notebooks.

iii) Throw the magnetic compass next to the equipment that they made. Let them note the direction where the north pole points compared to their equipment.

iv) Guide the pupils to recall the use of a magnetic compass in showing direction.

v) During the activities, keep on observing and understanding what pupils do and answer, and note their progress. At the end the lesson, ask again some questions on the performed activities or content learnt to assess the learning and summarize the content.



Answers to the application activity 14.6

1. When a magnet is situated on the surface of the Earth, it functions similarly to a magnetic compass, with its north pole being drawn toward the Earth's magnetic south pole.
2. The magnet behaved as a pointer of the compass and it settled in a direction that is always south of the Earth's magnetic force.
3. The magnetic compass is used to tell the direction where one is going or coming from.
4. Magnets are used in the refrigerators, electric bells, in speakers, in TVs, and telephones and radios, etc.

14.7. End unit assessment



I. Guidance to end unit assessment

1. This part provides the guidance and answers for end of unit assessment with cross reference to the textbook.
2. Tell the pupils that the last part of the unit is the end of unit assessment comprising the project where they apply what they learnt and questions that check pupils' knowledge, understanding, skills, attitudes and values related to the whole content of the unit.
3. **For the project:**
   - Request every pupil to do the project either at school or at home. As a teacher, you will decide if it is possible to do the project at school or home and give them guidance. They will use materials available in their community.
   - Request pupils to submit their work, organize a class display table or mini exhibition and allow pupils explain how they made their objects and how it is useful.
   - Reward the objects made or work of pupils by providing feedback and marks.
   - Select some of good objects made by pupils and display them in class.
4. **For the quiz,** choose some questions for the quiz depending on available time for the assessment. The quiz must be done individually whereby pupils will submit their responses on papers to be marked.
5. As all questions may not be done during the period(s) allocated to the end of unit assessment, request them to answer the questions during their self-study time or as homework.
6. Mark their answers and keep records of every student's achievement.
7. Provide the remedial, consolidation and extended activities when necessary.

II. Answers to the end of unit assessment

A. Knowledge and understanding

Choose the correct answer

1. b) An object that attracts materials made of iron or steel
2. b) Magnetite
3. a) Natural and artificial magnets
4. Magnet
5. Poles
6. Temporary, permanent
7.

Magnetic materials	Non-magneticmaterials
Iron	Paper
Steel	Cork
Nails	Wood
Pins	Plastic
Needle	Ceramic
Paper clips	Pieces of cloth

8. i
9. Magnetite or lodestone.

B. Skills

10. a) By rubbing a nail with a magnet in one direction
11. b) Iron nail
12. b) They repel
13. Iron, steel
14. Paper, plastic
15. When separating mixtures of magnetic and non-magnetic substances, in finding lost magnetic

    Items, in telling direction (magnetic compass), etc

C. Attitudes and values

16. True
17. False
18.. True
19. False

14.8. Additional activities

A. Remedial Activities

1. How would you differentiate magnetic from non-magnetic substances?
2. Look around you. List some magnetic materials that you can see.
3. Like charges attract while unlike charges repel. True or false?
4. What is a magnet?

❖ **Answers to the remedial activities**

1. By bringing a magnet close to the item. If it attracts, it is magnetic; if it doesn't, it is not.
2. Magnetic: Nail, pin, paper clip, door handle. Non-magnetic :wooden chair/table, biro pen case, book, etc)
3. False
4. An object/item that attracts or repels certain materials.

B. Consolidation activities

1. Why is it more appropriate to use a magnet to separate iron fillings from flour?
2. What is magnetic compass?
3. How can you make a temporary magnet using a permanent magnet?

❖ **Answers to the consolidation activities**

1. Because it will attract all the iron fillings living behind pure flour.
2. An instrument or a device that uses magnetized steel bar to indicate direction relative to the Earth's magnetic poles.
3. Moving the metal to be magnetized in the same direction, rather than back and forth. Using the same quick motion you would use to light a match and continue rubbing the material with the magnet about 50 times as quickly as you can.

C. Extended activities

1. Explain why it is possible to tell direction using a magnetic compass.
2. How are magnets used in 1
    a. Speaker?
    b. Refrigerator?
3. Explain why electromagnets are used in lifting of heavy machinery at the airport.

❖ **Answers to the extended activities**

1. The Earth has magnetic materials at its core. As such, it is kind of a huge magnet. Being a magnet, it has North and South poles. When a magnetic compass is placed on the Earth's surface. Its North pole is repelled by the Earth's magnetic North pole. Therefore, the north pole of the magnetic compass always settles southwards. Using this knowledge, we can drop a magnetic compass and look at the pointer. Where it points is the south of the earth hence we can use that information to tell the direction where we are going or coming from.

2. (a) A speaker converts electrical signals into sound energy. By moving the cone back and forth, the speaker increases and decreases the air pressure in front of it thus creating sound waves.

   When a fluctuating electric current (from the sound system such as radio or TV) flows through the coil, it becomes a temporary electromagnet, attracted and repelled by the permanent magnet. As the coil moves, it moves the cone back and forth, pumping soundwaves into the air. That is how we get to hear the sound.

   (b). In the doors to keep the doors locked.

3. Huge magnets are very powerful hence can be used to lift heavy loads. Additionally, electromagnets are made from magnetic cores and coiling, which use electricity. It is therefore possible to determine what amount of force is generated or even manipulate the amount of force generated, plus it can be switched on to lift the load and switched off to drop it once the point of offloading is reached.

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